Computational Emulation of Nuclear Dynamics

M1 Research Internship Report

Mike Silva

Supervised by Dr Chloë Hebborn
Co-supervised by Dr Patrick McGlynn
IJCLab and the Few Body Reaction Group, FRIB
Referee: Pr Guillaume Patanchon

IJCLab, Laboratoire de Physique des 2 Infinis Irène Joliot-Curie
UFR de Physique, Université Paris Cité
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Abstract

Theoretical nuclear physics needs scattering observables for nuclei far from the valley of stability. Those predictions rest on the optical potential, the effective nuclear interaction standing in for every reaction channel not treated explicitly, whose parameters are calibrated near stability while the predictions are wanted far from it. A prediction therefore has to carry a quantified uncertainty, and quantifying it takes between ten thousand and a million solves of a coupled-channel scattering problem, which puts the campaign out of reach. This internship addresses that cost by emulation.

A high-fidelity multichannel solver based on the Direct Boundary Matching Method was implemented, for single-channel and for coupled-channel problems, with distinct channel thresholds, closed channels and complex optical potentials, and validated against an independent calculable R-matrix code. Two emulators were built on it: a reduced basis method already benchmarked in the literature, and a learning reduced-order model whose coupled-channel version is produced here. They were measured on four benchmark systems, spanning a spin-orbit doublet, a rotational band with two thresholds inside the energy window, and two rigid rotors carrying quadrupole and hexadecapole deformations, with up to eight coupled channels and up to ten free parameters. A fifth system, a one-phonon vibrational excitation, carries the single test on which an undivided predictor budget fails.

Both emulators reproduce the solver below the tolerance adopted here, $\epsilon_{\mathrm{post}} = 5 \times 10^{-3}$, at the median over the parameter box, on all four systems, and both are faster than the solver exploiting the same affine split, by a factor that grows with the size of the system. The learning model is cheaper on all four, with a gain against the same baseline 1.3 to 1.9 times larger. Which of the two is the more accurate depends on the system and on the quantity scored. For a campaign limited by the number of forward evaluations I recommend the learning model. Every statement here holds over the parameter boxes of Appendix C, which run from ± 10 per cent on a radius to -50 to $+100$ per cent on a weakly constrained absorption depth, on one J^{π} block, and against this solver rather than against experiment.

Keywords: Optical potential; coupled-channel scattering; Direct Boundary Matching Method; reduced basis method; learning reduced-order model; reaction emulation; Bayesian uncertainty quantification.

Acknowledgements

My first thanks go to Chloë Hebborn and Patrick McGlynn, who supervised this internship. They gave me a real problem rather than an exercise, the freedom to get it wrong, and the time to understand why. Patrick's R-matrix code is the reference every result here is measured against, and the idea of splitting the predictor budget between the diagonal and the coupling blocks, at the heart of the coupled-channel LROM, came out of a conversation with Chloë and him. What I owe Chloë goes well beyond the physics. She taught me what it takes to work efficiently, consistently and seriously in research, spent a great deal of time explaining how that world actually functions, and gave me long conversations about academic and professional futures that were among the most useful hours of the summer. Beyond all of that, both were unfailingly kind, and they made these weeks the best I have spent in a laboratory.

I thank Guillaume Patanchon for agreeing to referee this report. I am grateful to the whole Few Body Reaction Group at FRIB[1] for making an outsider welcome in their discussions, and to Pablo Giuliani in particular, whose learning reduced-order model is the second half of this work and who helped me understand his single-channel construction, with a patience I did not expect from someone I had never met. Thanks to Leandro and Barabas, who shared the office with me during their L3 internship from the magistère de Saclay, for long conversations of wildly varying productivity, and to Denis Lacroix for an afternoon on what quantum information might and might not do for nuclear physics, which was the most interesting thing I did not work on this summer.

A few things belong here that will not appear anywhere else in this report. The quiet of Orsay, which turned out to be exactly the right setting for thinking. The interminable commute between the RER E and the RER B, which was not. The canteen, which was genuinely good. The heatwave that drove me to work from home for a stretch of it. The evenings spent oscillating between the code and the World Cup, and the disappointment of watching both Portugal and France go out. The stress of presenting this work in front of the whole laboratory, and then in front of the American Few Body team. And above all the struggle, longer than any of the physics, of writing code that is genuinely *clean* and still works.

Finally, to Julie, who has shared my life for five years and who put up with a summer of coupled channels: thank you.

[1] https://sites.google.com/view/frib-few-body-reaction-group/

Contents

Appendices

How to read this report. In five minutes, the abstract and Figure 18. In thirty, add Sections 3.4, 4.4 and 5. The appendices hold the detail behind the body and no step of the argument; nothing in Sections 1 to 6 depends on reading them.

1 Introduction

> Physical motivation and objectives of the internship. Nothing here is original. The status of optical potentials for exotic nuclei follows the review of Hebborn *et al.* [1], the astrophysical context follows Iliadis [2] and Thompson and Nunes [3], and the case for Bayesian uncertainty quantification follows Drischler *et al.* [4].

1.1 Cross sections and the effective interaction

At a rare-isotope beam facility such as FRIB, GANIL or RIKEN, a nuclear reaction is the instrument through which the structure of an unstable nucleus is measured. Elastic scattering measures the size and diffuseness of the matter distribution, inelastic scattering excites and identifies collective modes, transfer reactions extract single-particle occupancies, and breakup reveals the structure of weakly bound systems. The same quantities enter stellar models, where the rate of a reaction is the average of its cross section $\sigma(E)$ over a Maxwell-Boltzmann distribution of relative energies [2], [3]. A rate therefore requires $\sigma(E)$ over an energy range and not at a single energy, which is a constraint the emulators of this report have to meet. In both cases the observable is a cross section, and its interpretation requires a reaction model.

Figure 1 shows what these reactions have in common. A projectile meets a target, and the collision can end in any of several distinct final configurations, each called a *channel*. A few are of direct interest and are described explicitly. The rest still drain flux out of them and must be accounted for. That is the role of the *optical potential*, a complex, energy-dependent effective interaction standing in for every channel not treated explicitly (Section 2.3). It is the common ingredient of essentially all direct reaction theory: the same object underlies an astrophysical capture rate and the analysis of a breakup experiment [1], [5].

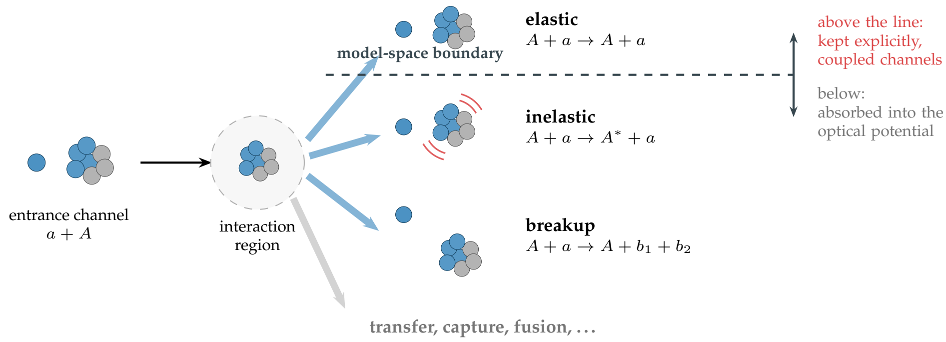

Figure 1: A binary nuclear reaction and its channels. The model-space boundary separates the channels kept explicitly from those absorbed into the optical potential; moving it down promotes a channel to explicit treatment and couples it to the others (Section 2.4). A promoted channel must then no longer be counted in the absorption, so the optical potential of a coupled-channel calculation is not the one of a single-channel calculation.

1.2 Why theory is needed far from stability

The reactions of greatest interest cannot be measured, for two reasons that compound each other. The first is energetic. Two nuclei of charges Z_1 and Z_2 must penetrate a Coulomb barrier of height $Z_1 Z_2 e^2/R_N$, which is 2.2 MeV for $p + {}^{12}\mathrm{C}$ with $e^2 = 1.44$ MeV·fm and $R_N \simeq 4$ fm, while stellar thermal energies are $kT \sim 1$–100 keV, one to three orders of magnitude below. Fusion then proceeds only by tunnelling and the cross section collapses exponentially, so the energy window that contributes to a stellar rate sits far below the reach of accelerators for short-lived species [2]. The accessible energies are measured higher up and extrapolated down, and the reaction model used for the extrapolation sets its accuracy. The second obstacle is that the nuclei of interest often cannot be made into a target at all. About 3340 nuclides have been observed and evaluated [6], against mass-model estimates of the bound species ranging from some 7000 upward depending on the functional used [7], and the unobserved ones lie predominantly on the neutron-rich side. Where a beam of them can be produced, it is weak and short-lived, which limits both the statistics and the number of observables that can be measured.

The missing predictions are therefore supplied by theory, through the optical potential. Its strengths and geometrical parameters, collected in a vector $\theta \in \mathcal{P} \subset \mathbb{R}^p$ with p typically between 2 and 10, are calibrated on elastic-scattering data. Figure 2 shows the difficulty this creates. The calibration data lie near stability, the predictions are wanted far from it, and the extrapolation carries an uncertainty that must be quantified [1].

Quantifying it is a problem of Bayesian inference [4]. One builds the posterior $p(\theta \mid \text{data})$ and integrates over it to obtain a probability distribution on the cross section, and hence on the reaction rate. The obstacle is arithmetic. A

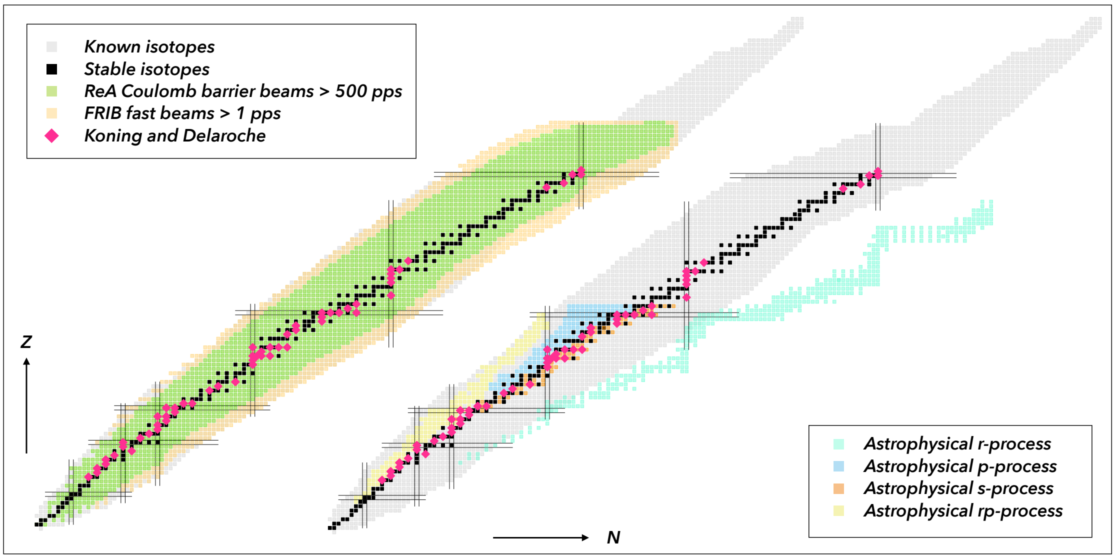

Figure 2: Chart of nuclides, with the estimated reach of rare-isotope facilities and the main astrophysical processes. The r-process path runs through neutron-rich nuclei far outside the region where elastic-scattering data, the calibration data of optical potentials, are available. Figure from Hebborn *et al.* [1].

Markov chain over the posterior needs 10^4 to 10^6 evaluations of the forward model, and one evaluation means solving a coupled-channel scattering problem, summed over partial waves, at every energy of the data set. A single J^π block at one energy costs under a millisecond to about a tenth of a second on the systems benchmarked here. Multiplied by the contributing blocks and the measured energies, one evaluation is seconds to minutes, and the campaign is 10^4 to 10^6 times that. The cost of a campaign comes from the number of solves and not from the cost of any one of them, and the whole campaign is re-run whenever the model or the data change. Computing a cross section is a solved problem. The problem addressed here is computing it often enough, and fast enough, for its uncertainty to be quantified.

1.3 Objectives and inherited material

The work was carried out in June and July 2026 at IJCLab, the Laboratoire de Physique des 2 Infinis Irène Joliot-Curie, in Orsay, within the theory group working on nuclear reactions. It was supervised by Chloë Hebborn and co-supervised by Patrick McGlynn. The group collaborates with the Few Body Reaction Group at FRIB, Michigan State University, and needs quantified uncertainties on reaction observables for nuclei far from stability, which the rare-isotope facilities now coming into operation will measure. Emulation is the answer the field has converged on over the last five years [4], [8], and its application to coupled channels, which is where nuclear reactions live, was still open when this internship began.

The subject as it was set is the emulation of nuclear dynamics by learning methods. The starting point is an intuition of C. Hebborn and P. McGlynn: the Direct Boundary Matching Method (DBMM) of Lei [9], and the reduced basis emulator he builds on it [10], should carry over to coupled channels and should scale with the number of channels.

The approach is *emulation*. The forward model is a smooth map $\theta \mapsto \mathbf{U}(\theta)$ from parameters to the collision matrix. A reduced-order model builds it from a small number of expensive solves computed once, then replaces each forward evaluation by a cheap query. The reduction acts on the governing linear system itself and not on a fit to its output. The equation is retained, and the accuracy comes from a subspace whose quality is measured offline.

Three things were asked for at the start, and a fourth was a standing constraint:

1. implement the DBMM for single-channel and for coupled-channel problems, with complex optical potentials and channel thresholds crossed inside the energy range, and validate it against an independent calculable R-matrix code;
2. build the reduced basis method (RBM) of Lei on top of it, and benchmark it;
3. cover a whole energy window rather than a single energy;
4. keep the construction compatible with an extension to the three-body problem, which is the subject of the group's parallel programme.

The work moved once the DBMM and the RBM were running. Discussions with P. Giuliani opened a second branch: his learning reduced-order model (LROM), built for a single channel. Extending it to coupled channels became the substance of the internship, and no coupled-channel version of it existed before (Section 4.3; Section 4.5 sets it against the coupled-channel emulators already published). The route first envisaged for the third objective was given up for it. A single emulator carrying E as a coordinate was implemented and not validated: the channel thresholds fall inside the window and change the number of open channels across it, which the construction does not handle (Appendix E.10). The window is covered by one emulator per energy node instead, so the third objective is met by twenty emulators rather than by one. Everything here is done on two-body problems, as a proof of concept for the three-body case of the fourth objective. McGlynn and Hebborn [11] solve two-neutron halo nuclei with the calculable R-matrix method, on the same Lagrange-Legendre mesh and with the same diagonal potential matrix, and their hyperradial equation is structurally the coupled-channel system of Section 2.4, with many more channels than any benchmark here (Section 6).

This work sits inside an active research effort, so the boundary between what was inherited and what was produced has to be explicit. Figure 3 traces it. The same equation is addressed by two solvers, by different means: the calculable R-matrix method, whose Python implementation in the group is due to P. McGlynn and predates the internship, and the DBMM, whose formulation is due to Lei and whose implementation is the first deliverable here. The emulators descend from the DBMM, because the DBMM exposes the algebraic structure they need. The RBM branch follows Lei and the ROSE package of Odell *et al.* The LROM branch descends from a single-channel construction of Giuliani, combined here with the DBMM to produce a coupled-channel version that did not previously exist.

Section 2 defines the quantity all of this predicts, and Section 3 the structure of the matrix the two emulators are built on.

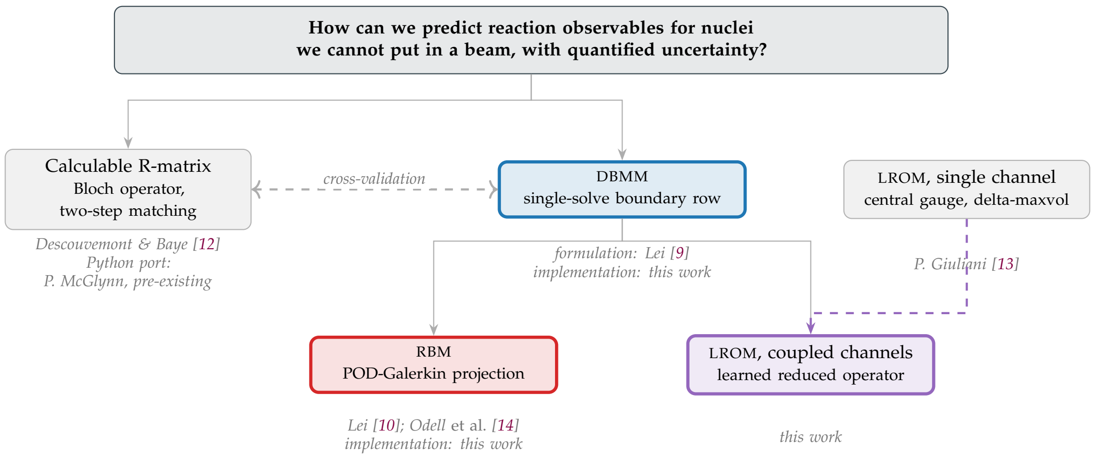

Figure 3: Lineage of the methods used and produced in this work. Grey boxes are inherited, from the literature or from code written before this internship. Outlined boxes are implemented during the internship. The dashed arrow marks the combination of Giuliani's single-channel construction with the DBMM, which gives the coupled-channel LROM.

2 From a collision to a linear system

A compressed account of standard scattering theory, retaining only what is needed to define the collision matrix **U** and to write the linear system that Section 3 discretises. Sources: Thompson and Nunes [3] for the general formalism, Glendenning [15] and Paetz gen. Schieck [16] as parallel textbook treatments, Descouvemont and Baye [12] for the notation and the interior/exterior construction, and the reviews of Hebborn *et al.* [1] and Holt and Whitehead [5] for the optical model. The R-matrix formalism itself is stated without derivation and referred to Appendix B.2.

2.1 Reduction to a radial system

Consider two nuclei of masses m_1, m_2, charges Z_1, Z_2 and mass numbers A_1, A_2, at positions $\mathbf{r}_1, \mathbf{r}_2$, interacting through a central potential, particle 1 being the projectile. Introducing the relative coordinate $\mathbf{R} = \mathbf{r}_1 - \mathbf{r}_2$ and the centre-of-mass coordinate, the Hamiltonian separates exactly into a free centre-of-mass term and a relative-motion term

$$H_{\text{rel}} = -\frac{\hbar^2}{2\mu}\vec{\nabla}^2_{\mathbf{R}} + V(R), \qquad \mu = \frac{m_1 m_2}{m_1 + m_2},$$

so the collision is equivalent to the scattering of a single fictitious particle of mass μ off a fixed potential $V(R)$. Everything below is written in the centre-of-momentum frame. The relative motion is governed by the stationary Schrödinger equation at positive energy $E = \hbar^2 k^2/2\mu$, related non-relativistically to the laboratory bombarding energy by $E = E_{\text{lab}}\, m_2/(m_1 + m_2)$.

The potential splits into two pieces of very different character,

$$V(R) = V_n(R) + V_C(R),$$

a short-ranged nuclear component, negligible beyond a nuclear range $R_n \simeq 1.2(A_1^{1/3} + A_2^{1/3})$ fm, and the Coulomb component $V_C = Z_1 Z_2 e^2/R$, which never vanishes. Every numerical method in this report is built on that distinction. The nuclear term does not stop abruptly at R_n: a Woods-Saxon still holds half its depth there and dies over the diffuseness a_0. A usable channel radius is therefore $a \gtrsim R_n + 6a_0$ to $R_n + 8a_0$, and more on a charged system, where the exterior solution is the analytic one only once the Coulomb tail alone remains: the benchmarks of Section 5 run at 14 to 20 fm against a rule of thumb that would allow eight. Such a *channel radius a* divides space into an interior region, where both forces act and no analytic solution exists, and an exterior region, where only the Coulomb tail survives and the radial equation is solved exactly by known special functions. Every scattering observable then follows from computing the interior solution numerically and matching it to the exterior solution at $R = a$ (Figure 4).

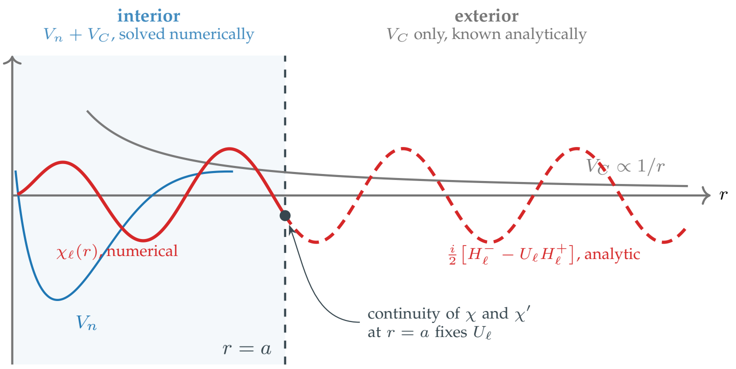

Figure 4: The interior/exterior construction. Inside the channel radius a both the nuclear and the Coulomb potential act and the radial equation is solved numerically; outside, only the Coulomb tail remains and the solution is a known combination of Coulomb-Hankel functions. Imposing continuity of the wave function and of its derivative at $r = a$ determines the single number U_ℓ that carries all the physics of the collision. Both solvers of Section 3 implement this picture, by structurally different means.

Because V is central, angular momentum is conserved. Choosing the polar axis along the beam, the wave function is azimuthally symmetric and expands on Legendre polynomials, $\psi(R,\theta) = \sum_\ell (2\ell+1) i^\ell P_\ell(\cos\theta)\, \chi_\ell(R)/kR$, which reduces the three-dimensional problem to an infinite set of decoupled radial equations, one per partial wave,

$$\left[-\frac{\hbar^2}{2\mu}\left(\frac{d^2}{dR^2} - \frac{\ell(\ell+1)}{R^2}\right) + V(R) - E\right]\chi_\ell(R) = 0. \tag{1}$$

The second term is the centrifugal barrier. With the Coulomb barrier it controls low-energy nuclear physics in two ways: it suppresses high partial waves, so the sum converges after a few terms at astrophysical energies, and the pocket it forms with the attractive nuclear well supports the quasi-bound states responsible for resonances. Equation (1) is second order and needs two boundary conditions: regularity at the origin, $\chi_\ell(0) = 0$, and an asymptotic condition fixed by the physics of the collision. The next section makes that second condition quantitative, and the observable enters there.

2.2 Coulomb asymptotics and the collision matrix

In the exterior region, Eq. (1) with $V_n = 0$ becomes the Coulomb wave equation, whose solutions are the regular and irregular Coulomb functions $F_\ell(\eta, kR)$ and $G_\ell(\eta, kR)$. Their behaviour is governed by the dimensionless Sommerfeld parameter

$$\eta = \frac{Z_1 Z_2 e^2}{\hbar v} = \frac{Z_1 Z_2 e^2}{\hbar}\sqrt{\frac{\mu}{2E}}, \tag{2}$$

which measures the Coulomb interaction relative to the kinetic energy: $\eta \ll 1$ is the high-energy, weakly distorted regime, $\eta \gg 1$ the astrophysical regime where barrier penetration dominates and the cross section becomes exponentially sensitive to energy. Incoming and outgoing flux are carried by the Coulomb-Hankel combinations $H_\ell^\pm = G_\ell \pm iF_\ell$, whose asymptotic phase $\rho - \ell\pi/2 + \sigma_\ell - \eta \ln 2\rho$ retains a logarithmic term, the permanent signature of the infinite range of the Coulomb force, with $\sigma_\ell = \arg \Gamma(\ell + 1 + i\eta)$ the Coulomb phase shift. The physical solution is regular at the origin and, beyond $R = a$, is a combination of the incoming wave H_ℓ^- and the outgoing wave H_ℓ^+. The incoming amplitude is fixed by the boundary condition, not by the interaction: an absorptive potential removes flux, but it cannot alter the incoming wave that was imposed. The entire effect of the interaction therefore lies in the amplitude and phase of the outgoing wave relative to the incoming one. Normalising the incoming coefficient defines the collision matrix element, also called the scattering-matrix element:

$$\boxed{\chi_\ell(R) \xrightarrow[R\to\infty]{} \frac{i}{2}\left[H_\ell^-(\eta, kR) - U_\ell\, H_\ell^+(\eta, kR)\right].} \tag{3}$$

This is the central definition of the report. Both solvers of Section 3 impose this condition at the edge of the interior region, and U_ℓ, or its multichannel generalisation $\mathbf{U}$, is the quantity every emulator of Section 4 predicts. It is the same object as the scattering matrix S; U is the notation of Descouvemont and Baye [12], which this report follows.

Both its modulus and its phase carry physical content. For a real potential and an isolated channel, flux is conserved partial wave by partial wave, $|U_\ell| = 1$, and $U_\ell = e^{2i\delta_\ell}$ defines a real phase shift δ_ℓ, which an elastic angular distribution measures. Nothing in Eq. (3) requires that unitarity. As soon as the potential acquires an absorptive imaginary part (Section 2.3) or the channel is coupled to others (Section 2.4), U_ℓ remains well defined but $|U_\ell| < 1$: some incoming flux never returns. The observables follow directly,

$$\sigma_{\rm el}^N = \frac{\pi}{k^2}\sum_\ell (2\ell+1)\,|U_\ell - 1|^2, \qquad \sigma_{\rm reac} = \frac{\pi}{k^2}\sum_\ell (2\ell+1)\left(1 - |U_\ell|^2\right), \tag{4}$$

the first measuring the flux that stays in the entrance channel, the second the flux that leaves it. The superscript on $\sigma_{\rm el}^N$ marks a restriction. On a charged system the Rutherford amplitude diverges at forward angles and its partial-wave series does not converge, so the pure Coulomb phase is divided out. The nuclear amplitude alone remains, and the observable elastic cross section is the modulus squared of its coherent sum with the Rutherford amplitude. $\sigma_{\rm el}^N$ is used here as a scalar functional of $\mathbf{U}$ and not as a measurable quantity. $\sigma_{\rm reac}$ carries no such restriction. It vanishes identically when $|U_\ell| = 1$ and is the direct signature of channels not treated explicitly.

2.3 The effective interaction: the optical model

So far V_n has been left unspecified. Solving the Schrödinger equation for the $A_1 + A_2$ interacting nucleons of a real collision is intractable beyond the lightest systems [1], [17], so the microscopic Hamiltonian is replaced by an effective interaction between the colliding fragments, calibrated so that the reduced problem reproduces the observables.

The formal justification is the Feshbach projection formalism. Partition the Hilbert space into a model subspace P, the channels kept explicitly, and its complement Q. Projecting the Schrödinger equation onto P yields an exact effective Hamiltonian acting in P alone,

$$H_{\text{eff}}(E) = H_{PP} + V_{PQ}\,\frac{1}{E^+ - H_{QQ}}\,V_{QP}, \qquad E^+ = E + i\epsilon. \tag{5}$$

The second term, the polarisation potential, is energy-dependent because H_{QQ} carries its own spectrum. It becomes complex as soon as at least one Q threshold lies below E, since the propagator then has continuum states at the energy at which it is evaluated and $1/(E^+ - H_{QQ})$ develops a negative imaginary part. This is the optical potential [1], [5], named by analogy with the complex refractive index of an absorbing medium: its imaginary part removes exactly the flux that escapes to the eliminated channels.

In practice it is not computed from Eq. (5). It is assembled from a small library of radial form factors whose strengths and geometries are fitted to data. Medium and heavy systems use Woods-Saxon shapes, light and cluster systems Gaussians; the Coulomb term is never fitted.

Table 1: Radial form factors of the effective interaction. Throughout, $f(r; R, a) = \left[1 + e^{(r-R)/a}\right]^{-1}$ is the Woods-Saxon shape and $R = r_0 A^{1/3}$. The last column names a system in which the term is dominant.

Term	Radial form	Physical content	Typical use
Real volume	$-V_0\, f(r; R_0, a_0)$	mean field seen by the projectile; sets the phase of the interior wave function	all systems
Imaginary volume	$-iW_0\, f(r; R_w, a_w)$	absorption throughout the nuclear interior; grows with energy as more channels open	$n + {}^{40}\text{Ca}$
Imaginary surface	$+4ia_d W_d\, \frac{d}{dr} f(r; R_d, a_d)$	absorption peaked at the surface; dominant at low energy, where the interior is Pauli-blocked	$n + {}^{40}\text{Ca}$
Spin-orbit	$+V_{\text{so}}\, \frac{1}{r}\frac{d}{dr} f(r; R_{\text{so}}, a_{\text{so}})\, \boldsymbol{\ell}\cdot\mathbf{s}$	splits $j = \ell \pm \frac{1}{2}$; surface-peaked because it follows the density gradient	$n + {}^{40}\text{Ca}$
Coulomb	$\frac{Z_1 Z_2 e^2}{2R_c}\left(3 - \frac{r^2}{R_c^2}\right)$, $r < R_c$; $\frac{Z_1 Z_2 e^2}{r}$, $r > R_c$	uniformly charged sphere; never fitted, fixed by the charges	all charged systems
Gaussian	$-V_0\, e^{-(r/\beta)^2}$	compact alternative to Woods-Saxon for light and cluster systems	$p + {}^{12}\text{C}$, $\alpha + d$

One rarely starts from scratch: *global* parametrisations such as CH89 [18] or Koning-Delaroche [19] fit one set of depths and geometries across many targets and energies, and it is their parameters that a Bayesian analysis treats as uncertain.

The free strengths and geometries, together with any deformation parameter of Section 2.4, form the parameter vector $\theta \in \mathcal{P} \subset \mathbb{R}^p$ introduced in Section 1.2, with $p = 5$ to 10 for the models used here. These are the quantities whose uncertainty must be propagated, and therefore the quantities with respect to which the forward model has to be evaluated 10^4 to 10^6 times.

2.4 Coupled channels

The optical potential captures the average, incoherent effect of the eliminated channels. When a few channels are strongly and *coherently* coupled to the elastic one, such as a low-lying collective excitation of the target or the D-wave component of the deuteron ground state, the coupling oscillates with energy and angle in a way no absorptive term can reproduce. Retaining those few channels explicitly, while folding the rest into the optical potential, is the minimal extension that recovers the coherent effects at tractable cost.

A *channel* α is the complete label of one asymptotic configuration: which pair of fragments is present, in which internal state, and with which relative orbital angular momentum ℓ_α. Its internal spins couple to a channel spin s, and $\mathbf{J} = \boldsymbol{\ell}_\alpha + \mathrm{s}$ with parity $\pi = \pi_1\pi_2(-1)^{\ell_\alpha}$ are conserved. Channels of different J^π never mix, so the problem block-diagonalises, and every calculation of Section 5 solves a single block, and every σ quoted there is that block's contribution at unit statistical weight.

Within a block, projecting the Schrödinger equation onto each channel replaces Eq. (1) by a coupled set,

$$\left[-\frac{\hbar^2}{2\mu_\alpha}\left(\frac{d^2}{dr^2} - \frac{\ell_\alpha(\ell_\alpha+1)}{r^2}\right) + V_{\alpha\alpha}(r) - (E - \epsilon_\alpha)\right] u_\alpha(r) = -\sum_{\beta\neq\alpha} V_{\alpha\beta}(r)\, u_\beta(r). \tag{6}$$

The left-hand side is the single-channel operator with two additions: the diagonal potential $V_{\alpha\alpha}$ is now the optical potential of Section 2.3 for that channel, and the available energy is reduced by the *threshold* ϵ_α, the internal excitation energy of the configuration. The right-hand side contains the off-diagonal couplings $V_{\alpha\beta}$, which are responsible for transferring amplitude between channels.

The couplings are the physical content of a coupled-channel model, and a small number of standard mechanisms produce them. In the collective mechanisms the potential is taken to follow a deformed or oscillating nuclear surface, and expanding it in that deformation generates a coupling proportional to $\mathrm{d}f/\mathrm{d}r$, peaked at the surface where the density gradient is largest. A permanently deformed target gives a *rotational* coupling between members of a band, as in $\alpha + {}^{12}\mathrm{C}$; a spherical target with a low-lying collective state gives a *vibrational* coupling, as in $n + {}^{58}\mathrm{Ni}$. A third mechanism, the *tensor* force, comes from the nucleon-nucleon interaction rather than from collective motion, and couples channels differing by two units of orbital angular momentum without changing the pair of fragments, as in the S to D mixing of $\alpha + d$. Appendix B.4, Table B.3, gives the explicit form factors.

Each channel has its own wave number $k_\alpha^2 = 2\mu_\alpha(E - \epsilon_\alpha)/\hbar^2$. A channel is *open* when $E > \epsilon_\alpha$, oscillating asymptotically as the Coulomb-Hankel functions above, and *closed* when $E < \epsilon_\alpha$, the only admissible asymptotic solution being the exponentially decaying Whittaker function $W_{-\eta_\alpha,\ell_\alpha+1/2}(2\kappa_\alpha r)$, written with the real modulus $\kappa_\alpha = |k_\alpha|$. Closed channels carry no flux but still influence the open ones through $V_{\alpha\beta}$, so their correct asymptotic treatment is a genuine implementation requirement. As the incident energy sweeps upward across a threshold, a channel changes character and the dimension of the observable collision matrix changes with it (Figure 5). One benchmark system of Section 5, $\alpha + {}^{12}\mathrm{C}$, crosses two thresholds inside its energy range; on the others every channel is open at every energy of the window.

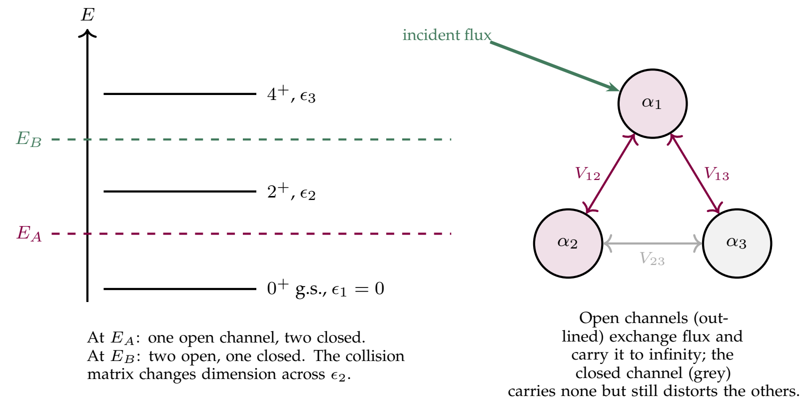

Figure 5: Channels and thresholds. *Left:* the internal states of the target define channel thresholds ϵ_α; a channel is open or closed according to whether the incident energy lies above or below its threshold, and crossing a threshold changes the number of open channels. *Right:* within one J^π block, channels are coupled pairwise by the off-diagonal potentials $V_{\alpha\beta}$. Closed channels take part in the coupled interior problem but carry no asymptotic flux.

Imposing an incoming wave in one entrance channel and purely outgoing waves in all other open channels, and matching at the channel radius exactly as in Eq. (3), defines the collision matrix $\mathbf{U}$, whose element $U_{\alpha\beta}$ is the amplitude for a particle incident in channel β to emerge in channel α. Cross sections follow as in Eq. (4), with the elastic and inelastic contributions read off the diagonal and off-diagonal elements.

Given a parameter vector θ, an energy E and a J^{π} block of N_c channels, solve the coupled radial system (6) on $[0, a]$ subject to regularity at the origin and the outgoing-wave condition (3) at $r = a$, then extract $\mathbf{U}(E, \theta)$. Discretising on N radial points turns this into a dense complex linear system of dimension $N_c N$. Section 3 solves that system exactly and examines the structure of its matrix, because that structure is what Section 4 exploits.

3 The solver and its system matrix

Inherited: the Lagrange-mesh discretisation, due to Baye [20]; the calculable R-matrix method of Descouvemont and Baye [12], [21], implemented in a Python code written by P. McGlynn before this internship; and the Direct Boundary Matching Method, formulated by Lei [9]. Produced here: the multichannel implementation of the DBMM with its closed-channel boundary treatment, and the reading of its system matrix given in Section 3.4. The solver is validated in Section 5.3.

3.1 Discretisation: the Lagrange-Legendre mesh

The interior region $[0, R]$ is discretised on an N-point Gauss-Legendre *mesh*, and the radial functions are expanded on the x-regularised Lagrange basis built from it. Writing $\{x_j\}_{j=1}^N \subset]0,1[$ for the quadrature nodes and $\{\lambda_j\}$ for the weights, the physical mesh points are $r_j = Rx_j$ and the regularised Lagrange functions are [20]

$$\hat{f}_j(x) = (-1)^{N-j}\sqrt{\frac{1-x_j}{x_j}}\,\frac{x\,P_N(2x-1)}{x-x_j}, \qquad \hat{f}_j(x_i) = \frac{\delta_{ij}}{\sqrt{\lambda_j}}. \tag{7}$$

Expanding the radial wave function as $\chi_\ell(r) = R^{-1/2}\sum_j c_j \hat{f}_j(r/R)$ makes each coefficient proportional to the wave function *value* at a mesh point, $c_j = \sqrt{R\lambda_j}\,\chi_\ell(r_j)$.

This basis is the natural choice for two properties, both of them used repeatedly downstream. The kinetic-energy operator has closed-form matrix elements T_{ij} (Appendix B.1), so no numerical differentiation is required. And the Gauss quadrature makes the potential matrix *diagonal*,

$$\langle \hat{f}_i | V | \hat{f}_j \rangle \simeq V(r_i)\,\delta_{ij}, \tag{8}$$

to the order of the quadrature [12]. The equality is not exact, since the integrand $\hat{f}_i\hat{f}_jV$ exceeds the degree $2N-1$ that the N-point rule integrates exactly, even before V is counted, so Eq. (8) carries a truncation error that falls with N for a smooth V and vanishes at no finite N. Baye [20] gives the bound, and everything downstream inherits that error. A local potential therefore contributes only to the diagonal of the system matrix, evaluated pointwise on the mesh.

3.2 Two solvers, one boundary condition

Both solvers impose the same physical condition, the outgoing-wave asymptotics of Eq. (3), through structurally different machinery.

The calculable R-matrix method restores hermiticity of the Hamiltonian on the finite interval $[0, a]$ by adding a surface (Bloch) operator, solves a real symmetric linear system, and forms the R-matrix, essentially the inverse logarithmic derivative of the interior solution at $r = a$; $\mathbf{U}$ follows in a second matching step. Appendix B.2 summarises the formalism. The reference code implements the algorithm published by Descouvemont [21] and supports coupled channels with distinct thresholds, closed-channel Whittaker asymptotics included, so the two solvers can be compared on the same coupled-channel configurations and not on a single-channel check.

The DBMM [9] avoids both the Bloch operator and the two-step matching. The wave function is split into a known incident part and an unknown scattered part,

$$\psi_{\alpha\beta_0}(r) = e^{i\sigma_{\ell_{\beta_0}}}\Big[\delta_{\alpha\beta_0}F_{\ell_{\beta_0}}(\eta_{\beta_0}, k_{\beta_0}r) + \psi^{\rm sc}_{\alpha\beta_0}(r)\Big], \tag{9}$$

which turns the homogeneous coupled system (6) into an *inhomogeneous* equation for $\psi^{\rm sc}$ whose source depends only on the potentials and on the known Coulomb function. That inhomogeneous structure makes the DBMM a single-solve method. Expanding $\psi^{\rm sc}$ on the Lagrange basis yields a block linear system $\mathbf{M}(E)\,\mathbf{c} = \mathbf{b}(E)$ of dimension N_cN, in which the first $N-1$ rows of each diagonal block encode the Schrödinger equation at the interior mesh points,

$$M^{(\alpha\alpha)}_{ij} = T_{ij} + \Big[\tfrac{\hbar^2\ell_\alpha(\ell_\alpha+1)}{2\mu_\alpha r_i^2} + V_C(r_i) + V_N(r_i) - E_\alpha\Big]\delta_{ij}, \qquad M^{(\alpha\beta)}_{ij} = V_{\alpha\beta}(r_i)\,\delta_{ij}, \tag{10}$$

and the last row of each diagonal block *replaces* the Schrödinger equation by the outgoing-wave condition itself,

$$M_{Nj}^{(\alpha\alpha)} = \left.\frac{d\hat{f}_j}{dx}\right|_{x=1} - R\,\gamma_{s,\alpha}\,\hat{f}_j(1), \qquad \gamma_{s,\alpha} = k_\alpha\,\frac{H_{\ell_\alpha}^{+\prime}(\eta_\alpha, k_\alpha R)}{H_{\ell_\alpha}^{+}(\eta_\alpha, k_\alpha R)}. \tag{11}$$

That row sets $\hat{\chi}_\alpha'(R) - \gamma_{s,\alpha}\hat{\chi}_\alpha(R)$ to zero, so the interior solution and the outgoing Coulomb-Hankel function share their logarithmic derivative at the edge: the boundary condition is satisfied by the solve itself and not imposed after it. For a closed channel $H_{\ell_\alpha}^{+}$ is replaced by the Whittaker function, which is real and exponentially decaying, so $\gamma_{s,\alpha}$ becomes real and negative, the off-diagonal boundary rows are set to zero, and channels do not mix at the boundary. Because $\gamma_{s,\alpha}$ is complex, $\mathbf{M}$ is complex and non-symmetric, where the R-matrix method solves a real symmetric system. Table 2 sets the two side by side. One consequence of the boundary row is worth stating here and is measured in Appendix B.3: $\mathbf{U}$ must be read from it with each channel normalised by its own velocity, and the two invariants that fix that normalisation, the symmetry of $\mathbf{U}$ and its unitarity with the absorption removed, hold whatever code produced it.

Table 2: Structural comparison of the two solvers. The kinetic matrix T_{ij} is common to both; the Bloch surface term and the boundary parameter γ_s are mutually exclusive treatments of the same matching problem. Section 4 builds on the last row.

	R-matrix (`python_rmatrix.py`)	DBMM (`core/dbmm.py`)
System matrix	$\mathbf{D}(E,B)$	$\mathbf{M}(E,\theta)$
Nature	real, symmetric	complex, non-symmetric
Rows of T_{ij} used	all N	first $N-1$ only
Boundary treatment	Bloch operator, free $B \in \mathbb{R}$	$\gamma_s \in \mathbb{C}$, physically fixed
Solve path	$\mathbf{D}^{-1} \to \mathcal{R} \to \mathbf{U}$, two steps	$\mathbf{M}^{-1}\mathbf{b} \to \mathbf{U}$, one step
Closed channels	energy-dependent Bloch parameter	Whittaker $W_{-\eta,\ell+1/2}$
Emulator interface	none	exact $\mathbf{K}(E) + \mathbf{V}(\theta)$ split

3.3 Implementation

The solver is the class `DBMMSolver`, organised so that a change of θ or of E redoes as little as possible (Figure 6). Everything depending only on the mesh and the channel list is assembled at construction: the kinetic matrix, the centrifugal barrier and the Coulomb potential. The four special functions the method needs are F_ℓ on the mesh, H_ℓ^+ and $W_{-\eta,\ell+1/2}$ with their derivatives at the boundary, and the Coulomb phase σ_ℓ. They depend on E and on the channel but never on θ, so they are evaluated once per energy node and cached. What remains inside the parameter loop is the potential on the mesh, the assembly of $\mathbf{M}$, and one LU factorisation serving all N_c incident channels at once, the right-hand side being a matrix with one column per incident channel. Appendix D describes the backend chain behind the special functions and its measured cost.

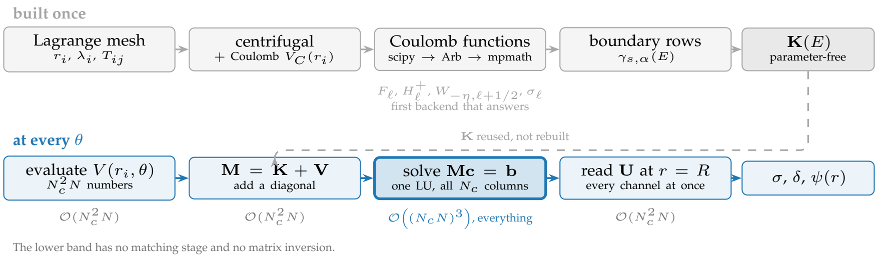

Figure 6: One evaluation, end to end. The upper band is assembled at construction and never entered again; the lower band runs at every parameter point. Its first two boxes are linear in the problem size and its third is cubic, which carries the whole cost of a campaign; Section 4 replaces that step. The only θ-dependent object is a diagonal.

3.4 The algebraic structure that makes emulation possible

The nuclear potential enters $\mathbf{M}$ only through the diagonal of the interior rows. It never enters the kinetic matrix, and it never enters the boundary row, which carries only $\gamma_{s,\alpha}(E)$, a function of the kinematics and not of θ. By the diagonality property (8) its contribution is the pointwise evaluation $V(r_i, \theta)$. The system matrix therefore splits *exactly* into

$$\boxed{\mathbf{M}(E,\theta) = \underbrace{\mathbf{K}(E)}_{\text{parameter-free}} + \underbrace{\mathbf{V}(\theta)}_{\text{parameter-dependent}}} \tag{12}$$

where $\mathbf{K}(E)$ contains the kinetic energy, the centrifugal barrier, the full Coulomb term and the boundary rows, and $\mathbf{V}(\theta)$ is block-diagonal in radial space, carrying $V_{\alpha\alpha}(r_i, \theta)$ and $V_{\alpha\beta}(r_i, \theta)$ on its diagonals (Figure 7).

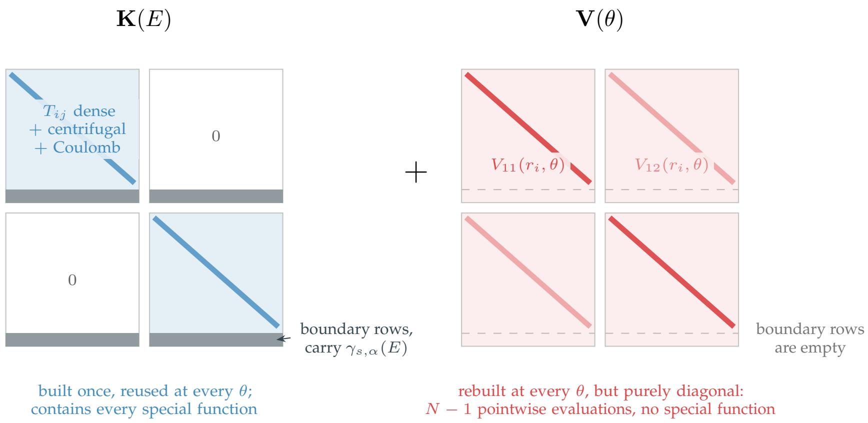

Figure 7: The DBMM system matrix for a two-channel problem, and the exact split of Eq. (12); each of the N_c^2 blocks is $N \times N$. In $\mathbf{K}(E)$ the diagonal blocks are dense, the kinetic operator coupling all mesh points, and the last row of each carries the outgoing-wave condition. In $\mathbf{V}(\theta)$ every block is purely diagonal by the Gauss quadrature (8): the parameters act on only $N-1$ isolated entries per block, which collapses the projection of Section 4 into a sum over mesh points.

$\mathbf{K}(E)$ is parameter-free in the sense of θ alone: E enters it, and an emulator treating the energy as one more coordinate would have to rebuild or tabulate it. Putting the Coulomb term there assumes its geometry is fixed, which it is here, the charge radius being held at its central value throughout. A calibration that fitted R_C would move a θ-dependent term into the parameter-free block, and the remedy costs nothing, since the uniform-sphere Coulomb potential is diagonal in radial space exactly like the nuclear terms and moves into $\mathbf{V}(\theta)$ at the price of one more diagonal per query. The reduction does not depend on which block a diagonal term sits in, only on the θ-dependent part staying diagonal. Two consequences follow. The expensive, special-function-laden part of the matrix does not depend on θ, so it is built once and projected once. And because $\mathbf{V}$ is diagonal in r, its projection onto any subspace collapses to a sum over mesh points instead of a dense matrix product, which is the property Eq. (17) exploits. Neither holds for a *non-local* potential $V(r, r')$ with $r \neq r'$, which would populate full off-diagonal blocks in $\mathbf{V}$, and the present implementation is restricted to local potentials.

3.5 Other approaches and the case for emulation

Neither method above is the only one in use. Lei [9] places the DBMM against the two families that dominate practice. Propagation methods march the radial solution outward from the origin by Numerov or log-derivative stepping and re-orthogonalise when a closed channel makes the solution stiff; the calculable R-matrix, which the DBMM is closest to, replaces that march by a single interior solve. What the DBMM changes with respect to the second is the boundary treatment: one row per channel instead of a Bloch operator and a second matching step. Within the BAND framework [22] the two codes this work sits nearest to are ROSE [14], [23], which emulates single-channel elastic scattering on a propagated solver, and jitR [24], a just-in-time R-matrix library written for calibration. The DBMM contributes no speed on a single solve, and contributes instead the shape of its system matrix, Eq. (12).

That shape is worth exploiting because the direct cost is prohibitive at campaign scale. Lei builds a reduced basis emulator on the DBMM for exactly that reason [10], and the arithmetic is that of Section 1.2: a single coupled-channel solve is under a millisecond to about a tenth of a second on the systems benchmarked here, and a posterior needs 10^4 to 10^6 of them at every energy of the data set. Section 4 builds two reductions on Eq. (12), and Section 5.7 measures what they return.

4 Emulation by reduced-order models

Both branches of Figure 3 meet here. *Inherited.* Reduced basis methods are standard model order reduction [25], [26], already applied to scattering by Lei [10], by Odell *et al.* in ROSE [14] and by Catacora-Rios *et al.* on the wave function [27], and reviewed for this community by Melendez *et al.* [4], [8]; the radial-sum projection of Eq. (17) is Lei's [10]; the learning reduced-order model is Giuliani's [13], [28], built for a single channel, and is operator inference [29], [30] under another name. *Produced here.* The coupled-channel version of that model, and the split of its predictor budget between the diagonal and the coupling blocks, an idea due to P. McGlynn and C. Hebborn.

4.1 The idea: a low-dimensional solution manifold

The affine split of Eq. (12) reduced the physics to a parametric linear system,

$$\mathbf{M}(E,\theta)\,\mathbf{c}(\theta) = \mathbf{b}(\theta), \qquad \mathbf{M} = \mathbf{K}(E) + \mathbf{V}(\theta), \qquad \mathbf{c} \in \mathbb{C}^{N_{\text{tot}}},\ N_{\text{tot}} = N_c N. \tag{13}$$

A Bayesian campaign solves it for 10^4 to 10^6 different θ. Solving independently every time ignores that the solutions are not independent of one another: the potential depends smoothly on θ and the solution operator $\mathbf{M}^{-1}$ depends smoothly on the potential, so the *solution manifold*

$$\mathcal{M} = \big\{\, \mathbf{c}(\theta)\ :\ \theta \in \mathcal{P} \,\big\} \subset \mathbb{C}^{N_{\text{tot}}}$$

does not fill the ambient space. Each $\mathbf{c}$ has $N_{\text{tot}} \sim 10^2$ to 10^3 components on the systems of this report, but the set reachable by varying θ is, to excellent approximation, a surface of dimension $n_b \ll N_{\text{tot}}$: a few fixed shapes carry every solution by linear combination (Figure 8).

(a) $\mathbf{c}(\theta) \in \mathcal{M} \subset \mathbb{C}^{N_{\text{tot}}} \ \longrightarrow\ X_r\, a(\theta) \in \mathcal{S}_{n_b}$

(b) a continuum of radial solutions $\to n_b$ fixed modes

Figure 8: What a reduced basis method replaces. *(a)* The solutions reachable by varying θ lie on a thin curved manifold $\mathcal{M}$; a few snapshots determine a flat subspace $\mathcal{S}_{n_b}$ of dimension $n_b \ll N_{\text{tot}}$, and a new solution is replaced by its representative in that subspace. The distance to $\mathcal{S}_{n_b}$ is bounded by the discarded singular values for a snapshot, and not for the query point drawn here. *(b)* The same statement on the radial solutions: a whole family, one per parameter, is carried by n_b fixed modes, whose coefficients follow from imposing Eq. (13) inside the subspace and not from interpolating previous outputs.

Both emulator families exploit this in two stages: an *offline* stage, paid once, in which the expensive solver is run at N_s training parameters and the results are compressed into a basis; and an *online* stage, paid at every query, in which a problem of size n_b is solved. Where they differ is in what happens online. The RBM *projects* the governing operator onto the basis at each query; the LROM *learns* a surrogate for that projected operator offline, so the online stage never touches the full mesh. Figure 9 sets the two chains side by side.

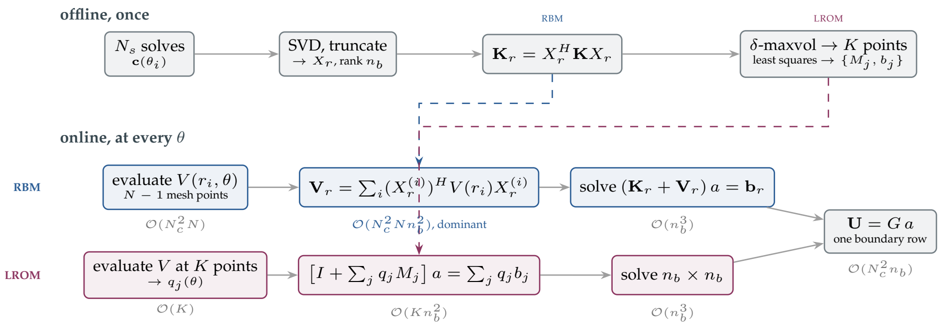

Figure 9: The two chains. They share the snapshots, the basis and the final $n_b \times n_b$ solve, and they differ in one place: the RBM rebuilds the reduced operator from the full mesh at every query, while the LROM evaluates one learned offline from K scalars. The operation count under each online box is what the difference costs. The term $\mathcal{O}(N_c^2 N n_b^2)$ is the only one that carries the mesh size, and removing it is the whole purpose of the second chain.

4.2 The reduced basis method

The training points $\{\theta_1, \ldots, \theta_{N_s}\}$ are drawn from the parameter box $\mathcal{P}$ by Latin hypercube sampling: each of the p dimensions is divided into N_s equiprobable strata and one point is drawn per stratum, with independent permutations across dimensions. Every one-dimensional projection is then stratified, at a cost growing linearly with p rather than exponentially, which matters here since p reaches 10. The construction gives no guarantee on the joint coverage of a 10-dimensional box, and no sampling of a few hundred points could.

For each training point the full system (13) is solved with the DBMM and the solutions are assembled column-wise into a snapshot matrix $\mathbf{C}_{\text{snap}} \in \mathbb{C}^{N_{\text{tot}} \times N_s}$. This is the only expensive step of the construction, and it is embarrassingly parallel. The snapshot matrix is then compressed by a truncated singular value decomposition $\mathbf{C}_{\text{snap}} = X\Sigma W^H$, whose left singular vectors are the POD modes, the orthonormal shapes the reduced basis is built from. By the Eckart-Young theorem, retaining the first n_b modes gives the best rank-n_b approximation of the snapshot set in the Frobenius norm. The rank is fixed by an energy criterion,

$$\frac{\sum_{i \le n_b} \sigma_i^2}{\sum_{i \le N_s} \sigma_i^2} > 1 - \epsilon_{\text{tol}}, \qquad X_r = [\, x_1 \cdots x_{n_b} \,] \in \mathbb{C}^{N_{\text{tot}} \times n_b}, \tag{14}$$

where X_r, the retained basis, carries the first n_b POD modes as its columns and is the object every emulator below projects onto. The tolerance is the square of the resulting accuracy, so with $\epsilon_{\text{tol}} = 10^{-8}$ the retained basis reproduces its own snapshots to $\sqrt{\epsilon_{\text{tol}}} = 10^{-4}$ in the relative Frobenius norm.

The two natural diagnostics say different things. The normalised spectrum σ_i/σ_1 measures how fast an *individual* mode dies. What Eq. (14) thresholds is the *accumulated* quantity, or equivalently the discarded energy fraction

$$r(k) \;=\; 1 - \frac{\sum_{i \le k} \sigma_i^2}{\sum_{i \le N_s} \sigma_i^2} \;=\; \frac{\sum_{i > k} \sigma_i^2}{\sum_{i \le N_s} \sigma_i^2}, \tag{15}$$

and n_b is the smallest k for which $r(k) < \epsilon_{\text{tol}}$. The distinction matters because the representation error is bounded by $r(k)$ and not by σ_k/σ_1. By the Eckart-Young theorem $r(k)$ is exactly the squared relative error of the best rank-k approximation of the snapshot set. On a common logarithmic axis (Figure 12) the criterion is a horizontal line whose crossing point is the selected rank.

Substituting $\mathbf{c} \approx X_r\alpha$ into Eq. (13) leaves N_{tot} equations for n_b unknowns. Requiring the residual to be orthogonal to the reduced subspace, that is projecting from the left by X_r^H, closes the system:

$$\underbrace{X_r^H\, \mathbf{M}(\theta)\, X_r}_{\mathbf{M}_r(\theta)}\; \alpha = \underbrace{X_r^H\, \mathbf{b}(\theta)}_{\mathbf{b}_r(\theta)}, \qquad \mathbf{M}_r(\theta) = \mathbf{K}_r + \mathbf{V}_r(\theta), \quad \mathbf{K}_r = X_r^H \mathbf{K} X_r. \tag{16}$$

The affine split (12) carries the reduction. $\mathbf{K}_r$ is parameter-independent and is computed once offline, so it never reappears at query time. The source term is not inert. In the DBMM the right-hand side is $b_\alpha \propto -V^{\text{eff}}_{\alpha\beta_0}(r)\, F_{\ell_{\beta_0}}(\eta_{\beta_0}, k_{\beta_0} r)$, which carries the potential through V^{eff} and therefore *does* depend on θ. That is the price of having split the wave function into an incident and a scattered part in Eq. (9). What is parameter-independent is the Coulomb factor $F_{\ell_{\beta_0}}$, precomputed once on the mesh, so no special function is evaluated online. Both $\mathbf{V}_r(\theta)$ and $\mathbf{b}_r(\theta)$ are rebuilt at each query, and both need the potential on the mesh, so the predictor-point construction of Section 4.3 has to eliminate the mesh loop in the source as well as in the operator.

Rebuilding $\mathbf{V}_r(\theta) = X_r^H \mathbf{V}(\theta) X_r$ naively would cost a dense $N_{\text{tot}} \times N_{\text{tot}}$ product. The diagonality property (8) removes it. Because $\mathbf{V}$ is block-diagonal in radial space (Figure 7), the projection collapses to a sum over mesh points: writing $X_r^{(i)}$ for the $N_c \times n_b$ sub-block of X_r associated with mesh point r_i,

$$\mathbf{V}_r(\theta) = \sum_{i=1}^{N-1} \big(X_r^{(i)}\big)^H\, V(r_i, \theta)\, X_r^{(i)}, \tag{17}$$

a sequence of $N-1$ rank-N_c updates. This is Eq. (23) of Ref. [10], written for the coupled case. No affine dependence of the potential on θ is assumed, hence no empirical interpolation, the extra approximation a reduced basis needs when the operator cannot be written as a parameter-independent sum. The potential may depend on its parameters in an arbitrarily non-linear way; only its locality in r is used.

The offline stage pays N_s full solves and one singular value decomposition, and stores X_r, $\mathbf{K}_r$ and the Coulomb factors. The online stage evaluates the potential on the mesh, forms $\mathbf{V}_r(\theta)$ by Eq. (17) and $\mathbf{b}_r(\theta)$ from the same mesh values, then solves an $n_b \times n_b$ system and lifts the result back through X_r to extract $\mathbf{U}$.

4.3 The learning reduced-order model

The RBM online stage evaluates the potential at all $N-1$ interior mesh points, a term linear in N that the $\mathcal{O}(n_b^3)$ solve does not carry. The LROM removes it by learning, offline, a surrogate for the reduced operator itself. Whether that term *dominates the cost* is a separate question, answered by measurement in Section 5.7.

Around a central parameter point θ_c the solution is written as a reference plus a deviation,

$$\mathbf{c}(\theta) \simeq \mathbf{c}_c + X_r\, a(\theta), \qquad \mathbf{c}_c = \mathbf{c}(\theta_c), \qquad a(\theta_c) = 0, \tag{18}$$

and instead of projecting the operator to obtain $a(\theta)$, one postulates that it satisfies a small implicit system whose coefficients are fitted:

$$\Big[I + \sum_{j=1}^{K} q_j(\theta)\, M_j \Big]\, a(\theta) = \sum_{j=1}^{K} q_j(\theta)\, b_j, \qquad q_j(\theta_c) = 0. \tag{19}$$

The $q_j(\theta)$ are scalar *predictors*, values of the potential at K selected radial points, while $M_j \in \mathbb{C}^{n_b \times n_b}$ and $b_j \in \mathbb{C}^{n_b}$ are learned once offline. An online query then costs K scalar potential evaluations instead of N, with K fixed independently of the mesh size.

Equation (19) is an assumption, and a different one from the projection the RBM makes. It postulates that the reduced operator is *affine in the predictors*: the matrix acting on $a(\theta)$ is $I + \sum_j q_j(\theta) M_j$ with the M_j fixed, so the operator depends on θ through K scalars and linearly in them. The dependence of $\mathbf{U}$ on the potential is not linear, and nothing forces the reduced operator to be. The RBM makes no such assumption, since it projects the true operator and its only approximation is the subspace. A projection touches the potential at every mesh point and an operator affine in K scalars does not. The same assumption therefore makes the LROM fast and carries its risk of failure.

For each training snapshot the exact reduced coordinate is directly computable, $a_{\mathrm{LS}}(\theta_i) = X_r^H(\mathbf{c}(\theta_i) - \mathbf{c}_c)$, independently of the unknowns. Inserting it into Eq. (19) gives a residual that is linear in $\{M_j, b_j\}$, so training is a single linear least-squares problem for $K(n_b^2 + n_b)$ unknowns, and the rows decouple. The equation for row i of every M_j involves only row i of the unknowns and shares its design matrix with all the others, so the whole fit is one least-squares system of size $N_s \times K(n_b + 1)$ with n_b right-hand sides instead of a block problem n_b times larger. Comparing those two sizes defines the *overfit ratio*

$$r_{\mathrm{fit}} \equiv \frac{N_s}{K\,(n_b + 1)}. \tag{20}$$

Above unity the system is overdetermined and the fit is a genuine least-squares problem. Below it, LSTSQ returns the minimum-norm solution instead. That solution reproduces the training snapshots exactly and selects, among the infinitely many operators that do so, the one of smallest norm, which is an implicit regularisation. Every configuration in this report sits in that regime, since the calibration rule $N_s = 5n_b$ makes $r_{\mathrm{fit}} \simeq 5/K$; Appendix D.5 reports the ratio per configuration, and no measurement in this report says what the regularisation is worth.

4.4 The predictor budget in coupled channels

The K predictor points are chosen offline, and this is where the coupled case departs from the single-channel construction. The potentials over the full mesh are stacked into a *signature vector* per training sample. The centred variations from θ_c form a delta matrix whose low-rank SVD exposes the dominant directions of parameter-induced variation, and a greedy *maxvol* pass takes the K rows of largest volume, that is K radial positions. Three rules restrict the candidates, and Appendix D.5.1 measures what each is worth.

A single pass is not enough on a coupled-channel system, because of the difference in amplitude. The diagonal blocks carry the optical potential and the off-diagonal blocks carry only the coupling. Measured over the box, the variation of the diagonal exceeds that of the coupling by a factor of 5 on $\alpha + {}^{12}\mathrm{C}$, 38 on $n + {}^{58}\mathrm{Ni}$, 65 on $n + {}^{238}\mathrm{U}$ and 220 on $\alpha + {}^{24}\mathrm{Mg}$. Where that ratio is large a single maxvol pass over the whole signature places essentially every point on the diagonal (Figure 10). A parameter that lives only in the coupling is then never sampled, and no increase of the rank recovers it, because the fit does not see the direction at all. Physically, the emulator is being asked to learn the effect of a deformation while looking only at radii and channels where the deformation does not act. The extension implemented here runs two *independent* passes with separately fixed budgets,

$$K = K_{\mathrm{diag}} + K_{\mathrm{coup}},$$

one maxvol selection on the diagonal signature and one on the coupling signature.

The split is not always needed, and the condition is a property of the interaction. Diagonal points carry information on every parameter the diagonal depends on, so the test is whether a parameter is absent from the diagonal, not whether it is written in the coupling. A deformed level of spin I with $2I \geq \lambda$ couples to itself through the $3j$ symbol $(I\,\lambda\,I; 000)$, so its deformation reaches the diagonal of its own channel through that self-coupling, the reorientation term, and a joint budget suffices. A one-phonon vibrational excitation of a 0^+ ground state has no such term, because $\langle 0|Y_\lambda|0\rangle$ vanishes identically (Appendix B.4), so its deformation is invisible from the diagonal whatever the rank. The budget is therefore split when a parameter is absent from the diagonal of every channel, and kept joint otherwise. The rule is tested on five systems in Section 5.6.

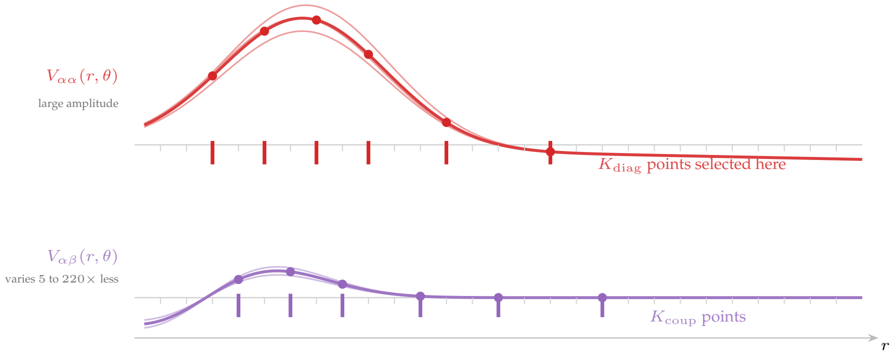

Figure 10: Predictor-point selection for a coupled-channel system. The potential is evaluated online only at the marked radial positions, chosen by a maxvol pass on the dominant directions of parameter-induced variation. Because the variation of the diagonal block exceeds that of the coupling block by up to a factor of 220, a single joint selection concentrates its points on the diagonal; running one selection per block with its own budget is what makes the construction usable beyond a single channel.

4.5 The two branches side by side

The coupled case adds three changes to Giuliani's single-channel construction, all forced by the channel structure: the signature the selection acts on becomes a stack of channel blocks, the reduced operator becomes one block per open channel, and the predictor budget has to be divided between diagonal and coupling. Table 3 sets the two branches side by side. The RBM makes a single approximation, the Galerkin projection, while the LROM makes a second one, the least-squares fit of the reduced operator, and removes in exchange the only term that carries the mesh size.

Table 3: The two emulator families. Both take the same snapshot set, the same truncation criterion and the same rank n_b, and both end on an $n_b \times n_b$ online solve. The RBM decomposes the snapshots themselves and the LROM their deviations from $\mathbf{c}_c$, so the two subspaces differ by the reference solution. N is the number of mesh points, N_c the number of channels, n_b the basis rank, K the number of predictor points and N_s the number of training snapshots.

	- - - RBM	-·-·· LROM
Reduced operator	projected from the full mesh at each query	learned offline, evaluated from K scalars
Online potential step	$N-1$ mesh evaluations	$K \ll N$ scalar evaluations
Online cost	$\mathcal{O}(N_c N n_b^2) + \mathcal{O}(n_b^3)$	$\mathcal{O}(K n_b^2) + \mathcal{O}(N_c^{\text{open}} n_b^3)$
Offline cost	N_s DBMM solves + SVD	idem + delta-maxvol + least squares
Validity condition	POD energy criterion	POD criterion; r_{fit} reported, not required
Coupled channels	automatic; one projected operator for all channels	split budget where the coupling has its own parameter; one learned operator per open channel
Approximation made	Galerkin projection only	projection and operator fit

Coupled-channel scattering has been emulated by a reduced basis before this work. Catacora-Rios *et al.* [27] emulate the wave function of a coupled-channel problem whose interaction is parametrised non-affinely, on a target that overlaps the reduced basis branch of Section 4.2. What this report adds sits on the other branch. The learning reduced-order model existed for a single channel, and its coupled-channel form is the construction of Section 4.3, with the three changes listed above. The novelty claimed here is that construction and the budget rule of Section 4.4. Emulating coupled channels is itself not new.

The same caution applies to both, since they size their basis by the same criterion. Eckart-Young bounds the distance from the *training snapshots* to the subspace, in the norm of the coefficient vector. It does not bound the error at an unseen θ, and it does not bound the error on $\mathbf{U}$, which is a functional of the coefficients whose relative error grows without limit near one of its zeros. The discarded singular values size the basis; they are not an error estimate, and Section 5.4 measures the gap.

The Galerkin projection of Section 4.2 carries a stability constant, which governs that gap as well, and the error at a query point is the best-approximation error of the subspace divided by it. A subspace can therefore improve while the projected solve on it does not, and Section 5.4 measures two systems on which it does.

5 Benchmarks and results

> Everything measured during the internship. The optical potentials are published and are written out with their sources in Appendix C. The reference the solver is measured against is the calculable R-matrix code of P. McGlynn, and the emulator is measured against the single-channel implementation of P. Giuliani. Every figure and every number comes from one reproducible pipeline (Appendix D); the supporting tables are in Appendix E.

5.1 The five systems

The comparison rests on four systems, and each of them exercises one difficulty: a complex potential with ten parameters, closed channels that open inside the energy window, two deformation multipoles on a light target, and a compressed band with five open channels. A fifth, $n + {}^{58}$Ni, carries a single measurement, for a reason given in Section 5.6.

Table 4: The five systems. The four benchmarks carry every measurement of this section; $n + {}^{58}$Ni carries one, the budget test of Section 5.6, and no accuracy or cost figure of the comparison. f is the Woods-Saxon shape and f' its radial derivative, each with the radius and diffuseness of its own term, and $\mathcal{F}^{(\lambda)}$ is the recoupling coefficient of Eq. (54). Central values, boxes and sources are in Appendix C.

	$n + {}^{40}$Ca	$\alpha + {}^{12}$C	$\alpha + {}^{24}$Mg	$n + {}^{238}$U	$n + {}^{58}$Ni
Channels	$\ell = 1, j = \frac{3}{2}, \frac{1}{2}$, uncoupled	band $0^+2^+4^+$ at $J^\pi = 3^-$, $\ell = 1, 3, 5, 7$	band $0^+2^+4^+$ at $J = 0, \ell = 0, 2, 4$	band 0^+ to 8^+ at $J = 0, \ell = 0$ to 8	0^+ and 2_1^+, one-phonon
Thresholds (MeV)	none	4.44, 14.08; both inside the window	1.369, 4.123	0.045 to 0.518	1.454
$N_c \times N$	2×64	8×120	3×100	5×80	2×60
Interaction	$-(V_{v0}+iW_{v0})f_v + 4ia_{d0}W_{d0}f_d' + V_{so}\frac{1}{r}f_{so}'\,\boldsymbol{\ell}\cdot\mathbf{s}$	$-(V_0+iW_0)f$, rigid rotor $\lambda = 2$	as $\alpha + {}^{12}$C, plus surface absorption, $\lambda = 2, 4$	as $\alpha + {}^{24}$Mg	CH89 volume and surface, spin-orbit, vibrational $\lambda = 2$
Parameters	4 depths, 3 radii, 3 diffusenesses ($p = 10$)	$V_0, W_0, R_0, a_0, \beta_2$ ($p = 5$)	$+W_{d0}, \beta_4$ ($p = 7$)	as $\alpha + {}^{24}$Mg ($p = 7$)	V_0, W_s, W_v, δ_2 ($p = 4$)
Difficulty	highest p, smallest matrix, no coupling	largest matrix, closed channels that open	two multipoles on a light deformed target	most compressed band, five open channels	δ_2 absent from the diagonal
Figures	11–E.3	11–E.3	11–E.3	11–E.3	Table 9 only

Every error below is measured at validation parameters never seen in training, against the DBMM rather than the R-matrix code, inside the boxes of Appendix C. Those boxes are not uniform. They are ± 20 per cent on every parameter of $n + {}^{40}$Ca, which is Giuliani's own choice, and elsewhere they run from ± 10 per cent on a radius to -50 to $+100$ per cent on an absorption depth, following how tightly each parameter is determined in the reference the system is taken from.

5.2 The measurement protocol

Accuracy is quoted on the cross section, except where the off-diagonal element $S_{\alpha\beta}$ is named. A coupled-channel campaign propagates that element, and it is the harder of the two to score: on $\alpha + {}^{12}$C those elements span five decades within a single draw, so a relative error of order 1 on the smallest is 10^{-8} in absolute terms. Every benchmark of this suite carries an absorptive potential, so the cross section scored is the reaction cross section of Eq. (4); the two control systems of Section 5.3 have real interactions and are scored on the elastic one. Table 5 fixes the definitions.

The threshold below which an emulator counts as accurate enough is fixed by choice here. An elastic angular distribution is measured to a few per cent at best, so an emulator error a decade inside that is invisible in a likelihood built on such data. This report adopts $\epsilon_{\text{post}} = 5 \times 10^{-3}$ on that reasoning and uses it as the single threshold every configuration is judged against. Nothing here establishes that it is the right number for a given experiment: a campaign with sharper data would have to lower it, and Section 5.4 shows what that costs, since the rank needed grows as the tolerance falls.

Table 5: What each error measures. The reference is the converged DBMM at the same θ and the same energy, except on the last row, where the two solvers are the two things compared. An element of $\mathbf{U}$ counts as existing when its largest value over the whole validation ensemble exceeds 10^{-8} of the largest element, since deciding draw by draw would discard a coupling exactly where it passes through zero.

Quantity	Metric	Evaluated at
σ	relative	the 20 nodes of the energy window
$U_{\alpha\alpha}$, $U_{\alpha\beta}$	worst relative error over the diagonal, and over the off-diagonal elements that exist	the demonstration energy, all validation draws
$\psi(r)$	pointwise $\max\|\Delta\psi\|/\max\|\psi\|$; Appendix D.8 uses a relative L^2 norm, as reported by the reference implementation	the interior mesh, after calibrating the overall constant of one solver onto the other
$\delta(E)$	absolute, in degrees, since a phase shift passes through zero	the 20 nodes of the energy window
solver residual	relative, on σ and on the elements of $\mathbf{U}$	all validation draws and all 60 energies, against the independent R-matrix code

Autocalibration. Four numbers configure an emulator, and none is chosen by hand or by looking at an error. The number of predictor points follows the intrinsic rank of the *potential* variation, exactly as the basis rank follows the intrinsic rank of the *solution* variation. On a small probe sample, K_{diag} and K_{coup} are the ranks at which the normalised spectra of the diagonal and coupling potential signatures fall below $\epsilon_{\text{pot}} = 5 \times 10^{-3}$; n_b follows from the cumulative energy criterion (14) at $\epsilon_{\text{tol}} = 10^{-8}$; the training size is $N_s = 5n_b$. Whether any of that budget goes to the coupling is then decided by the rule of Section 4.4 rather than by the spectrum, for the reason Section 5.6 measures. The code that does this is `core/calibration.py` (Appendix D.5), run before any training and frozen to disk, so every session configures every block identically.
On this suite the procedure returns $n_b = 104, 53, 37, 22$ and $K = 15, 17, 16, 10$ on $\alpha + {}^{12}$C, $n + {}^{238}$U, $\alpha + {}^{24}$Mg and $n + {}^{40}$Ca, with all four budgets spent on the diagonal. The choice $N_s = 5n_b$ makes the overfit ratio of Eq. (20) collapse to $r_{\text{fit}} \simeq 5/K$, so every LROM fit here is underdetermined and LSTSQ returns the minimum-norm operator among those that reproduce the snapshots exactly.

5.3 The two floors

The DBMM is placed by two comparisons, which also set the level below which a residual stops carrying information about physical accuracy. The first runs against the R-matrix code, on two control systems that isolate one solver feature each. $p + {}^{12}$C, an s-wave proton on a central Gaussian well, tests the single-channel Coulomb matching, with a median relative residual against the R-matrix code of 4×10^{-7} over 40 parameter draws and 60 energies. $\alpha + d$ in the $J^\pi = 1^+$ block, whose tensor term mixes $\ell = 0$ and $\ell = 2$, tests the off-diagonal coupling, at 1×10^{-4}. On the four benchmarks the median residual on the cross section is 4×10^{-6} to 3×10^{-5}, and weaker on the individual elements of $\mathbf{U}$ (Table 6).

Table 6: The two solvers element by element, median and 95th percentile of the relative difference over 40 parameters and 60 energies. Each reading is the worst element of its group at one parameter and one energy, the convention of Table 5; the median and the percentile are then taken over the 2400 pairs. The last column is the geometric mean of a reciprocal pair, and it is optimistic against the elements themselves on all three systems.

	diagonal		off-diagonal		$\sqrt{\|U_{\alpha\beta}U_{\beta\alpha}\|}$	
System	median	p95	median	p95	median	p95
$\alpha + {}^{12}$C	2.9×10^{-4}	2.9×10^{-3}	3.1×10^{-3}	4.1×10^{-1}	1.3×10^{-3}	1.3×10^{-1}
$\alpha + {}^{24}$Mg	5.4×10^{-3}	2.6×10^{-2}	2.1×10^{-3}	1.2×10^{-2}	1.2×10^{-3}	6.8×10^{-3}
$n + {}^{238}$U	2.6×10^{-4}	3.6×10^{-3}	1.1×10^{-2}	1.5×10^{0}	3.1×10^{-3}	4.2×10^{-1}

The floor is read on the elements, since a scalar built from $\mathbf{U}$ can hide compensating errors on the elements it sums. On the diagonal the two codes agree to a few 10^{-4} on $\alpha + {}^{12}$C and $n + {}^{238}$U and to 5×10^{-3} on $\alpha + {}^{24}$Mg. On the worst off-diagonal element they agree to 2×10^{-3} to 1×10^{-2} at the median and no better than order one at the p95, because that reading is carried by the smallest element of the matrix. The emulator errors below are measured against the DBMM and not against the R-matrix code, so this floor does not bound them. It bounds instead how far either solver can be read as physics, which on a single off-diagonal element is 10^{-2}.

The $\alpha + {}^{24}$Mg figure is a property of the mesh, and it is the one place where the solver floor reaches the tolerance the emulators are judged against. At the working $N = 100$ the DBMM sits 4.9×10^{-3} from its own converged answer against $\epsilon_{\text{post}} = 5 \times 10^{-3}$, where the R-matrix code is already at 3.0×10^{-5}, so one solver pays it alone and an emulator that clears the tolerance there tracks this solver at this mesh rather than the converged answer. Table E.2 gives the price of closing the gap, at a cost in $N_c N$ cubed paid by the offline stage alone. The cost comparison of Section 5.7 is unaffected, since baseline and emulator are timed on the same mesh.

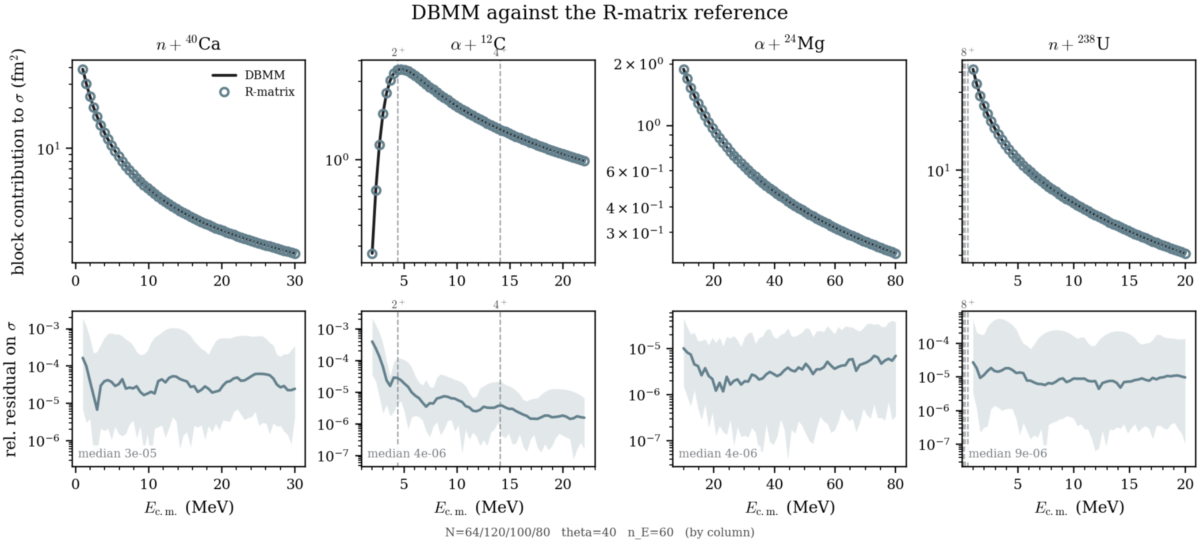

Figure 11: The four benchmarks against the R-matrix code. Upper row: the cross section from both solvers over each energy window. Lower row: their relative difference, median and p5–p95 band over 40 parameter draws.

The comparison cannot reach the recoupling coefficient $\mathcal{F}^{(\lambda)}$ of Eq. (54). Both solvers are handed the same coupling function, so an error in the $3j$, the $6j$ or the phase cancels in their difference while moving every inelastic element and leaving the diagonal intact. It was therefore checked on its own, four ways and without a reference calculation (Table B.4), and none of the four finds a fault.

The second floor comes from a different code and a different method, and it places the emulator as well as the solver. ROSE propagates the radial equation where the DBMM inverts a matrix, and $n + {}^{40}$Ca runs Giuliani's own configuration so that both codes solve one problem. The two solvers agree to 1.1×10^{-4} at the production mesh and to 2×10^{-7} once the DBMM is converged, three decades below the R-matrix floor, once one convention is aligned.

The two reduced masses differ by 1.4 per cent, which moves k by 0.7 per cent and $\mathbf{U}$ by five per cent, more than any other difference between the two codes. The difference is in the kinematics and not in the mass table: ROSE applies the relativistic correction of Ingemarsson [31], so its reduced mass depends on the energy, where the non-relativistic $m_1 m_2/(m_1 + m_2)$ used here does not (Appendix D.8). The shift is common to solver and emulator on a given system, so it cancels in every comparison scored here and bounds only what the report says about physics.

5.4 Thinness of the solution manifold

The POD spectrum tests the premise of Section 4.1. Depending on the system, 22 to 104 modes reach $\epsilon_{\mathrm{tol}} = 10^{-8}$, against parameter vectors of five to ten components and an ambient dimension of 128 to 960.

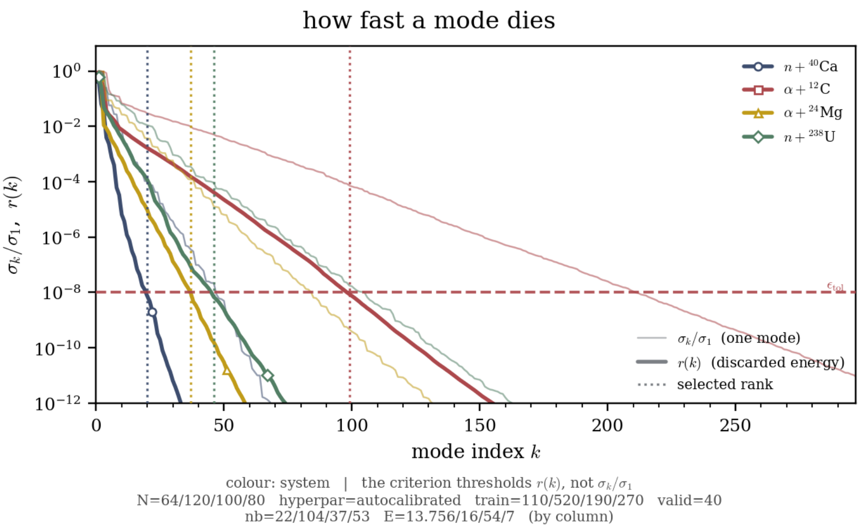

Figure 12: The POD spectrum of each system. Thin: the individual mode σ_k/σ_1. Thick: the discarded energy $r(k)$, the quantity the criterion thresholds. The vertical rule marks the rank each system's own criterion selects.

The criterion thresholds the accumulated $r(k)$ and not the individual mode σ_k/σ_1: the thin curves fall faster, so reading the rank off them would select a basis two to three times too small. That n_b exceeds p by a factor of 2 to 21 does not contradict thinness, since p bounds the *intrinsic* dimension of the manifold while n_b is the rank of the best *linear* subspace covering it. The gap measures curvature, and $\alpha + {}^{12}\mathrm{C}$, whose spectrum falls slowest, is the most curved of the suite. More RBM snapshots at fixed rank change nothing (Appendix E), so its offline stage is a fixed cost sized in advance.

Figure 13 puts the measured out-of-sample error beside the Eckart-Young bound $\sqrt{r(n_b)}$.

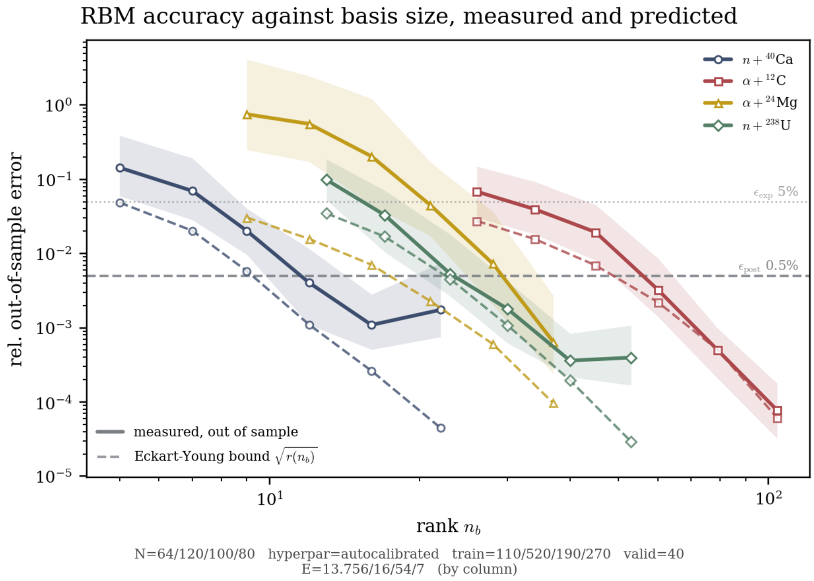

Figure 13: Measured out-of-sample error against retained rank, solid, with the p5–p95 envelope over validation draws, beside the Eckart-Young bound $\sqrt{r(n_b)}$, dashed. The bound is a statement about the training snapshots; the solid curve gives the error at an unseen parameter.

The two do not meet, and that gap forbids relaxing ϵ_{tol}: a subspace discarding a fraction r of the snapshot energy sits at about $\sqrt{r}$ from those snapshots, so $\epsilon_{\mathrm{tol}} = 10^{-5}$ should return 3×10^{-3}, and the measured median runs from 3.9×10^{-3} to 6.9×10^{-2} (Appendix E). The pattern is not followed on two systems. $n + {}^{238}\mathrm{U}$ tracks the bound down to $n_b = 40$, then floors and rises, 3.6×10^{-4} against 4.0×10^{-4} at $n_b = 53$, and $n + {}^{40}\mathrm{Ca}$ does the same between $n_b = 16$ and $n_b = 22$, while the bound falls a further decade on both.

The distance from an unseen solution to the subspace, $\|\mathbf{c} - X_r X_r^H \mathbf{c}\|/\|\mathbf{c}\|$, separates the two possibilities, and the Eckart-Young bound does not cover it. Over the step where both systems floor that distance falls by a factor of six (Appendix E), so the added modes carry what the solution needs and the projected solve fails to use it. The ratio of the two stays nearly constant along each ladder and jumps by an order of magnitude at those two rungs alone. Conditioning is the first suspect for that jump and does not suffice: the median $\mathrm{cond}(\mathbf{M}_r)$ jumps there too, from 49 to 364 and from 100 to 506, while $\alpha + {}^{12}\mathrm{C}$ reaches 534 with both quantities still falling. The floor therefore sits in the step from the subspace to $\mathbf{U}$, and a projection in a norm that weights the boundary rows would say which part of that step carries it.

5.5 Fidelity to the solver

Figure 14 shows the cross section end to end: $\sigma(E)$ across the full window at a parameter never seen in training, with the R-matrix code marked on the solver's own curve, so both steps of the chain are visible at once.

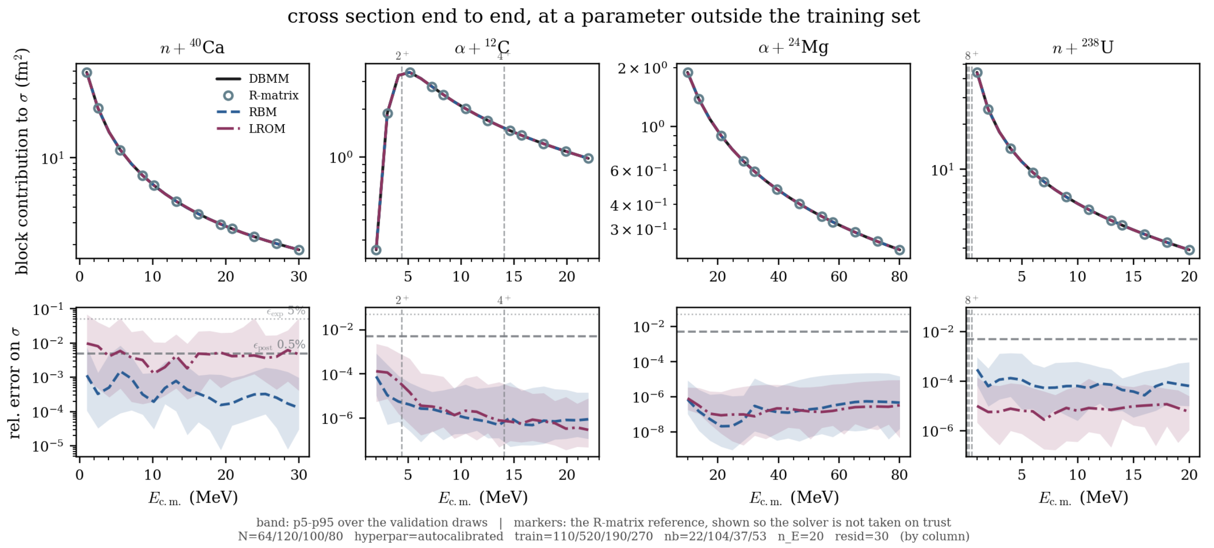

Figure 14: The cross section end to end, at a parameter outside every training set. Markers: the R-matrix reference, so the solver is not taken on trust. Lower row: the relative error of each emulator, median and p5–p95 band over the validation draws.

The median error over the window runs from 1.8×10^{-7} to 4.0×10^{-3}, inside ϵ_{post} on every system and for both methods, and the two are not separated in one direction: the RBM is ahead on $n + {}^{40}$Ca by a factor of twelve and the LROM on $n + {}^{238}$U by a factor of ten, while on the other two the two medians differ by less than a sixth, which thirty validation draws cannot resolve (Table 7). Neither tracks the solver less well where the cross section turns over or where a threshold opens, which is the behaviour a fit to the output would not guarantee and a projection of the equation does.

Table 7: The manifold, and what the emulators achieve on it. The rank is the one the truncation criterion selects at $\epsilon_{\text{tol}} = 1 \times 10^{-8}$, and the three error columns are all quoted at it. Column 5 is obtained at the demonstration energy alone, over 40 validation parameters; columns 6 and 7 over 30 validation parameters and all 20 energies of the window.

		retained rank		median relative error on σ, at that rank		
				at E_{demo}	over the energy window	
System	$N_c N$	retained rank n_b	$n_b/N_c N$	RBM	RBM	LROM
n+^{40}Ca	128	22	17%	8.0×10^{-4}	3.4×10^{-4}	4.0×10^{-3}
α+^{12}C	960	104	11%	6.7×10^{-7}	1.6×10^{-6}	1.8×10^{-6}
α+^{24}Mg	300	37	12%	3.3×10^{-7}	1.8×10^{-7}	2.0×10^{-7}
n+^{238}U	400	53	13%	7.4×10^{-5}	7.0×10^{-5}	6.9×10^{-6}

A scalar can hide compensating errors, so the same comparison is made on the object it is drawn from. Figure 15 shows the interior solution.

The emulator deviations on $\psi(r)$ run from 9.7×10^{-6} to 3.1×10^{-3} of the peak amplitude (Table E.5), so the agreement is not confined to the scalar drawn from $\mathbf{U}$. The value of that agreement depends on the system: the two solvers themselves disagree on ψ by 6×10^{-8} on $n + {}^{40}$Ca and by 9×10^{-3} on $\alpha + {}^{12}$C, five decades apart, and three of the eight emulator readings sit below their own system's solver disagreement. On the interior solution the solver floor therefore decides what can be measured.

The two LROM implementations, scored on the radial wave function against their own solvers, differ by a factor of 2 in favour of the construction here at matched rank and matched budget, and converge once the budget is large enough (Table 8). It is the only comparison in this report against an emulator built elsewhere, on a problem neither was tuned for; Appendix D.8 gives the conventions and the conditions.

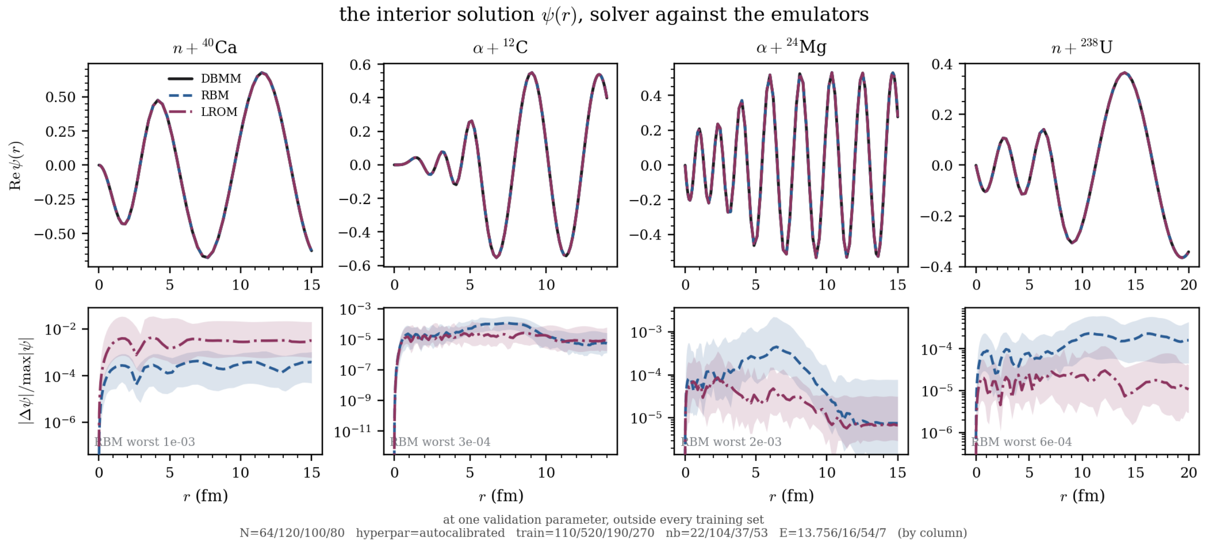

Figure 15: The interior solution $\psi(r)$, solver against both emulators, at a validation parameter. Lower row: the pointwise deviation as a fraction of the peak amplitude.

Table 8: The two LROM implementations on one problem, $n + {}^{40}$Ca. Median relative L^2 error on the radial wave function, each against its own solver, over 100 training and 30 validation draws, the same on both sides. Left: rank matched at $K = 10$. Right: predictor budget swept at $n_b = 22$.

matched rank, $K = 10$			budget swept, $n_b = 22$		
n_b	repository	this work	K	repository	this work
4	8.1×10^{-3}	5.4×10^{-3}	6	5.0×10^{-3}	3.5×10^{-3}
8	5.6×10^{-4}	3.4×10^{-4}	10	4.9×10^{-4}	2.2×10^{-4}
16	4.9×10^{-4}	2.2×10^{-4}	16	4.7×10^{-5}	4.1×10^{-5}
22	4.9×10^{-4}	2.2×10^{-4}	24	5.5×10^{-5}	5.5×10^{-5}

5.6 Test of the predictor budget rule

Section 4.4 predicts that splitting the budget matters exactly when a parameter is absent from the diagonal of every channel, and the suite contains both kinds. The comparison holds the total K fixed, so the split reallocates a budget: at equal K the online cost of Figure 9 is identical either way, and the test bears on where the points are spent.

Table 9: Joint against split predictor budget, at fixed total K: *split* divides K as shown, *joint* spends it all on the diagonal, and everything else is held equal over 40 validation draws. Column 4 is how far the diagonal potential travels, relative to its own size, when the coupling parameter of column 2 crosses its box; it is the quantity the rule of Section 4.4 is read on. Columns 5 and 6 are the median error of the joint budget divided by that of the split one, on the elastic and on the inelastic element: $\times n$ means the split is n times better, $\div n$ that it is n times worse, and a dash that the two lie within a factor of 1.5. The last column is the allocation the autocalibration applies, which follows the rule rather than a choice: split where the diagonal does not move, joint everywhere else. $n + {}^{58}$Ni is the only system whose coupling parameter is absent from the diagonal exactly, by the selection rule of Section 2.4.

				gain from splitting		
System	param.	$K_{\mathrm{diag}}+K_{\mathrm{coup}}$	diagonal travel	elastic	inelastic	applied
$\alpha+{}^{12}$C	β_2	8+7	0.03	–	–	*joint*
$\alpha+{}^{24}$Mg	β_2	6+10	0.0005	$\div 1.9$	$\times 2$	*joint*
$n+{}^{238}$U	β_2	6+11	0.001	$\div 6.8$	–	*joint*
$n+{}^{58}$Ni	δ_2	2+2	0	$\times 47$	$\times 2987$	*split*
$\alpha + d$	V_{t0}	2+2	0.4	–	–	*joint*

On $n + {}^{58}$Ni the coupling parameter is absent from the diagonal exactly, by the selection rule of Section 2.4. There the split is worth $\times 47$ on the elastic element and $\times 2987$ on the inelastic one, and swept across δ_2 the joint prediction does not move at all. A selection rule imposes the condition here, so the test does not depend on a preset chosen for it.

On the other four the parameter reaches the diagonal, by the static reorientation on the three rotational systems and by the diagonal element of the tensor operator in the $\ell = 2$ channel on $\alpha + d$, and there the split costs more than it returns. On $\alpha + {}^{24}$Mg and $n + {}^{238}$U, where the deformation reaches the diagonal but only barely, moving ten and eleven of the sixteen and seventeen points into the coupling block costs a factor of 1.9 and 6.8 on the elastic element while returning 2 and nothing on the inelastic one. At fixed K every point spent on the coupling is a point not spent on the diagonal, so the reallocation loses accuracy wherever the diagonal already carries the parameter. On $\alpha + {}^{24}$Mg the two directions are of the same size, and the joint allocation prescribed there improves the elastic element and degrades the inelastic one, which a coupled-channel campaign propagates. The diagnostic reads the diagonal travel, so it does not arbitrate that trade.

The fourth column is the diagnostic the rule needs, and it separates the suite without a threshold to choose: exactly zero on $n + {}^{58}$Ni, between 5×10^{-4} and 4×10^{-1} on the other four. The autocalibration evaluates it before any training and spends the whole budget on the diagonal wherever the diagonal moves, which is the last column and the configuration every LROM number in this report was produced with.

5.7 The cost of one evaluation

The cost of one forward evaluation can be defined in three ways, and they differ by more than the reduction itself does: the DBMM rebuilt from scratch at each θ; the DBMM keeping $\mathbf{K}(E)$ and re-evaluating only $\mathbf{V}(\theta)$, which is the strongest baseline available without any emulator and the denominator used throughout; and the R-matrix code, whose column bounds from above what an unrefined implementation costs, since it rebuilds a θ-independent kinetic matrix at every call. Figure 16 places every validation draw against all three.

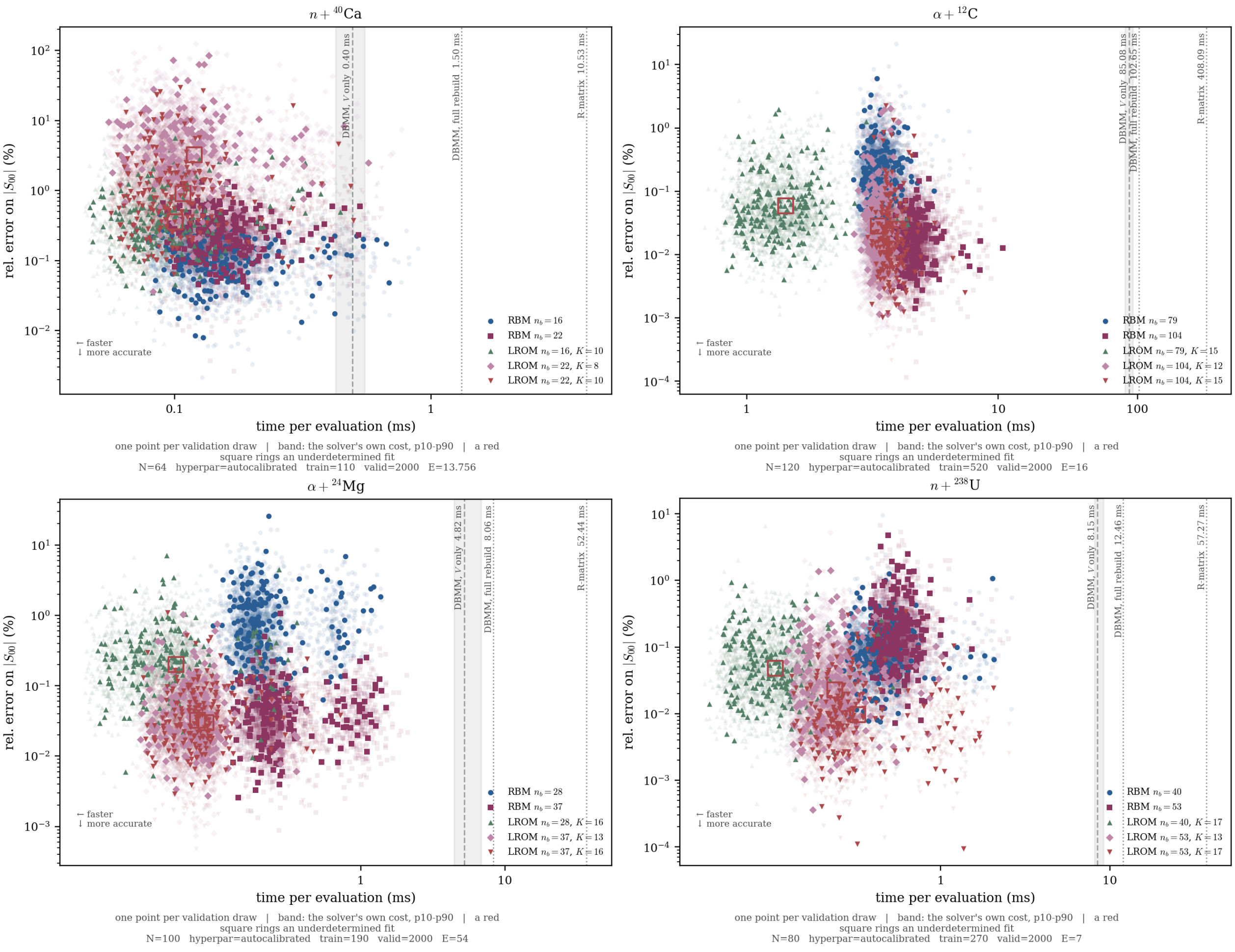

Figure 16: Cost against accuracy, one point per validation draw, on the four benchmarks. Dotted rules: the DBMM rebuilt and the R-matrix code; the grey band is the split DBMM, the baseline every gain in Table 10 is measured against. A red square rings an underdetermined residual fit, which on this suite is every LROM configuration. The error is the elastic element on all four systems, where Table 10 and Figure 18 score the worst off-diagonal element wherever the system is coupled; Figure E.4 is the same plane on that element.

Each point is one draw carrying its own measured time, so the vertical spread is a result, the error varying by up to two decades over the box, while the horizontal spread is the scatter of the timing alone. Both emulators sit half a decade to two decades left of the strongest solver baseline and about two decades left of the R-matrix code, and the LROM cloud sits left of the RBM cloud on every system. The two error distributions nevertheless overlap almost everywhere, so the separation between the methods holds over the distributions and not at any single draw.

Table 10: The cost of one forward evaluation, and what it amortises to. Medians over the timed draws, each the minimum of at least seven repetitions, on one core. The two error columns are the metric of Table 5 on the elements of **U**: the worst off-diagonal element that exists where the system is coupled, the elastic element where it is not. They are read off 2000 validation draws, none of them seen in training; the p95 in particular needs a sample of that size to be an estimate rather than close to the largest value observed. Absolute times reproduce to between 5 and 30 per cent between runs, so the gain against the V-only baseline is *paired*, baseline and prediction timed on the same θ microseconds apart; the other two are unpaired and rounded. It is a median of per-draw ratios, so it does not equal the ratio of the two median columns. **The R-matrix column is an upper bound, not the cost of the method**: that implementation rebuilds a θ-independent kinetic matrix on every call and inverts where N_c right-hand sides would do. The gain against V only is the one to defend. Each configuration is the cheapest the autocalibration proposes that clears the posterior tolerance. Two things this table is not. The offline cost is paid *per energy node and per J^π block*: a campaign over twenty energies pays it twenty times, so the training minutes quoted here are not the training cost of a full campaign. And every time in it is Python against Python. It is the ratio that carries over to another implementation, not the absolute values: a compiled solver would move both the baseline and the emulator columns.

			solver, ms per evaluation			emulator		gain against			error	
System	method	n_{train}	R-matrix	rebuild	split	operating config.	ms	R-matrix	rebuild	split	median	p5–p95
$n{+}^{40}$Ca	RBM	110	10.5	1.5	0.40	n_b 16	0.13	×83	×12	×3.5	1.1×10^{-3}	2.7×10^{-4}–2.6×10^{-3}
	LROM					K 10	0.09	×113	×16	×4.4	3.6×10^{-3}	7.0×10^{-4}–1.9×10^{-2}
$\alpha{+}^{12}$C	RBM	520	408.1	102.6	85.08	n_b 79	2.49	×164	×41	×34	2.0×10^{-3}	3.1×10^{-4}–1.1×10^{-2}
	LROM					K 15	1.18	×345	×87	×64	7.3×10^{-4}	1.5×10^{-4}–4.4×10^{-3}
$\alpha{+}^{24}$Mg	RBM	190	52.4	8.1	4.82	n_b 37	0.37	×143	×22	×15	6.9×10^{-4}	1.4×10^{-4}–3.4×10^{-3}
	LROM					K 16	0.16	×337	×52	×27	4.5×10^{-3}	9.6×10^{-4}–2.5×10^{-2}
$n{+}^{238}$U	RBM	270	57.3	12.5	8.15	n_b 40	0.56	×103	×22	×15	4.0×10^{-3}	7.1×10^{-4}–2.1×10^{-2}
	LROM					K 17	0.26	×217	×47	×29	3.4×10^{-3}	5.4×10^{-4}–2.2×10^{-2}

Table 10 carries two results, and they do not multiply one another. The affine split alone moves the DBMM from the *rebuilt* column to the V *only* column, a factor 3.7 on $n + {}^{40}$Ca falling to 1.2 on $\alpha + {}^{12}$C, since the larger the system the more the $\mathcal{O}((N_c N)^3)$ solve dominates the assembly the split saves; the reduction returns more the larger the system, up to ×64.3, so the two gains run in opposite directions with size. Timing the online stage step by step identifies what the LROM removes (Table D.5): on $\alpha + {}^{12}$C, forming the reduced operator and its source is four fifths of the RBM query, and it is where the mesh size enters, through the $\mathcal{O}(N_c^2 N n_b^2)$ of Figure 9; the LROM replaces it by K scalar evaluations.

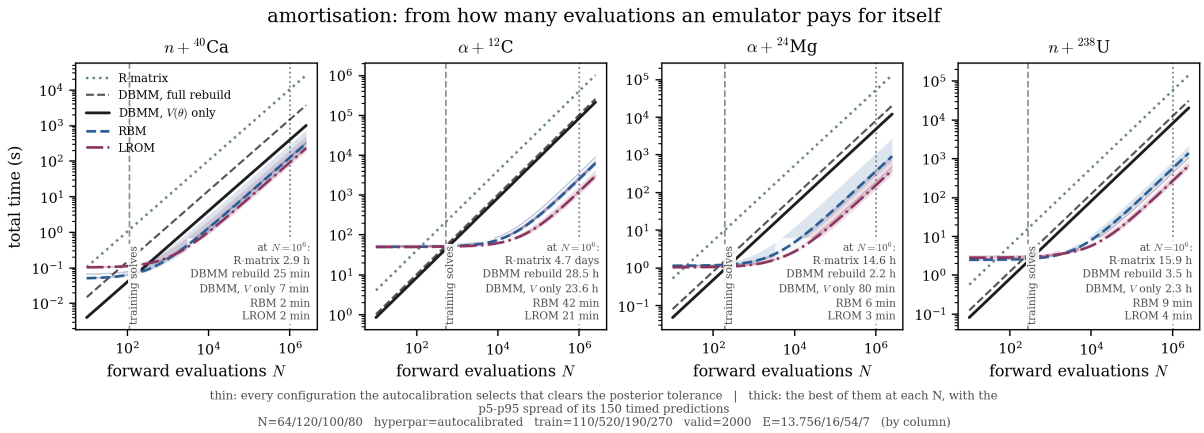

Figure 17: Total time against the number of forward evaluations. Three solver baselines, straight lines through the origin differing only in slope, against an emulator's offline constant plus a shallower slope; the dashed rule is the training size. Below the crossing with the split DBMM emulating is a loss. All times are Python against Python, and only the ratio between the lines carries over to another implementation.

An emulator is a loss until it has been queried enough to repay its training (Figure 17). Against the split DBMM the break-even falls at $N^* = 191$, 612, 274 and 358 evaluations for the RBM, against training sizes of 110, 520, 190 and 270: the ratio stays between 1.18 and 1.74 while the training size varies by a factor of nearly five. The break-even is therefore of the order of the training size on these four systems. The LROM starts behind and overtakes the RBM at 1600, 250, 120 and 2000 evaluations, in the order of the table, far below the 10^4 a posterior needs. Per energy node and per J^π block, a chain of 10^5 samples on $\alpha + {}^{12}$C costs 142 minutes with the split DBMM against 2.8 with the LROM, training included, so over the twenty nodes of the window that one block is 47 hours against 0.9 hour. The number of blocks multiplies both identically, so only the ratio carries over.

5.8 Robustness outside the training box

Outside the parameter box the degradation is gradual. Half again the half-width costs a factor of 1.4 to 13 at the median and 2 to 90 at the p95 across the eight configurations, nothing diverges, all eight medians still clear ϵ_{post} at 1.2 times the half-width and seven of the eight at 1.5. The limit of employment is therefore at least twenty per cent beyond the box the emulator was trained on (Appendix E).

Over the same widening the LROM loses more than the RBM on all four systems, a factor of 4.8 to 13 at the median against 1.4 to 4.6, and 10 to 90 at the p95 against 2 to 18. The one configuration whose median leaves the tolerance at 1.5 times the half-width is the LROM on $n + {}^{40}$Ca, which is also the system where it is furthest behind inside the box. On robustness the two therefore rank the other way round from cost.

5.9 Verdict and consequences for a Bayesian campaign

Figure 18 gathers the eight configurations on one axis. Of the three criteria, only cost ranks the two methods consistently. The LROM is cheaper online on all four systems, and cheaper amortised beyond the crossings of Section 5.7. Its gain against the split DBMM is the larger of the two on each: $\times 4.4$ against $\times 3.5$ on the smallest system and $\times 64.3$ against $\times 33.7$ on the largest, a factor of 1.3 to 1.9 between the two methods. The medians of the two online times give 1.4 to 2.3 instead; the two readings separate only on $\alpha + {}^{12}$C, and by less than the run-to-run spread Table 10 quotes.

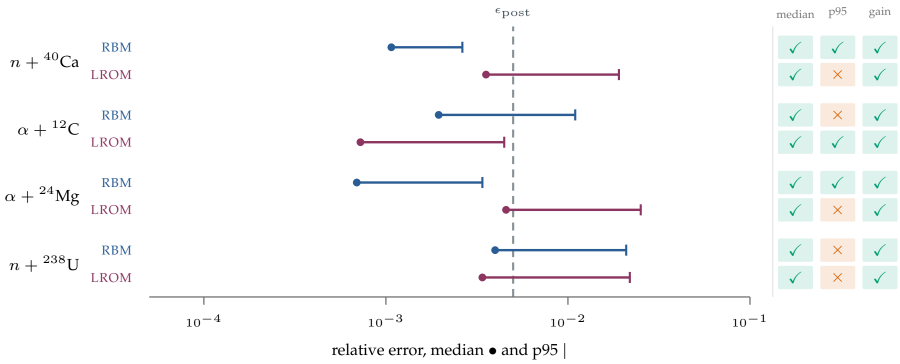

Figure 18: The eight configurations of Table 10 against the posterior tolerance. The error is the metric of Table 5 on the elements of $\mathbf{U}$, the worst off-diagonal element that exists where the system is coupled and the elastic element where it is not: the dot is its median over the box, the line runs to its p95, and everything left of ϵ_{post} clears the tolerance. On the right, ✓ for a criterion met and × otherwise; the *gain* is measured against the split DBMM, ✓ beyond $\times 1.5$ and × below $\times 1.05$.

All eight medians clear the posterior tolerance, so accuracy admits both, and beyond that the ranking depends on the quantity scored and on the system: the RBM leads on the cross section and on the phase shift, the LROM on the interior solution, and the two split the suite on the worst matrix element. In the tail the RBM clears the tolerance on two systems and the LROM on one, and nothing in the calibration constrains that tail, because it never measures an error.

The LROM is furthest behind on $n + {}^{40}$Ca, the system with ten free parameters and a spectral budget of ten predictor points. An operator affine in K scalars has one scalar per parameter there and no margin, which is a limit of reading K off a spectrum rather than off the parameter count. On that system alone the two methods are separated by more than a factor of ten on the cross section, on the interior solution and on the phase shift at once.

For the problem this report addresses, where the campaign is limited by the number of forward evaluations and both methods clear the tolerance at the median, I recommend the LROM. It is cheaper on every system, by the factor above, and it costs nothing in median accuracy on three systems of four. The RBM is the safer choice where the tail matters more than the median, or on a system with as many parameters as $n + {}^{40}$Ca has, since it makes one approximation where the LROM makes two.

A likelihood built on σ evaluated by either emulator is faithful at the level the data demand, and that cannot be turned into a statement about a posterior. The emulator error is a deterministic function of θ, so it enters a likelihood as a systematic distortion of the surface the chain walks, correlated exactly where the chain spends its time, and 10^5 samples do not average it away. Declaring it harmless would mean showing the error is small *where the posterior has mass*, which presupposes a posterior nobody sampled here. The training cost is paid once, and a campaign then runs at the ratio quoted above.

6 Conclusion and perspectives

6.1 Results

On these four systems a coupled-channel scattering calculation can be made cheap enough for its uncertainty to be quantified, by a factor of $\times 3.5$ to $\times 64.3$ against the strongest solver-side baseline.

The Direct Boundary Matching Method was implemented for single-channel and for coupled-channel problems, with distinct channel thresholds, closed channels and complex optical potentials, and validated against an independent calculable R-matrix code. Non-local potentials are not supported. The symmetry of $\mathbf{U}$ and its unitarity with the absorption removed hold whatever code produced it, and they fix the normalisation the collision matrix must be read with.

The boundary condition occupies one row of $\mathbf{M}$ and the potential stays diagonal on a Lagrange mesh, which splits the matrix exactly into a parameter-free $\mathbf{K}(E)$ and a diagonal $\mathbf{V}(\theta)$. The reduced basis method is built on that split, and the learning reduced-order model of Giuliani is extended from one channel to several. The extension required splitting the predictor budget between the diagonal and the coupling blocks, an idea due to P. McGlynn and C. Hebborn. The split is needed exactly when a parameter is absent from the diagonal of every channel. Section 5.6 tests the rule on five systems, none of them built to make it hold. Its diagnostic is how far the diagonal travels when the coupling parameter crosses its box, the autocalibration evaluates it before any training, and every configuration in this report is the allocation it prescribes.

Both emulators clear the posterior tolerance at the median on all four benchmarks, over the boxes of Appendix C. The LROM is cheaper online on all four systems, with a gain against the same baseline 1.3 to 1.9 times larger than the RBM's. Which of the two is the more accurate depends on the system and on the quantity scored. Section 5.9 states the recommendation that follows and the conditions on it. The break-even against the split DBMM tracks the training size on all four systems. The sample is too small to turn that into a design rule.

6.2 Limits

- No posterior was sampled, and Section 5.9 says what that costs. One experiment would settle it and is the first thing to do: a single chain on a synthetic likelihood, run once with the solver and once with the emulator, with the two posteriors compared marginal by marginal. Everything it needs is already in place, since the emulator exposes the same interface as the solver.
- The boxes are built by scaling published potentials, so the thinness of the manifold is untested for a genuinely different family of interactions.
- The implementation is restricted to local potentials, since a non-local $V(r, r')$ destroys the diagonality the whole construction depends on. Transfer and continuum-discretised coupled channels call for new physics input more than for numerical development.
- Only integrated cross sections are tested, and on one J^π block. A measurable cross section sums the blocks, and an angular distribution makes them interfere.
- No residual-based error estimator is implemented, which would turn the POD spectrum into a bound at a query point. The error stops falling at the last rank of the ladder on two systems while the distance from an unseen solution to their subspace keeps falling, so the floor is in the step from the subspace to $\mathbf{U}$ and not in the basis; the conditioning of the reduced solve jumps at the same rung on both without being sufficient to explain it, since a third system reaches the same conditioning with its error still falling. A projection in a norm that weights the boundary rows is the test that remains.
- The budget rule has one positive case, $n + {}^{58}$Ni, so the branch that prescribes a split is exercised once and every configuration reported here is the other branch.
- The LROM calibration is not tested against itself. The overfit ratio is below unity on every configuration, so each is the minimum-norm interpolant of its own snapshots and what that regularisation is worth is not measured; and the predictor budget is read off a potential spectrum, which on $n + {}^{40}$Ca returns ten points for ten free parameters, the system where the LROM is furthest behind the RBM (Appendix D.5).

6.3 Perspectives

The group is already pursuing the extension to the three-body problem. McGlynn and Hebborn [11] solve it for two-neutron halo nuclei on the same Lagrange-Legendre mesh, with the same diagonality of the potential, and their hyperradial system is structurally that of Eq. (6), with 610 partial waves at $K_{\max} = 40$ against eight channels here. Sweeping the channel count within one physical family, the gain of both emulators grows with that count over the range tested (Appendix E.8). Two costs of the LROM grow with it for a structural reason: its learned operator differs from one entrance channel to the next, so the reduced solve is N_c^{open} factorisations of an $n_b \times n_b$ matrix where the RBM factorises once and applies N_c right-hand sides (Table D.5), and the stored operator, $KN_c^{\text{open}}n_b^2$ complex numbers, is 21 MB here and tens of gigabytes at 610. That structure already shows in the online cost, and not only in the memory: over the same sweep the LROM grows as $N_c^{1.75}$ against $N_c^{1.50}$ for the RBM, and at the four largest channel counts tested the RBM is the cheaper of the two. The mesh axis runs the other way, and there the LROM gains with N (Appendix E.7), which is the axis the four benchmarks sit on. Section 5.9 recommends the LROM on a suite of large meshes and few channels; a three-body extension moves along the other axis. The measurement covers one physical family over the sizes tested and is not an asymptotic law. It establishes only that emulating should gain more over solving at 610 channels than at eight. The algebraic conditions of Section 3.4 are already satisfied in that problem, and its authors need quantified uncertainties.

The implementation leaves two engineering gains unused. The offline stage is a trivially parallel loop whose batch of LU factorisations maps onto batched GPU routines, which would make retraining on the fly practical. Online, batching queries over θ into a single GEMM would return more than any improvement to the reduced solve itself. A quadratic-manifold reduction is a further direction, aimed at the curvature Section 5.4 measures on $\alpha + {}^{12}\text{C}$. No measurement in this report points at it.

Beyer and Nunes [32] survey the tools the community has converged on within the BAND framework [22], among them ROSE [14], [23] and jitR [24]. The emulators built here apply the same idea to a different solver, and the next step is to measure them against jitR and not only against the solver they are built on. In the terms of that survey, the work makes parametric uncertainty affordable to explore and leaves model uncertainty unchanged, since an emulator that is exact with respect to its solver carries every assumption that solver makes.

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A Scattering theory: notation and complements

Section 2 keeps only what defines **U** and writes the linear system the solver discretises. This appendix carries the notation the whole report uses, and the three derivations that section states without proof: how the partial-wave sum assembles into an observable, what the Coulomb functions do at the two ends of their range, and how a block of one J^π relates to a measurable cross section.

A.1 Notation and conventions

Acronyms

DBMM	Direct Boundary Matching Method (Section 3.2)
RBM	reduced basis method (Section 4.2)
LROM	learning reduced-order model (Section 4.3)
POD	proper orthogonal decomposition (Section 4.2)
SVD	singular value decomposition (Section 4.2)
CH89	Chapel Hill 89, a global optical potential (Section 2.3)
ROSE	Reduced Order Scattering Emulator, of the BAND framework (Section 1.3)
jitR	just-in-time R-matrix library, of the BAND framework (Section 6)
BAND	Bayesian Analysis of Nuclear Dynamics, a software framework (Section 6)
GEMM, GEMV	general matrix-matrix and matrix-vector product, BLAS level 3 and 2 (Appendix D.6)
LU	lower-upper factorisation of a matrix (Section 3.2)
GIL	global interpreter lock, of the Python interpreter (Appendix D.6)

Symbol	Meaning
The collision, and what is measured	
ℓ, I, J^π	orbital, target-spin and total angular momentum; $\mathbf{I}+\boldsymbol{\ell}=\mathbf{J}$, and J^π labels the block solved
α, β	channel indices, i.e. one full asymptotic configuration each; β_0 is the entrance channel
$v_\alpha = \hbar k_\alpha/\mu_\alpha$	channel velocity; the ratio $\sqrt{v_\alpha/v_\beta}$ is the flux factor of Eq. (53)
$\chi_\alpha(r)$, $\psi^{\rm sc}$	radial wave function in channel α, and the scattered part of Eq. (9)
$f(r; R_0, a_0)$	Woods-Saxon shape $[1+e^{(r-R_0)/a_0}]^{-1}$; f' is its radial derivative
β_2, β_4	quadrupole and hexadecapole deformation. On every rigid-rotor system of the suite they reach the diagonal of each excited channel through the static reorientation term of Eq. (54), and not the coupling blocks alone (Appendix B.4)
T_{ij}, $\hat{f}_j$, λ_j	kinetic matrix elements, regularised Lagrange functions and Gauss-Legendre weights
σ	cross section, elastic or reaction as stated; not to be confused with the singular values σ_k
$r(k)$	discarded energy of the POD spectrum, the quantity $\epsilon_{\rm tol}$ thresholds
$n_{\rm train}$, N^*	training snapshots, and the number of evaluations at which an emulator overtakes the solver
$t_{\rm on}$, $t_{\rm off}$	online cost of one prediction and total offline cost of building the emulator
Kinematics and asymptotics	
μ	reduced mass, MeV$/c^2$
E	centre-of-mass energy of relative motion, MeV
k	wave number, $k=\sqrt{2\mu E}/\hbar$, fm^{-1}
κ_α	its real modulus in a closed channel, $\kappa_\alpha = \lvert k_\alpha \rvert$, fm^{-1}
η	Sommerfeld parameter, $\eta = Z_1 Z_2 e^2/\hbar v$
a, R	channel radius (matching radius), fm

Symbol	Meaning *(continued)*
F_ℓ, G_ℓ	regular and irregular Coulomb wave functions
$H_\ell^\pm = G_\ell \pm iF_\ell$	outgoing / incoming Coulomb-Hankel functions
$W_{-\eta,\ell+1/2}$	Whittaker function (closed-channel asymptotics)
σ_ℓ	Coulomb phase shift, $\arg\Gamma(\ell+1+i\eta)$
δ_ℓ	nuclear (elastic) phase shift
U_ℓ	collision-matrix element, single channel
$\mathbf{U}$	collision matrix, coupled channels
$\mathcal{R}$, $\mathbf{R}$	R-matrix, single channel and coupled channels
ϵ_α	threshold energy of channel α, MeV
$\gamma_{s,\alpha}$	logarithmic derivative of the outgoing wave at R, fm^{-1}
Discretisation and reduction	
N	number of Lagrange-Legendre mesh points. The number of forward evaluations of a campaign is written N_{eval}, and the break-even between them N^*
N_c	number of coupled channels in a J^π block
$N_{\text{tot}} = N_c N$	dimension of the full linear system
$\theta \in \mathcal{P} \subset \mathbb{R}^p$	optical-potential parameter vector, p parameters
θ_c	central (gauge) parameter point
N_s	number of training snapshots
n_b	reduced basis rank
$K = K_{\text{diag}} + K_{\text{coup}}$	number of predictor points
$\mathbf{M}(E,\theta)$	DBMM system matrix, $N_{\text{tot}} \times N_{\text{tot}}$, complex
$\mathbf{K}(E)$, $\mathbf{V}(\theta)$	parameter-free and parameter-dependent parts of $\mathbf{M}$
X_r	POD basis, $N_{\text{tot}} \times n_b$
$a(\theta)$	reduced coordinates, in both emulators
$q_j(\theta)$	the K scalar predictors of the LROM, values of the potential at the selected radial points
ϵ_{tol}	POD truncation tolerance, 10^{-8}; thresholds $r(k)$ and fixes n_b
ϵ_{pot}	spectral threshold on the potential signature, 5×10^{-3}; fixes the predictor budget K. Unrelated to ϵ_{post} despite the equal value
r_{fit}	overfit ratio, $N_s/[K(n_b+1)]$
Collisions of notation, and the constants	
Two symbols carry more than one meaning, and are kept:	
σ	a cross section without a subscript, the singular values with σ_k (Section 4.2), the Coulomb phase with σ_ℓ (Appendix B). Each is the standard symbol of its own literature and each carries its own subscript
K	the predictor budget throughout, except in the single sentence of Section 6.3 quoting a hyperspherical model, where $K_{\max}$ is that model's hyperangular momentum
The tolerance every configuration is judged against:	
ϵ_{post}	5×10^{-3}: the emulator tolerance adopted here, one decade inside the experimental uncertainty on an elastic angular distribution
$\hbar c$	197.327 MeV·fm
e^2	1.44 MeV·fm

The two calibration constants

ϵ_{pot} shares its numerical value with ϵ_{post} by coincidence only. The first is a threshold on a potential spectrum and fixes how many predictor points to spend; the second is an accuracy the emulator is required to reach. Every rank, budget and training size quoted in this report follows from ϵ_{tol} and ϵ_{pot} and from nothing else. The three ranks the tables keep apart are the *selected* rank, where the criterion of Section 4.2 crosses the discarded-energy curve; the *retained* rank, the one a given fit keeps, swept from a quarter of the selected rank up to it; and the *operating* rank, the cheapest of those rungs whose error clears ϵ_{post}, which is the rank Table 10 runs at. When the emulator fits are run, the truncation inside them is set to 10^{-14}, loose enough to retain every mode available, so the rank used is the calibrated one and not one the fit re-derived for itself.

Conventions

Throughout, r and R both denote the radial relative coordinate; R is reserved for the channel radius when the distinction matters. Channel indices are written α, β and basis-function indices i, j. The collision matrix is written $\mathbf{U}$ following Descouvemont and Baye [12]; it is the same object as the S-matrix of much of the literature, and the symbol S is used for its individual elements where that is clearer.

A.2 From the plane wave to the collision matrix

Figure A.1 is the skeleton of this subsection and the next two: which step follows from which, and what each one assumes.

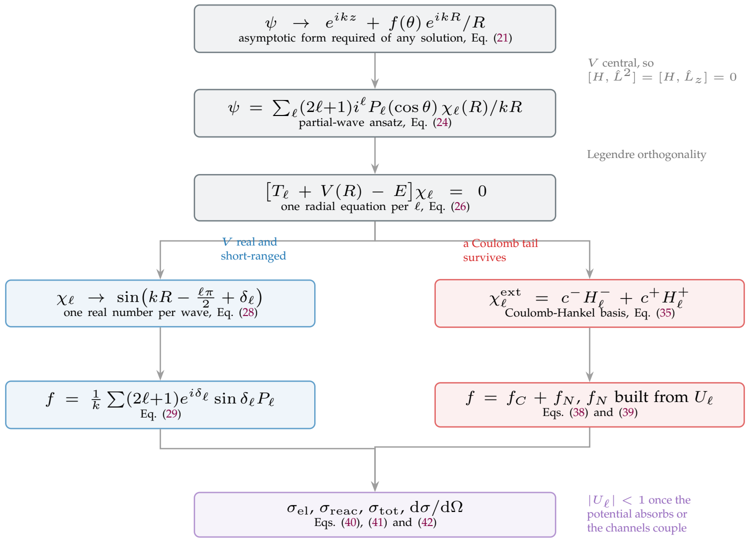

Figure A.1: How the observables are reached from the asymptotic form, and what each step assumes. The trunk holds for any interaction. A central potential separates the problem by ℓ, and the Legendre orthogonality turns the three-dimensional equation into one radial equation per partial wave. The two branches are the two regimes this report meets. On the left the potential is real and short-ranged and the whole effect of the interaction is one real phase shift per wave; on the right a Coulomb tail survives, the asymptotic phase acquires a logarithmic term and δ_ℓ becomes the nuclear part of a collision-matrix element. The two rejoin at the observables, where $|U_\ell| < 1$ separates the flux that returns to the elastic channel from the flux that does not.

The asymptotic picture

Before decomposing the problem into partial waves, it is useful to state the generic asymptotic picture that any scattering solution must reproduce. Far from the range of the interaction, where $V(R) \to 0$, the wave function separates into two physically distinct pieces: an incident plane wave, representing the beam of projectiles before the collision, and an outgoing spherical wave, representing the flux scattered by the target into all directions, illustrated in Figure A.2.

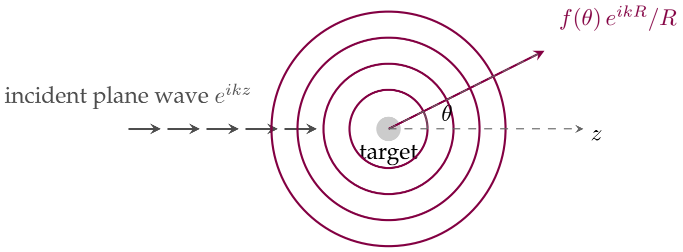

Figure A.2: Asymptotic geometry of a scattering process: an incident plane wave along z produces, far from the interaction region, an outgoing spherical wave of amplitude $f(\theta)$ modulating the angular distribution of the scattered flux.

Far from the target, the wave function is required to take the asymptotic form

$$\psi(\mathbf{R}) \xrightarrow{R\to\infty} e^{ikz} + f(\theta)\, \frac{e^{ikR}}{R} \tag{21}$$

The first term describes the unperturbed incident beam; the second describes an outgoing spherical wave whose amplitude is modulated by the scattering amplitude $f(\theta)$, depending on the scattering angle θ but not, by symmetry about the beam axis, on the azimuthal angle ϕ. This is the picture appropriate to a short-ranged potential acting alone; its Coulomb-distorted generalisation, needed for the charged systems of this report, is developed in Sections A.3 and A.4. The structure here nonetheless already fixes the central relation between theory and experiment: the differential cross section is obtained directly as the modulus squared of the scattering amplitude,

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 \tag{22}$$

a relation made explicit by computing the ratio of the outgoing radial probability flux to the incident flux at large R, and which holds, with f replaced by its Coulomb-distorted generalisation, throughout the rest of this appendix. Determining $f(\theta)$ from the potential $V(R)$ for a given partial wave is the object of the expansion developed next.

The partial-wave expansion

Because $V(R)$ is a central potential (Section 2.1), depending on R alone, $[H, \hat{L}^2] = [H, \hat{L}_z] = 0$ and total angular momentum is conserved. The resulting expansion is developed first for a neutral or short-ranged projectile, with no Coulomb tail, for which the incident beam is the bare plane wave e^{ikz}; the Coulomb-distorted generalisation follows in Section A.3. This plane wave is an eigenstate of $\hat{L}_z$ with eigenvalue $M = 0$, and spherical symmetry cannot generate components with $M \neq 0$, so the full wave function is independent of the azimuthal angle ϕ, with the $M = 0$ spherical harmonic $Y_\ell^0 \propto P_\ell(\cos\theta)$. The incident plane wave itself admits the standard expansion in spherical Bessel functions and Legendre polynomials,

$$e^{ikz} = \sum_{\ell=0}^{\infty} (2\ell+1)\, i^\ell\, j_\ell(kR)\, P_\ell(\cos\theta) \tag{23}$$

which motivates seeking the full wave function in the same angular basis, with the radial dependence promoted from the free spherical Bessel function $j_\ell(kR)$ to an unknown radial function $\chi_\ell(R)/(kR)$ that will carry the entire effect of the potential on partial wave ℓ,

$$\psi(R,\theta) = \sum_{\ell=0}^{\infty} (2\ell+1)\, i^\ell\, P_\ell(\cos\theta)\, \frac{\chi_\ell(R)}{kR} \tag{24}$$

equivalently obtained by looking for simultaneous eigenstates of H, $\hat{L}^2$ and $\hat{L}_z$. Using the identity

$$\nabla_R^2 \left[P_\ell(\cos\theta)\, \frac{\chi_\ell(R)}{R} \right] = \frac{P_\ell(\cos\theta)}{R} \left(\frac{d^2}{dR^2} - \frac{\ell(\ell+1)}{R^2} \right) \chi_\ell(R) \tag{25}$$

and the orthogonality of the Legendre polynomials, $\int_0^\pi P_\ell(\cos\theta) P_{\ell'}(\cos\theta) \sin\theta\, d\theta = \frac{2}{2\ell+1}\delta_{\ell\ell'}$, substituting the partial wave sum Eq. (24) into the Schrödinger equation Eq. (1) yields, for each ℓ independently, the radial wave equation

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dR^2} - \frac{\ell(\ell+1)}{R^2} \right) + V(R) - E \right] \chi_\ell(R) = 0 \tag{26}$$

The three-dimensional Schrödinger equation has been reduced to an infinite set of decoupled one-dimensional equations, one per partial wave ℓ, each governed by an effective potential

$$V_{eff}(R) = V(R) + \frac{\hbar^2}{2\mu} \frac{\ell(\ell+1)}{R^2} \tag{27}$$

where the second term is the centrifugal barrier. Together with the Coulomb barrier, it controls much of low-energy nuclear physics. It suppresses the contribution of high partial waves, so that only the lowest few ℓ contribute significantly at astrophysical energies and the sum Eq. (24) converges rapidly in practice. The pocket it forms with the attractive nuclear well supports the quasi-bound configurations responsible for resonances.

Equation (26) is a second-order ODE and requires two boundary conditions, carrying different information. At the origin, regularity of ψ at $R = 0$ requires $\chi_\ell(0) = 0$; near the origin the centrifugal term dominates and the regular solution grows as $\chi_\ell \propto R^{\ell+1}$, the quantum expression of centrifugal repulsion. At large R (the asymptotic condition), the wave function must describe the physical situation: an incoming wave and an outgoing scattered wave, in the generic sense already stated in Eq. (21).

This derivation already assumes a short-ranged potential with no surviving Coulomb tail. The general solution of Eq. (26) at large R then oscillates as a combination of $\sin(kR - \ell\pi/2)$ and $\cos(kR - \ell\pi/2)$, and the entire effect of the potential reduces to a single real number per partial wave. That number is the phase shift δ_ℓ, defined by the asymptotic behaviour of the regular solution,

$$\chi_\ell(R) \xrightarrow{R\to\infty} \sin\left(kR - \frac{\ell\pi}{2} + \delta_\ell\right) \tag{28}$$

real because, for a potential that is both real-valued and short-ranged, flux is conserved separately in every partial wave. Under these two conditions, real and short-ranged, the phase shifts already determine both the scattering amplitude of Eq. (22) and the total cross section,

$$f(\theta) = \frac{1}{k}\sum_{\ell=0}^{\infty}(2\ell+1)\, e^{i\delta_\ell}\sin\delta_\ell\, P_\ell(\cos\theta) \qquad \sigma = \frac{4\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)\sin^2\delta_\ell \tag{29}$$

with nothing yet said about how flux lost to other processes would be modelled, a generalisation taken up once the optical potential is introduced in Section 2.3 and the general scattering matrix in Section A.2. The Coulomb tail of the charged systems considered in this report modifies this picture in two ways developed next: the asymptotic phase acquires a logarithmic correction, and δ_ℓ itself becomes only the nuclear part of a larger collision matrix element.

A.3 Coulomb functions and their limits

For $R > a$, the radial equation Eq. (26) with $V_n = 0$ and $V_C = Z_1 Z_2 e^2/R$ becomes, in terms of $\rho = kR$,

$$\left[\left(\frac{d^2}{d\rho^2} - \frac{\ell(\ell+1)}{\rho^2}\right) - \frac{2\eta}{\rho} + 1\right]\chi_\ell^{ext}(R) = 0 \tag{30}$$

the Coulomb wave equation, where the Sommerfeld parameter

$$\eta = \frac{Z_1 Z_2 e^2}{\hbar v} = \frac{Z_1 Z_2 e^2 \mu}{\hbar^2 k} = \frac{Z_1 Z_2 e^2}{\hbar}\sqrt{\frac{\mu}{2E}} \tag{31}$$

measures the strength of the Coulomb interaction relative to the kinetic energy of relative motion, with v the relative velocity. When $\eta \ll 1$ (high-energy regime) the particle is fast and Coulomb effects are weak; when $\eta \gg 1$ (astrophysical regime, Section 1) the Coulomb repulsion dominates and the cross section becomes extremely sensitive to barrier penetration.

Equation (30) is a standard second-order ODE with two linearly independent solutions (see Box 3.1 & 3.2, Chapter 3, of [3]):

1. $F_\ell(\eta,\rho)$, the regular Coulomb function, which vanishes at the origin, $F_\ell(\eta, 0) = 0$, and is finite everywhere;
2. $G_\ell(\eta,\rho)$, the irregular Coulomb function, which diverges at the origin.

Their Wronskian is normalised to

$$W(G_\ell, F_\ell) = G_\ell\frac{dF_\ell}{d\rho} - F_\ell\frac{dG_\ell}{d\rho} = 1 \tag{32}$$

which ensures that F_ℓ and G_ℓ form a proper basis for every solution of Eq. (30). In the free case, $\eta = 0$, they reduce to spherical Bessel functions, $F_\ell(0,\rho) = \rho j_\ell(\rho)$ and $G_\ell(0,\rho) = -\rho y_\ell(\rho)$, consistent with the neutral-particle limit of Section A.2. For incoming and outgoing waves it is natural to work in the Coulomb-Hankel basis

$$H_\ell^{\pm}(\eta,\rho) = G_\ell(\eta,\rho) \pm iF_\ell(\eta,\rho) \tag{33}$$

which behaves asymptotically ($\rho \to \infty$) as

$$H_\ell^\pm(\eta,\rho) \approx e^{\pm i\Theta_\ell(\rho)} \qquad \Theta_\ell(\rho) = \rho - \frac{\ell\pi}{2} + \sigma_\ell - \eta\ln(2\rho) \tag{34}$$

where $\sigma_\ell = \arg\Gamma(\ell+1+i\eta)$ is the Coulomb phase shift and the logarithmic term is the remnant of the infinite range of the interaction. In the Coulomb-Hankel basis, the exterior solutions of Eq. (26) are

$$\chi_\ell^{\rm ext}(R) = c_\ell^- H_\ell^-(\eta,\rho) + c_\ell^+ H_\ell^+(\eta,\rho) \tag{35}$$

Table A.1 makes the two changes promised at the end of Section A.2 explicit, term by term against the neutral case.

Table A.1: Asymptotic radial solutions in the absence and in the presence of a Coulomb tail.

	Short-range potential only	With Coulomb tail
Regular solution	$kR\, j_\ell(kR)$	$F_\ell(\eta, kR)$
Irregular solution	$-kR\, y_\ell(kR)$	$G_\ell(\eta, kR)$
Asymptotic phase	$kR - \ell\pi/2$	$kR - \ell\pi/2 - \eta\ln(2kR) + \sigma_\ell$
Nuclear phase shift	δ_ℓ alone	δ_ℓ superposed on σ_ℓ

Figure A.3 illustrates F_ℓ, G_ℓ and H_ℓ^+ for several angular momenta. Two limiting behaviours of these functions will be used repeatedly. At small ρ (deep under the barrier), the regular and irregular solutions behave as

$$F_\ell(\eta,\rho) \propto C_\ell(\eta)\,\rho^{\ell+1} \qquad G_\ell(\eta,\rho) \propto \frac{\rho^{-\ell}}{(2\ell+1)C_\ell(\eta)} \tag{36}$$

where the entire energy scale of barrier penetration is carried by the normalisation constant $C_\ell(\eta)$, which for s waves takes the exact form

$$C_0^2(\eta) = \frac{2\pi\eta}{e^{2\pi\eta}-1} \xrightarrow[\eta\gg 1]{} 2\pi\eta\, e^{-2\pi\eta} \tag{37}$$

the same Gamow factor that sets the strength of barrier penetration in the thermonuclear reaction rates of Section 1.

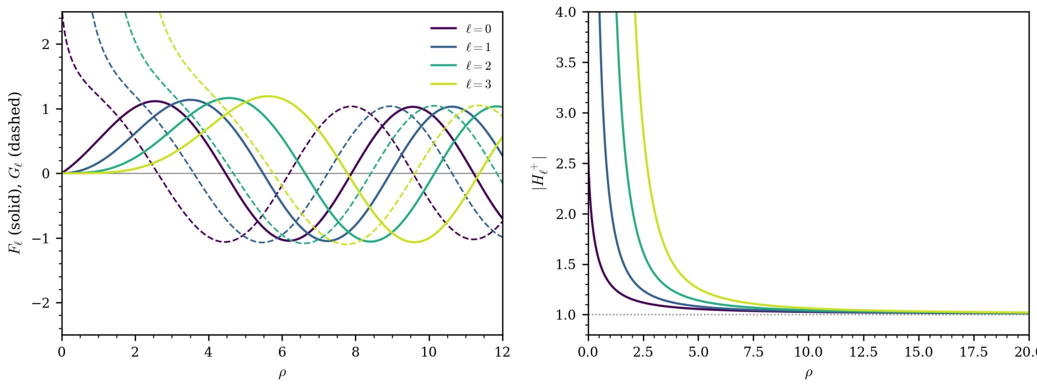

Figure A.3: Coulomb wave functions for multiple angular momenta ($\eta = 0.5$). (Left) Comparison between regular F_ℓ (solid lines) and irregular G_ℓ (dashed lines) solutions near the origin. (Right) Convergence of the Coulomb-Hankel modulus $|H_\ell^+|$ toward the asymptotic limit of 1, highlighting the centrifugal barrier effect at lower ρ for higher values of ℓ.

In the absence of any nuclear potential, $V_n \equiv 0$ everywhere, so that χ_ℓ is regular at the origin and proportional to $F_\ell(\eta, kR)$ at every radius, with no admixture of G_ℓ and no nuclear phase shift. The partial wave series of Eq. (29) can then be summed in closed form, since the Coulomb phase $e^{2i\sigma_\ell}$ alone is responsible for the deviation from a plane wave, yielding the Rutherford scattering amplitude and cross section,

$$f_C(\theta) = -\frac{\eta}{2k\sin^2(\theta/2)}\exp\bigl[-i\eta\ln\sin^2(\theta/2) + 2i\sigma_0\bigr] \qquad \frac{d\sigma_R}{d\Omega} = |f_C(\theta)|^2 = \frac{\eta^2}{4k^2\sin^4(\theta/2)} \tag{38}$$

the pure-Coulomb counterpart of the neutral-particle closure of Eq. (29). The divergence as $\theta \to 0$ reflects the infinite range of the interaction, for which the total cross section is formally infinite: no matter how large the impact parameter, a charged particle is deflected by a nonzero angle. Restoring a short-ranged nuclear potential inside $R < a$ superposes a genuine nuclear correction on this pure-Coulomb result, governed by the same collision matrix element introduced next.

A.4 Cross sections from the collision matrix

To connect with the measured cross section, the full wave function must be assembled. Matching the partial wave sum of Eq. (24), with each χ_ℓ in the form of Eq. (3), onto the Coulomb-distorted version of the scattering ansatz splits the scattering amplitude into the closed-form Coulomb part of Eq. (38) and a nuclear correction,

$$f(\theta) = f_C(\theta) + f_N(\theta) = f_C(\theta) + \frac{1}{2ik}\sum_{\ell=0}^{\infty}(2\ell+1)\, e^{2i\sigma_\ell}\,(U_\ell - 1)\, P_\ell(\cos\theta) \tag{39}$$

reducing, when $U_\ell \to 1$ for every ℓ (no nuclear potential at all), to the pure Rutherford amplitude of Eq. (38), and recovering the neutral-particle form of Eq. (29) when $\eta \to 0$. The total elastic cross section follows as

$$\sigma_{\rm el} = \frac{\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)\,|U_\ell - 1|^2 \overset{|U_\ell|=1}{=} \frac{4\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)\sin^2\delta_\ell \tag{40}$$

the second equality, in terms of the real phase shift δ_ℓ, holding only when the potential is real and unitarity $|U_\ell| = 1$ is exact, exactly as anticipated in Section A.2.

When $|U_\ell| < 1$, because of an absorptive optical potential (Section 2.3) or of coupling to other channels (Section 2.4), Eq. (40) measures only the flux that returns to the elastic channel, and a second, independent cross section is needed for the flux that does not,

$$\sigma_{\rm reac} = \frac{\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)\left(1 - |U_\ell|^2\right) \tag{41}$$

the reaction (or absorption) cross section, which vanishes identically when $|U_\ell| = 1$ for every ℓ and is otherwise the direct signature of channels not treated explicitly. The two add up to the total cross section, related to the imaginary part of the full amplitude at $\theta = 0$ by the general optical theorem,

$$\sigma_{\rm tot} = \sigma_{\rm el} + \sigma_{\rm reac} = \frac{4\pi}{k}\,{\rm Im}\, f(0) \tag{42}$$

valid independently of whether V_n is real or complex, single-channel or coupled, but for a short-ranged interaction only: with a Coulomb tail both $f(0)$ and $\sigma_{\rm el}$ diverge, as Eq. (38) shows, and $\sigma_{\rm reac}$ is the quantity that stays finite. Equations (40), (41) and (42) are the three relations against which every solver in Section 3 is ultimately checked: a real diagonal potential with no coupling must return $|U_\ell| = 1$ to machine precision, and any departure measures either an intended absorptive term or an unintended numerical error.

Resonances

The effective potential $V_{\rm eff}(R)$ of Eq. (27) can form a pocket between the centrifugal barrier at small R and the attractive nuclear interior. If the pocket is deep enough, it supports quasi-stationary states at positive energies E_r, resonances, that decay by tunnelling through the barrier with a lifetime $\tau = \hbar/\Gamma$, where Γ is the total width. Near such a state, the nuclear phase shift δ_ℓ passes rapidly through $\pi/2$, and the collision matrix element takes the Breit-Wigner single-level form,

$$U_\ell(E) = e^{2i\sigma_\ell}\,\frac{E - E_r - i\Gamma/2}{E - E_r + i\Gamma/2}, \tag{43}$$

which satisfies $|U_\ell| = 1$ identically (a resonance on a real potential involves no net absorption, only temporary trapping). The elastic partial cross section develops a characteristic Lorentzian peak at $E = E_r$:

$$\sigma_\ell^{\rm el}(E) = \frac{\pi}{k^2}(2\ell+1)\,\frac{\Gamma^2}{(E - E_r)^2 + (\Gamma/2)^2}, \tag{44}$$

with maximum $4\pi(2\ell+1)/k_r^2$ at the resonance energy. For multichannel systems, each exit channel c contributes a partial width Γ_c with $\Gamma = \sum_c \Gamma_c$, and the amplitude in Eq. (43) is modulated by the branching ratio Γ_c/Γ. The R-matrix theory of Appendix B.2 provides a systematic multi-level generalisation of Eq. (43) that handles overlapping resonances and non-resonant background contributions within a single, energy-independent parametrisation.

B Numerical methods

The discretisation the two solvers share, the reference method the DBMM is measured against, the normalisation the collision matrix must be read with, and the angular-momentum coefficients that build the coupling blocks.

B.1 The Lagrange-Legendre basis

Four properties of this basis are what the two solvers are built on, and Table B.1 states them together before each is derived.

Table B.1: What the Lagrange-Legendre basis gives, and what each property is used for. The last is the one the whole reduction rests on: it is what makes $\mathbf{V}(\theta)$ diagonal, and therefore what makes the projection of Eq. (17) a sum over mesh points instead of a dense product.

Property	Statement	Used for
cardinal at the nodes	$\hat{f}_j(x_i) = \delta_{ij}/\sqrt{\lambda_j}$	the coefficient vector is a direct sample of the wave function, $c_j = \sqrt{R\lambda_j}\,\chi(r_j)$
kinetic matrix in closed form	Eq. (45)	no numerical differentiation; built once per system
boundary values in closed form	$\hat{f}_j(1)$, $\hat{f}'_j(1)$ from Eq. (7)	the DBMM boundary row (11), and the Bloch term of the R-matrix
Gauss quadrature	$\int \hat{f}_i \hat{f}_j V \simeq V(r_i)\,\delta_{ij}$	the potential matrix is diagonal, Eq. (8)

The N-point Gauss-Legendre nodes $\{x_j\} \subset\,]0,1[$ and weights $\{\lambda_j\}$ are defined by

$$P_N(2x_j - 1) = 0, \qquad \lambda_j = \frac{1}{4x_j(1-x_j)\,[P'_N(2x_j-1)]^2},$$

and the physical mesh points are $r_j = Rx_j$. The x-regularised Lagrange functions, which vanish at the origin, are given in Eq. (7) and satisfy the cardinal property $\hat{f}_j(x_i) = \delta_{ij}/\sqrt{\lambda_j}$: each function is non-zero at exactly one mesh point. The coefficient vector is therefore a direct sample of the wave function, $c_j = \sqrt{R\lambda_j}\,\chi(r_j)$, and what makes the potential matrix diagonal under Gauss quadrature.

The operator $T = -\hbar^2 d^2/2\mu\, dr^2$ has closed-form matrix elements on this basis [9], [20]:

$$T_{ij} = \frac{\hbar^2}{2\mu R^2}\begin{cases} (-1)^{i-j}\dfrac{x_i + x_j - 2x_i^2}{x_j(x_j - x_i)^2}\sqrt{\dfrac{x_j(1-x_j)}{x_i(1-x_i)^3}}, & i \neq j, \\[2ex] \dfrac{N(N+1)\,x_i(1-x_i) - 3x_i + 1}{3\,x_i^2(1-x_i)^2}, & i = j. \end{cases} \tag{45}$$

No numerical differentiation is involved, and the matrix is built once per system. It is not symmetric, and the asymmetry is structural rather than a slip. Eq. (45) is the *collocation* representation of $-d^2/dr^2$ on the x-regularised basis, the operator that returns the second derivative sampled at the mesh points; the symmetric form tabulated for the Lagrange-Legendre mesh is the *variational* matrix element completed by the Bloch surface term, and neither reduces to the other. Applied to a function with a known second derivative, Eq. (45) reproduces it to 10^{-11} at $N = 20$, which is the check that distinguishes them; the boundary term the symmetric form carries in an added surface operator is carried here by the last row of $\mathbf{M}$. The R-matrix code uses the standard Lagrange-Legendre form, whose T_{ij} *is* symmetric because the Bloch operator restores hermiticity, so the two solvers carry different matrices on a shared mesh.

The DBMM boundary row (11) requires the basis functions and their derivatives evaluated at the channel radius, $\hat{f}_j(1)$ and $\hat{f}'_j(1)$; the R-matrix method requires $\hat{f}_j(a)$ and $\hat{f}'_j(a)$ for the Bloch surface term and the R-matrix formula. Both are computed analytically from Eq. (7) once at construction.

The N-point Gauss rule integrates polynomials of degree $2N-1$ exactly, which the integrand $\hat{f}_i\hat{f}_jV$ of Eq. (8) is not: the diagonal approximation is therefore convergent rather than exact, rapidly so for a smooth V. In practice $N = 40$ suffices for the light Gaussian systems, while the heavy-ion cases require $N = 120$. What drives the difference is the number of oscillations the interior wave function performs across $[0, R]$, and not the potential. The depth of the nuclear well raises the local wave number above the asymptotic k, so that count grows faster than kR alone. All results in Section 5 were checked for stability under an increase of N.

The solver was also compared against the published $\alpha + {}^{12}\mathrm{C}$ values of the reference package, computed there at a channel radius of 11 fm against the 14 fm used here. Above the highest threshold the two agree on the entrance column of $\mathbf{U}$ to 4.6×10^{-4} at 16 MeV and 1.0×10^{-3} at 20 MeV. Below that threshold the published values are not converged in channel radius, moving by factors of two to three across the three radii the reference reports, where this calculation is stable to 3×10^{-4} from 14 to 30 fm. That is a convergence check the internal comparison of Section 5.3 cannot provide, since it runs both solvers on one mesh and one radius.

B.2 The calculable R-matrix method

This appendix summarises the method underlying the reference code used in Section 5.3. It is given for completeness and to make the structural contrast with the DBMM explicit; the presentation follows Descouvemont and Baye [12], the original idea is due to Wigner and Eisenbud [33], and a modern account of its mathematical footing is given by Behrndt *et al.* [34].

B.2.1 Interior, exterior and the Bloch operator

The interior/exterior separation of Section 2.1 is exploited systematically. The complicated interior problem is solved numerically, then matched to the known exterior solution. The R-matrix is the object that encodes the matching: it is the inverse of the logarithmic derivative of the radial wave function at $r = a$, and knowing it is sufficient to determine the collision matrix. Inside, the wave function is expanded on N square-integrable basis functions $\{\varphi_i\}$ supported on $[0, a]$; outside, it is the combination $\chi_\ell^{\rm ext} = A\,[\,I_\ell - U_\ell O_\ell\,]$ of incoming and outgoing Coulomb-Hankel functions.

Determining the expansion coefficients requires the partial-wave Hamiltonian H_ℓ to be Hermitian on the finite interval $[0, a]$. Integrating the kinetic operator by parts leaves a surface term that does not vanish in general,

$$\langle \varphi_j | T | \varphi_i \rangle - \langle \varphi_i | T | \varphi_j \rangle = -\frac{\hbar^2}{2\mu}\big[\varphi_j(a)\varphi_i'(a) - \varphi_i(a)\varphi_j'(a)\big],$$

and imposing $\varphi_i(a) = 0$ would remove it at the price of preventing any matching. The solution is to add a surface (Bloch) operator

$$\mathcal{L}(B) = \frac{\hbar^2}{2\mu}\,\delta(r-a)\left(\frac{d}{dr} - \frac{B}{r}\right),$$

with B an arbitrary real constant. The operator $H_\ell + \mathcal{L}(B)$ is Hermitian on $[0, a]$, hence has a discrete spectrum and a complete set of eigenfunctions there. The interior Schrödinger equation is replaced by the inhomogeneous Bloch-Schrödinger equation

$$\big(H_\ell + \mathcal{L}(B) - E\big)\chi_\ell^{\rm int} = \mathcal{L}(B)\,\chi_\ell^{\rm ext}, \tag{46}$$

in which the Bloch operator plays a double role: it restores hermiticity on the left, and through the right-hand side it enforces continuity of the derivative at $r = a$.

B.2.2 From the R-matrix to the collision matrix

Projecting Eq. (46) onto the basis yields an $N \times N$ linear system with the symmetric matrix

$$D_{ij}(E, B) = H_{ij} + \frac{\hbar^2}{2\mu}\,\varphi_i(a)\Big(\varphi_j'(a) - \frac{B}{a}\varphi_j(a)\Big) - E\,N_{ij},$$

whose inversion gives the R-matrix

$$\mathcal{R}_\ell(E, B) = \frac{\hbar^2}{2\mu a}\sum_{i,j}\varphi_i(a)\,\big(\mathbf{D}^{-1}(E, B)\big)_{ij}\,\varphi_j(a), \tag{47}$$

a finite-basis approximation of the Green-function expression $\mathcal{R}_\ell = (\hbar^2/2\mu a)\,G_\ell(a, a)$. Continuity at $r = a$ then gives the collision matrix element

$$U_\ell(E) = e^{2i\phi_\ell}\,\frac{1 - (L_\ell^* - B)\,\mathcal{R}_\ell(E, B)}{1 - (L_\ell - B)\,\mathcal{R}_\ell(E, B)}, \qquad L_\ell = ka\,\frac{O_\ell'(ka)}{O_\ell(ka)} = S_\ell + iP_\ell, \tag{48}$$

with ϕ_ℓ the hard-sphere phase shift, S_ℓ the shift factor and P_ℓ the penetration factor. U_ℓ is *independent of B*, since R-matrices built with different B are related by a transformation that leaves the collision matrix unchanged. B therefore carries no physical content and may be chosen for numerical convenience.

For N_c coupled channels the matrix $\mathbf{D}$ acquires both basis and channel indices,

$$D_{in,jm} = \langle \varphi_n \big|\,\big(T_i + \mathcal{L}(B_i) + \epsilon_i - E\big)\delta_{ij} + V_{ij}(r)\,\big|\varphi_m\rangle, \tag{49}$$

of dimension $(NN_c) \times (NN_c)$, whose inversion at $\mathcal{O}((NN_c)^3)$ dominates the cost. The R-matrix becomes a symmetric $N_c \times N_c$ matrix, and the collision matrix follows from $\mathbf{U} = (\mathbf{Z}^O)^{-1}\mathbf{Z}^I$ restricted to the open-channel subblock, with

$$Z_{ij}^O = (k_j a)^{-1/2}\big[O_{\ell_i}(k_i a)\,\delta_{ij} - k_j a\, R_{ij}\, O'_{\ell_j}(k_j a)\big], \tag{50}$$

$$Z_{ij}^I = (k_j a)^{-1/2}\big[I_{\ell_i}(k_i a)\,\delta_{ij} - k_j a\, R_{ij}\, I'_{\ell_j}(k_j a)\big]. \tag{51}$$

For a real coupling potential $\mathbf{Z}^O = (\mathbf{Z}^I)^*$ and $\mathbf{U}$ is unitary; for a complex optical potential this no longer holds and $\mathbf{U}$ is symmetric but not unitary, the deficit measuring the flux lost to channels outside the model space.

The boundary parameter must be chosen channel by channel:

$$B_i = \begin{cases} 0 & \text{open channels,} \\ 2\kappa_i a\,\dfrac{W'_{-\eta_i,\ell_i+1/2}(2\kappa_i a)}{W_{-\eta_i,\ell_i+1/2}(2\kappa_i a)} & \text{closed channels,} \end{cases} \tag{52}$$

this being precisely the value for which the source term on the right of Eq. (46) vanishes identically in a closed channel, consistent with the absence of incident flux there. Closed channels then influence the interior problem through the couplings V_{ij} but do not act as external sources.

B.2.3 Contrast with the DBMM

Figure B.1 sets the two chains side by side.

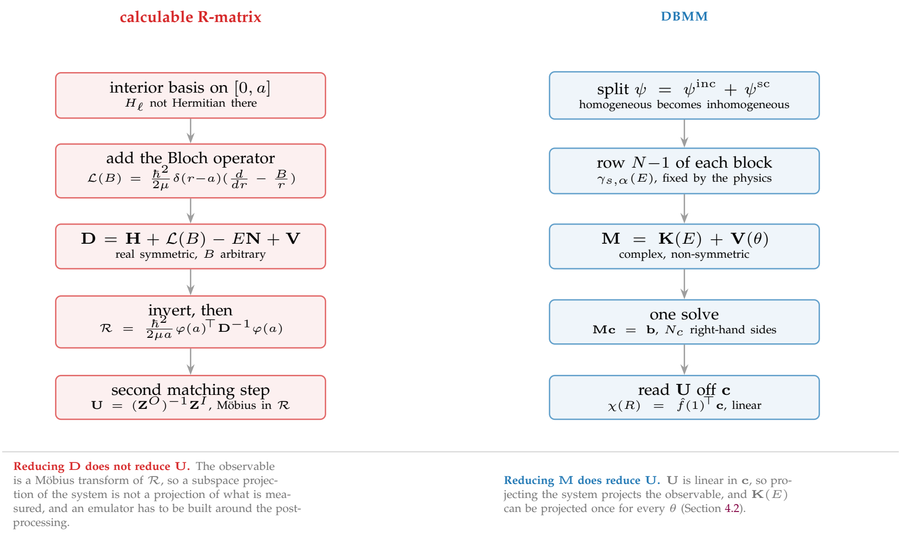

Figure B.1: Where the boundary condition enters, in each method, and what follows for a reduced-order model. The two differ in one place only. The R-matrix adds a surface operator with a free parameter B and recovers $\mathbf{U}$ in a second matching step, while the DBMM writes the outgoing condition directly into one row per channel and reads $\mathbf{U}$ off the solution vector.

The separation into a parameter-free part and a diagonal potential is not what distinguishes them: $\mathbf{D}$ splits the same way, into $\mathbf{H} + \mathcal{L}(B) - E\mathbf{N}$ and a potential the same quadrature makes diagonal. They differ in what follows the solve, the Möbius transform of Eq. (48) and its multichannel form (50)–(51), and it is a difference of degree and not of kind: reduced-order emulators of R-matrix codes exist, built around the post-processing rather than through it. It is the reason this work started from the DBMM, and it leaves the other route open.

B.3 The normalisation of U and its invariants

The collision matrix has two properties that hold whatever code produced it. Time reversal makes $\mathbf{U}$ symmetric, and with the absorptive part of the potential removed it is unitary. Neither requires a reference calculation, so both can be evaluated on the solver alone, and both fail if the asymptotics are read with the wrong normalisation.

Two normalisations are in use, side by side in the same reference [21, Sec. 2.1]. The first normalises each channel by its own velocity and gives $\mathbf{U}$; the second normalises every channel by the entrance velocity and gives $\tilde{\mathbf{U}}$. They differ by $\tilde{U}_{\alpha\beta} = (k_\beta/k_\alpha)^{1/2} U_{\alpha\beta}$, which is unity on the diagonal and unity whenever two channels share a threshold. Only the first is symmetric, only the first is unitary, and only the first makes $\sigma_{\rm reac} = 1 - |U|^2$ of Eq. (4) correct, so the three statements this report relies on select it. Recovering it from the scattered wave restores the velocity factor,

$$U_{\alpha\beta_0} = \delta_{\alpha\beta_0} + 2i\sqrt{\frac{v_\alpha}{v_{\beta_0}}}\,\frac{\psi^{\rm sc}_{\alpha\beta_0}(R)}{H^+_{\ell_\alpha}(\eta_\alpha, k_\alpha R)}, \qquad v_\alpha = \frac{\hbar k_\alpha}{\mu_\alpha}. \tag{53}$$

Written with the velocity rather than with k_α alone, Eq. (53) also covers channels of different reduced mass, which $(k_\beta/k_\alpha)^{1/2}$ does not.

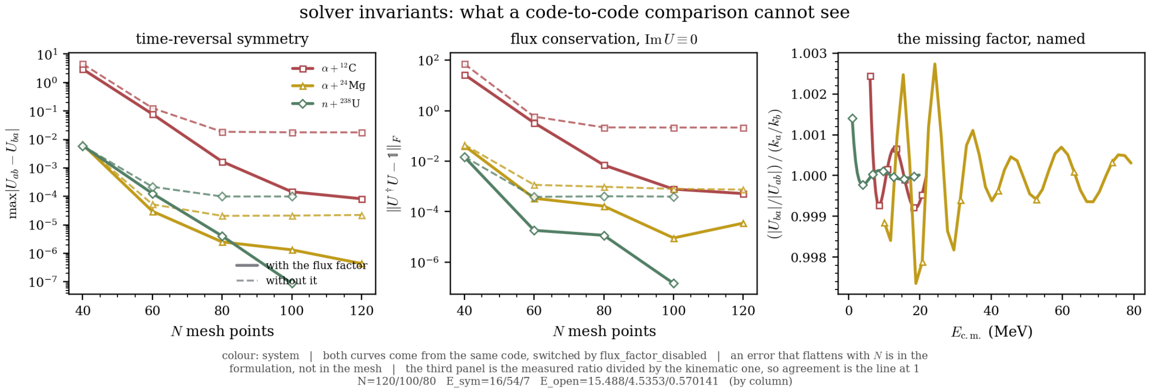

Figure B.2: Symmetry and unitarity against mesh refinement, for the three coupled systems, with and without the velocity factor of Eq. (53). With the factor the violations fall with N like a discretisation error; without it they reach a floor and stay on it. The third panel divides the measured ratio $|U_{\beta\alpha}|/|U_{\alpha\beta}|$ by the kinematic ratio, so agreement is the line at one. $n + {}^{40}$Ca is absent because it is uncoupled and the factor is identically unity there.

Figure B.2 separates a formulation error from a discretisation error by its behaviour under mesh refinement, which is what makes the two invariants useful as instruments: a discretisation error falls with N, a wrong normalisation does not.

The invariants are necessary and not sufficient, and the gap is measurable. On $\alpha + {}^{24}$Mg the DBMM violates unitarity by 9×10^{-6} at the working mesh $N = 100$ while sitting 4.9×10^{-3} from its own converged answer (Table E.2), so selecting a mesh from the invariants alone selects one that passes rather than one that has converged. The asymmetry also runs one way. The calculable R-matrix obtains $\mathbf{U}$ from a real symmetric $\mathcal{R}$ through a Möbius transform that is unitary identically, at every mesh, so a unitarity check can find a fault in the DBMM and can never find one in the code the DBMM is measured against.

The factor is placed in one function, which every reconstruction of $\mathbf{U}$ calls.

Listing 1: The factor lives in one function, which every reconstruction of $\mathbf{U}$ calls, in the solver and in each emulator. The context manager `flux_factor_disabled()` selects the other normalisation, so that Figure B.2 measures both from one code base.

```
def channel_velocities(channels, E, open_channels):
    # v = hbar k / mu for each open channel. The caller forms
    # sqrt(v_a / v_b), which is unity on the diagonal and unity whenever
    # two channels share a threshold: that is why omitting it is invisible
    # on elastic observables and on benchmarks without a real inelastic
    # threshold. Every reconstruction of U goes through here, in the
    # solver and in each emulator; see tests/test_invariants.py.
    if not _FLUX_FACTOR:                        # flux_factor_disabled()
        return {a: 1.0 for a in open_channels}
    return {a: _channel_kinematics(channels[a], E)[0].real / channels[a]['mu']
            for a in open_channels}
```

Table B.2 names the factor, where the invariants only detect a fault. The last row is decisive. The ratio of the two reciprocal elements equals k_α/k_β to four digits, which is the factor itself.

Table B.2: The two normalisations, measured on $\alpha + {}^{12}$C from one code base. The geometric mean of a reciprocal pair is identical under both, so symmetrising $\mathbf{U}$ by hand would leave the interior wave function wrong and the scalar right. At $N = 60$ neither normalisation is converged and the two differ by a factor of two, where at the converged mesh they differ by four hundred: the invariant separates them only once the mesh does not dominate it.

Quantity	$\tilde{\mathbf{U}}$	$\mathbf{U}$
Unitarity violation, absorption removed, converged mesh	22%	5×10^{-4}
Unitarity violation at $N = 60$	58%	33%
Asymmetry of $\mathbf{U}$, $N = 100$	1.8×10^{-2}	1.5×10^{-4}
Asymmetry of an emulator's $\mathbf{U}$, calibrated rank	4.3×10^{-2}	1.2×10^{-4}
$\sqrt{\lvert U_{\alpha\beta} U_{\beta\alpha} \rvert}$ against the R-matrix	10^{-3}	10^{-3}
$\lvert U_{\beta\alpha}\rvert / \lvert U_{\alpha\beta}\rvert$	k_α / k_β	1

The fourth row carries the statement into the emulators. An RBM built on the wrongly normalised solver leaves $\mathbf{U}$ asymmetric at 4×10^{-2} and no increase of the rank moves it, where the same emulator on the correct one sits at the solver's own floor. That is the assertion Appendix D.7 keeps as a test.

No code-to-code comparison can distinguish the two normalisations on the systems available here: $p + {}^{12}$C has one channel, and the two channels of $\alpha + d$ share a mass partition, so $k_0 = k_1$ and the factor is unity on both. Both invariants are therefore unit tests, guarded by an assertion that the system they run on has distinct thresholds. The same scan sets the working mesh of the two α-induced systems at $N = 120$ and $N = 100$; $N = 60$ is not converged for either.

The last row is measured on $\alpha + {}^{12}$C alone and on the geometric mean of a reciprocal pair, which is the combination the velocity factor leaves invariant, so it is identical under both normalisations by construction and detects nothing. Section 5.3 carries the element-by-element measurement on the three coupled systems, and it is the one to read: the geometric mean is optimistic against the elements it is built from by a factor of 1.8 to 3.5.

B.4 The effective interaction: reference tables

Table B.3 below is the reference material behind Section 2.4: the mechanisms that generate the off-diagonal couplings of a coupled-channel model. It complements Table 1, which lists the radial form factors of the optical potential itself and is given in the body. Both are drawn from the reviews of Hebborn *et al.* [1], Holt and Whitehead [5] and Koning and Delaroche [19], and from Thompson and Nunes [3] for the collective couplings.

Table B.3: Coupling mechanisms and their form factors. $\delta_\lambda = \beta_\lambda R_0$ is the deformation length and β_λ the deformation parameter of multipolarity λ; the deformation itself is written β_λ throughout, and δ_λ only where the length is what enters. $\mathcal{F}^{(\lambda)}$ is the angular-momentum recoupling coefficient of Eq. (54). All the collective form factors share the derivative shape $\mathrm{d}f/\mathrm{d}r$, which localises the coupling at the nuclear surface.

Mechanism	Form factor $V_{\alpha\beta}(r)$	When it applies	Example
Vibrational	$-\delta_\lambda \dfrac{\mathrm{d}f}{\mathrm{d}r}(r; R_0, a_0)$	spherical target with a low-lying collective state; one-phonon excitation of a surface vibration	$n + {}^{58}$Ni, 2_1^+
Rotational	$-\delta_\lambda\, \mathcal{F}^{(\lambda)} \dfrac{\mathrm{d}f}{\mathrm{d}r}(r; R_0, a_0)$, with $\mathcal{F}^{(\lambda)}$ of Eq. (54)	permanently deformed target; members of a rotational band are coupled by the static deformation	$\alpha + {}^{12}$C, ^{16}O+^{44}Ca
Tensor	$\propto S_{12}$, recoupled to the $\{\ell\}$ basis; e.g. $\frac{6\sqrt{J(J+1)}}{2J+1} V_t(r)$	non-central nucleon-nucleon force; couples ℓ and $\ell \pm 2$ at fixed J^π	$\alpha + d$, S to D
Spin-orbit	$\boldsymbol{\ell}\cdot\mathbf{s}$ term of Table 1	diagonal in channel but distinguishes $j = \ell \pm \frac{1}{2}$, so it doubles the channel count	$n + {}^{40}$Ca

The rotational coefficient is the one the code evaluates, for an arbitrary multipolarity λ and a genuine J-coupled basis in which one target spin I generates several orbital momenta ℓ:

$$\mathcal{F}^{(\lambda)}_{I_a\ell_a,\, I_b\ell_b}(J) = (-1)^{J+\lambda}\,(-1)^{|\ell_a-\ell_b|/2}\sqrt{\frac{(2\ell_a{+}1)(2\ell_b{+}1)(2I_a{+}1)(2I_b{+}1)(2\lambda{+}1)}{4\pi}}$$
$$\times \begin{pmatrix} I_b & \lambda & I_a \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_a & \lambda & \ell_b \\ 0 & 0 & 0 \end{pmatrix} \begin{Bmatrix} I_a & \ell_a & J \\ \ell_b & I_b & \lambda \end{Bmatrix}. \tag{54}$$

Equation (54) is what `core/potentials.py` evaluates for all three rigid-rotor systems of the suite. At $J^\pi = 3^-$ on $\alpha + {}^{12}$C one target spin generates several ℓ, and at $J = 0$ on $\alpha + {}^{24}$Mg and $n + {}^{238}$U the triangle rule gives $\ell = I$ for every band member, so each channel carries its own centrifugal barrier.

Its diagonal does not vanish whenever the $3j$ symbol survives, that is whenever $2I \geq \lambda$ with λ even, and not whenever $I \geq \lambda$, a condition that would wrongly exclude the $\lambda = 4$ reorientation of a 2^+ state. That diagonal term is the static reorientation. A level of spin I couples to itself through it, so on a rotational band the deformation reaches the diagonal of its own channel, and dropping the term would remove real physics from every excited channel. It is retained on all three. Its reach is described by two quantities, and they answer different questions. Its *amplitude*, the peak of the reorientation term over the peak of the diagonal potential in the same channel, runs from 1.3 to 11 per cent on $\alpha + {}^{12}$C, is 0.2 per cent on $\alpha + {}^{24}$Mg and 0.7 to 0.9 per cent on $n + {}^{238}$U. Its *travel*, how much the diagonal potential moves when the deformation crosses its box, is the quantity the budget rule is read on and is reported in Table 9. The first says the term exists; the second says whether a predictor point spent on the coupling recovers anything the diagonal does not already carry.

A one-phonon vibrational excitation of a 0^+ ground state has no such term. Its diagonal matrix element carries $\langle 0|Y_\lambda|0\rangle$, which vanishes identically for $\lambda > 0$, so the deformation appears in the coupling blocks alone. That is a selection rule and not a modelling choice, and it is the property that makes $n + {}^{58}$Ni the one system of the suite on which the budget rule of Section 4.4 prescribes a split.

Provenance and checks of the coefficient

Equation (54) is Eqs. (49) and (50) of the reference R-matrix package [21, Sec. 4.3], which takes them from Rhoades-Brown, Macfarlane and Pieper [35]. The factor written $(-1)^{|\ell_a - \ell_b|/2}$ here is the $i^{\ell_b - \ell_a}$ of their Eq. (49); the two agree because $\ell_b - \ell_a$ is even within a J^π block, and writing it with an absolute value makes the symmetry of the coefficient manifest.

Stating this matters because the code-to-code comparison of Section 3 does not test the coefficient. Both solvers are handed the same coupling function, so an error in the $3j$, the $6j$ or the phase is common to the two and cancels in their difference. A wrong phase would be invisible on the diagonal of $\mathbf{U}$ and would change every inelastic element, which the emulators are asked to reproduce. Four checks were therefore run directly on `rotational_xfac`, and Table B.4 gives what each constrains and what it returned.

Table B.4: What has been checked on the recoupling coefficient, and what each check constrains. None of the four needs a reference calculation, because the code-to-code comparison hands both solvers the same coupling function and cannot see an error in it.

Check	What it constrains	Result
the special-function chain	the $3j$ and $6j$ routines themselves, evaluated against exact rational arithmetic over the whole $\alpha + {}^{12}$C matrix	largest difference 1.4×10^{-15}
symmetry	the phase. Applying $i^{\ell_b - \ell_a}$ to one index only, or dropping the absolute value, breaks the symmetry at the first pair with $\lvert\ell_a - \ell_b\rvert = 2$	symmetric to 3.3×10^{-16}
the selection rule	the condition the budget rule of Section 4.4 is read on, as a measurement rather than as an argument	$\mathcal{F}^{(\lambda)}_{aa}$ vanishes for the 0^+ member and for it alone, on every channel of the band
the channel list	the construction of the coupled basis at a non-trivial J^π	at $J^\pi = 3^-$ it returns $\ell = 3$ for $I = 0$, $\ell = 1, 3, 5$ for $I = 2$ and $\ell = 1, 3, 5, 7$ for $I = 4$, the list Example 4 of the reference package specifies channel for channel

Beyond those, the entrance column of $\mathbf{U}$ at the central parameters agrees with the published output of that example to 4.6×10^{-4} at 16 MeV and 1.0×10^{-3} at 20 MeV, both above the highest threshold (Appendix B.1). One thing none of this can see: the overall $1/\sqrt{4\pi}$ rescales $\mathcal{F}^{(\lambda)}$ uniformly, and a uniform rescaling of $\mathcal{F}^{(\lambda)}$ is exactly degenerate with a rescaling of β_λ. It would change what the fitted deformation means physically and would leave every emulator measurement in this report unchanged.

One deviation from the example is deliberate. It places the 4^+ at 14.40 MeV; the value used here is the physical 14.08 MeV, which moves the upper threshold by 0.32 MeV and therefore moves where the third group of channels opens inside the energy window.

C The systems

Central values, parameter boxes and sources for the systems of Table 4, then the presets the code defines and this report does not use. Every parameter set is taken from the literature; where a value departs from its source, the departure is stated. Figure C.1 and Table C.1 put the five side by side before the fiches that follow.

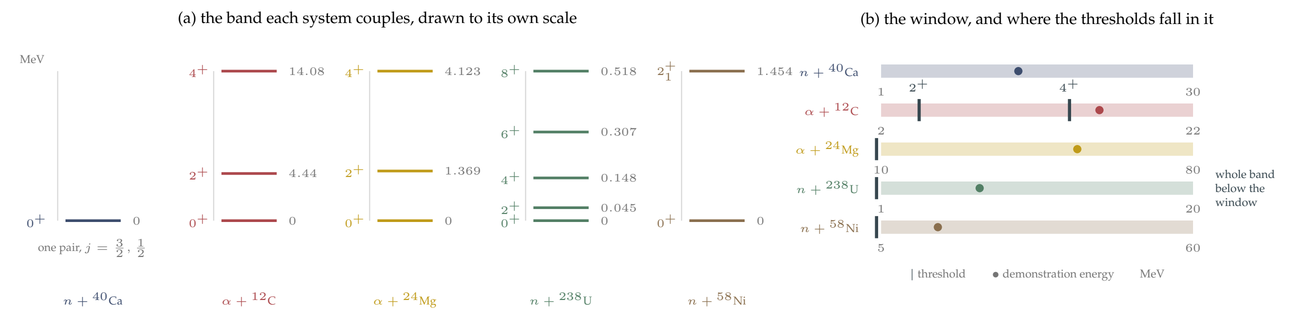

Figure C.1: The bands the five systems couple, and where their thresholds fall in the window each is studied over. *(a)* Each band is drawn to its own scale, with its top labelled, so the panel compares the spacing inside a band: $n + {}^{238}$U puts five levels below 0.52 MeV where $\alpha + {}^{12}$C spreads three over 14 MeV. *(b)* $\alpha + {}^{12}$C is the only system whose thresholds lie inside its window, so it is the only one on which the number of open channels changes as the energy is swept, from one to four to eight. On the other four every channel is open at every node, and the closed-channel treatment of Section 3.2 is exercised by $\alpha + {}^{12}$C alone.

Table C.1: The five systems, in one sheet. μ is the reduced mass and $N_c N$ the dimension of the full linear system. The angular momenta are those the triangle rule allows at the stated J^π: on the two $J = 0$ systems it forces $\ell = I$, so each band member carries its own centrifugal barrier. Sources are given with each parameter table below.

System	J^π, channels	N_c	N	R (fm)	μ (MeV/c^2)	Z_1Z_2	window (MeV)	E_{demo} (MeV)
$n + {}^{40}$Ca	$\ell = 1, j = \frac{3}{2}, \frac{1}{2}$	2	64	15	916.0	0	1–30	13.8
$\alpha + {}^{12}$C	$3^-; \ell = 3 \,/\, 1,3,5 \,/\, 1,3,5,7$	8	120	14	2816.8	12	2–22	16.0
$\alpha + {}^{24}$Mg	$0^+; \ell = 0, 2, 4$	3	100	15	3219.1	24	10–80	54.0
$n + {}^{238}$U	$0^+; \ell = 0, 2, 4, 6, 8$	5	80	20	935.0	0	1–20	7.0
$n + {}^{58}$Ni	$\ell = 1$; 0^+ and 2^+_1	2	60	15	923.0	0	5–60	15.0

C.1 The four benchmarks

$n + {}^{40}$Ca: ten parameters on the smallest matrix

Two channels, the spin-orbit partners $j = \frac{3}{2}$ and $\frac{1}{2}$ at $\ell = 1$, degenerate in threshold and coupled by nothing: the system matrix is block-diagonal and the difficulty is the parameter count. Mesh $N = 64$, $R = 15$ fm, $N_c N = 128$; $\mu = 916.0$ MeV$/c^2$, $Z_1 Z_2 = 0$; demonstration energy $E = 13.756$ MeV.

Table C.2: $n + {}^{40}$Ca, $p = 10$. Depths in MeV, radii and diffusenesses in fm. Central values are those of `cross_section_recompute.py` in Giuliani's repository [13], that is Koning-Delaroche evaluated on ^{40}Ca at $E_{\text{lab}} = 14.1$ MeV; the box is ± 20 per cent, his, sampled by Latin hypercube. V_{so} carries the Thomas factor $\lambda_\pi^2 \simeq 2$ fm^2 that his equation writes explicitly and `thomas_so` does not (Appendix D.8).

Parameter	central	box		role
V_{v0}	46.72	37.4	56.1	real volume depth
W_{v0}	1.723	1.38	2.07	volume absorption
W_{d0}	7.236	5.79	8.68	surface absorption
V_{so}	12.2	9.76	14.6	spin-orbit strength
R_{v0}	4.054	3.24	4.86	volume radius
R_{d0}	4.405	3.52	5.29	surface radius
R_{so}	3.454	2.76	4.14	spin-orbit radius
a_{v0}	0.6718	0.537	0.806	volume diffuseness
a_{d0}	0.5379	0.43	0.645	surface diffuseness
a_{so}	0.6	0.48	0.72	spin-orbit diffuseness

Four of the ten parameters, the depths V_{v0}, W_{v0}, W_{d0} and V_{so}, enter U linearly, so the affine decomposition of Section 3.4 splits it into four terms plus a nonlinear remainder carried by the six geometry parameters. It is the smallest system of the suite and therefore the least favourable case for emulation. The suite needs a case where the answer is close to negative.

$\alpha + {}^{12}$C: eight coupled channels across two thresholds

The ground-state rotational band of ^{12}C, $0^+_{\text{g.s.}}$, 2^+_1 at $E_x = 4.44$ MeV and 4^+_1 at 14.08 MeV, coupled to an α particle, of spin-parity 0^+, in the $J^\pi = 3^-$ partial wave. The spinless projectile reduces the channel spin to the target spin I, so that $\mathbf{I} + \boldsymbol{\ell} = \mathbf{J}$ suffices to label a channel. The coupling with $J = 3$ admits

$I = 0$	$\ell = 3$	threshold 0 MeV
$I = 2$	$\ell = 1, 3, 5$	threshold 4.44 MeV
$I = 4$	$\ell = 1, 3, 5, 7$	threshold 14.08 MeV

so $N_c = 8$. Mesh $N = 120$, $R = 14$ fm, $N_c N = 960$: the largest system of the suite. Reduced mass $\mu = 2816.8$ MeV/c^2, $Z_1 Z_2 = 12$. Over the energy window the two excited bands open in turn, so the number of *open* channels changes from 1 to 4 to 8; below each threshold the corresponding channels are closed and their boundary rows use the Whittaker logarithmic derivative of Section 3.2. The demonstration energy $E = 16$ MeV lies above both thresholds.

The interaction is a deformed Woods-Saxon. On the diagonal, $V_{\alpha\alpha}(r) = -(V_0 + iW_0) f(r; R_0, a_0)$; off it, the rotational coupling of order $\lambda = 2$

$$V_{\alpha\beta}(r) = -\beta_2 R_0\, \mathcal{F}^{(2)}_{I_\alpha \ell_\alpha, I_\beta \ell_\beta}(J)\, f'(r; R_0, a_0), \tag{55}$$

where $\mathcal{F}^{(2)}$ is Eq. (54) at $\lambda = 2$, the geometric factor built from the reduced matrix element of Y_2 between the band members and the $6j$ recoupling of ℓ and I to J (Thompson and Nunes [3]). β_2 is the only parameter of the coupling, but it is not confined to it: $\mathcal{F}^{(2)}_{I\alpha, I\alpha}$ does not vanish for the 2^+ and 4^+ members, so the deformation also reorients those states and reaches the diagonal of their own channels, which is why the split predictor budget of Section 5.6 recovers nothing here.

Table C.3: $\alpha + {}^{12}$C, $p = 5$. Central values reproduce Example 4 of the reference R-matrix package [21], the one system here whose published output can be compared value by value.

Parameter	central	box		role
V_0	110.0	77.0	143.0	real depth (MeV), $\pm 30\%$
W_0	20.0	10.0	35.0	volume absorption (MeV)
R_0	4.65	3.95	5.35	radius (fm)
a_0	0.50	0.38	0.62	diffuseness (fm)
β_2	0.58	0.40	0.72	quadrupole deformation; diagonal reorientation

$\alpha + {}^{24}$Mg: two deformation multipoles on a light target

Three channels, the 0^+, 2^+_1 (1.369 MeV) and 4^+_1 (4.123 MeV) members of the ^{24}Mg ground-state band, coupled to a spinless α particle at $J^\pi = 0^+$. The triangle rule then forces $\ell = I$, so the three channels carry $\ell = 0, 2$ and 4 and each has its own centrifugal barrier; all three are open over the whole window, whose lower edge at 10 MeV lies above the 4.123 MeV threshold. Mesh $N = 100$, $R = 15$ fm, $N_c N = 300$; $\mu = 3219.1$ MeV/c^2, $Z_1 Z_2 = 24$; demonstration energy $E = 54$ MeV.

The interaction carries a surface absorption on the diagonal in addition to the volume terms,

$$V_{\alpha\alpha}(r) = -(V_0 + iW_0)\, f(r; R_0, a_0) + 4 i a_0 W_{d0}\, f'(r; R_0, a_0), \tag{56}$$

and *two* multipoles off it, $\lambda = 2$ and $\lambda = 4$, through the reduced coefficient $\mathcal{F}^{(\lambda)}$ of Eq. (54), diagonal included,

$$V_{\alpha\beta}(r) = -R_0 \sum_{\lambda \in \{2,4\}} \beta_\lambda\, \mathcal{F}^{(\lambda)}_{I_\alpha \ell_\alpha,\, I_\beta \ell_\beta}(J)\, f'(r; R_0, a_0), \tag{57}$$

with the same coefficient on the diagonal, where it is the static reorientation. Two multipoles on a light deformed target is the difficulty this system contributes; the deformation reaches the diagonal through the reorientation term, so the split predictor budget of Section 5.6 recovers nothing here either.

Table C.4: $\alpha+{}^{24}$Mg, $p = 7$. The deformations are the quasi-elastic determination of Gupta *et al.* [36], $\beta_2 = +0.43\pm0.02$ and $\beta_4 = -0.11\pm0.02$. The optical depths and geometry are of the order of the Saxon-Woods set Neu *et al.* [37] fit at $E_\alpha = 54.1$ MeV, $V_0 = 103.78$ MeV, $r_0 = 1.437$ fm, $a = 0.670$ fm, and do not reproduce it: the surface term used here has no counterpart there, where the imaginary part is a single volume term with its own geometry, $W = 8.71$ MeV, $r_I = 1.921$ fm, $a_I = 0.349$ fm. The demonstration energy here is 54 MeV in the centre of mass, which is the frame every energy in this report is written in, so the two coincide in value and not necessarily in frame.

Parameter	central	box		role
V_0	100.0	70.0	140.0	real depth (MeV)
W_0	10.0	5.0	20.0	volume absorption (MeV)
W_{d0}	20.0	8.0	30.0	surface absorption (MeV)
R_0	3.46	2.94	3.98	radius (fm)
a_0	0.65	0.50	0.80	diffuseness (fm)
β_2	0.43	0.35	0.51	quadrupole; diagonal reorientation
β_4	-0.11	-0.17	-0.05	hexadecapole; diagonal reorientation

$n + {}^{238}$U: the most compressed band of the suite

Five channels, the 0^+ through 8^+ members of the ^{238}U ground-state rotational band, coupled to a projectile carried at spin zero, so the triangle rule at $J^\pi = 0^+$ gives $\ell = I$ and the five channels carry $\ell = 0$ to 8. Dropping the neutron spin is a modelling choice, and it is the only place in the suite where the block solved is not a physical partial wave: a real $n + {}^{238}$U calculation would couple the channel spin $s = I \pm \frac{1}{2}$ and populate $J^\pi = \frac{1}{2}^+$ with more channels. The coupling structure, the threshold pattern and the placement of the parameters are retained, and the emulators are tested on those; the channel count and the rank are unaffected by the omission. The couplings use Eq. (54) at $J = 0$, as on $\alpha + {}^{24}$Mg. Mesh $N = 80$, $R = 20$ fm, $N_c N = 400$; $\mu = 935.0$ MeV$/c^2$, $Z_1 Z_2 = 0$, so the asymptotics are free spherical waves. The demonstration energy is $E = 7$ MeV.

This benchmark tests a strongly coupled band at the largest channel count of the neutral systems. It is compressed: 2^+ at 44.9 keV, 4^+ at 148.0 keV, 6^+ at 307.2 keV and 8^+ at 518.3 keV, so all four excitations lie below 0.6 MeV, and therefore below the 1 MeV lower edge of the window studied, in which all five channels are open at every node. The interaction has the same form as $\alpha + {}^{24}$Mg, deformed Woods-Saxon with surface absorption and $\lambda = 2, 4$ couplings within the band.

Table C.5: $n+{}^{238}$U, $p = 7$. The deformations are the FRDM values tabulated for ^{238}U by Soukhovitskii *et al.* [38], $\beta_2 = 0.215$ and $\beta_4 = 0.093$. The depths are round values of the order of the global nucleon-actinide potential of Soukhovitskii *et al.* [39], whose real depth is near 48 MeV at this energy, and the radius is the CH89 form and not their $r_R = 1.245$ fm. Theirs is dispersive and carries a separate geometry for each term, where this one is a fixed-energy Woods-Saxon with a single geometry.

Parameter	central	box		role
V_0	50.0	35.0	65.0	real depth (MeV)
W_0	3.0	1.5	6.0	volume absorption (MeV)
W_{d0}	8.0	4.0	14.0	surface absorption (MeV)
R_0	7.52	6.77	8.27	radius (fm), $1.250A^{1/3} - 0.225$
a_0	0.65	0.50	0.80	diffuseness (fm)
β_2	0.215	0.17	0.26	quadrupole; diagonal reorientation
β_4	0.093	0.05	0.14	hexadecapole; diagonal reorientation

C.2 The auxiliary systems

Three systems appear once each in the body, and Table C.6 gives them. Two are the solver controls of Section 5.3 and one carries the budget test of Section 5.6.

Table C.6: The three auxiliary systems. None carries an accuracy or cost figure of the comparison, and none appears in the rank ladders of Appendix E.

System	Configuration	Interaction	Role, and source
$p + {}^{12}\mathrm{C}$	one channel, $\ell = 0$, $N = 40$, $R = 15$ fm	central Gaussian well, $-V_0 e^{-(r/\beta)^2}$, plus a uniform-sphere Coulomb term, $p = 2$	tests the single-channel Coulomb matching, the one place where $\eta \neq 0$ and $N_c = 1$ meet
$\alpha + d$	two channels, $J^\pi = 1^+$, $\ell = 0$ and 2, $N = 60$	central plus tensor, the S to D mixing of the deuteron ground state, $p = 2$	tests the off-diagonal coupling with no threshold difference, so the velocity factor is unity and cannot mask an error
$n + {}^{58}\mathrm{Ni}$	two channels, 0^+ and 2_1^+ at 1.454 MeV, $\ell = 1$, $N = 60$, $R = 15$ fm	CH89 volume and surface absorption, spin-orbit, one-phonon vibrational coupling $-\delta_2\, \mathrm{d}f/\mathrm{d}r$, $\theta = (V_0, W_s, W_v, \delta_2)$, $p = 4$	the only system whose coupling parameter is absent from the diagonal for a reason of physics. Geometry at its CH89 values for $A = 58$, $E_{\mathrm{lab}} = 15$ MeV; Varner *et al.* [18], with $\beta_2 = 0.18$ from Raman *et al.* [40]

The vanishing that makes $n + {}^{58}\mathrm{Ni}$ useful is a selection rule and not a modelling choice (Appendix B.4), and a rotational band member of spin I with $2I \geq \lambda$ does reorient, which is why the three rigid-rotor systems of the suite behave the other way. Its two channels carry the same $\ell = 1$, $s = \frac{1}{2}$ and $j = \frac{3}{2}$ and differ only by their threshold, so the block is no more a physical partial wave than the one of $n + {}^{238}\mathrm{U}$; it carries the placement of δ_2, on which the budget rule is read. The projectile keeps its spin, so the spin-orbit term of Table C.6 acts.

C.3 The presets this report does not use

`pipeline/presets.py` defines eleven systems and this report measures seven, the five of Table 4 and the two solver controls. The remaining four are listed here because they are part of the same library, they are taken from the same literature, and a reader extending the work will find them there. Table C.7 says what each is and why it is absent.

Table C.7: The presets defined in `pipeline/presets.py` and not used in this report, with the reference each is built from and the reason it is absent.

Preset	Configuration	Source	Why it is not in the suite
$p + {}^{40}\mathrm{Ca}$	one channel, $N = 60$, $p = 3$	CH89 global parametrisation, Varner *et al.* [18]	a charged single-channel case that adds nothing $p + {}^{12}\mathrm{C}$ does not already control
$n + {}^{40}\mathrm{Ca}$, coupled	four channels, one threshold, $N = 60$, $p = 9$	Koning-Delaroche [19] with an added collective state	duplicates the parameter dimension of the uncoupled version without a distinct difficulty
$\alpha + {}^{208}\mathrm{Pb}$	one channel, $N = 120$, $p = 4$	Woods-Saxon volume, near the Coulomb barrier; Goldring *et al.* [41]	single-channel, and its interest is the Coulomb barrier rather than the reduction
${}^{16}\mathrm{O}+{}^{44}\mathrm{Ca}$	two channels, one threshold, $N = 120$, $R = 22$ fm, $p = 5$	rotational coupling of a heavy-ion pair, Rhoades-Brown *et al.* [35]	would extend the suite towards heavier projectiles; not run for want of time

C.4 The interaction, term by term

Figure C.2 decomposes each interaction into the terms the affine split of Section 3.4 produces, and Figure C.3 shows how far the observable moves when one parameter crosses its box against how far it moves when all of them do.

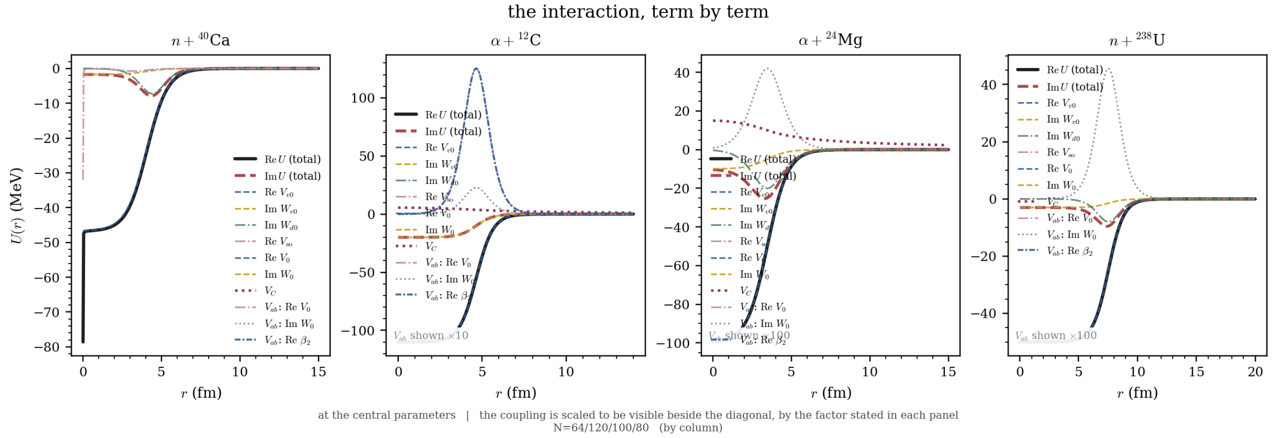

Figure C.2: The optical potential of each benchmark at its central parameters, term by term. The decomposition is the affine one of Section 3.4: each thin curve is $\theta_j\,\partial V/\partial\theta_j$ for a parameter the interaction is linear in, so the terms sum exactly to the total, and the geometry parameters, which enter non-linearly, carry no term.

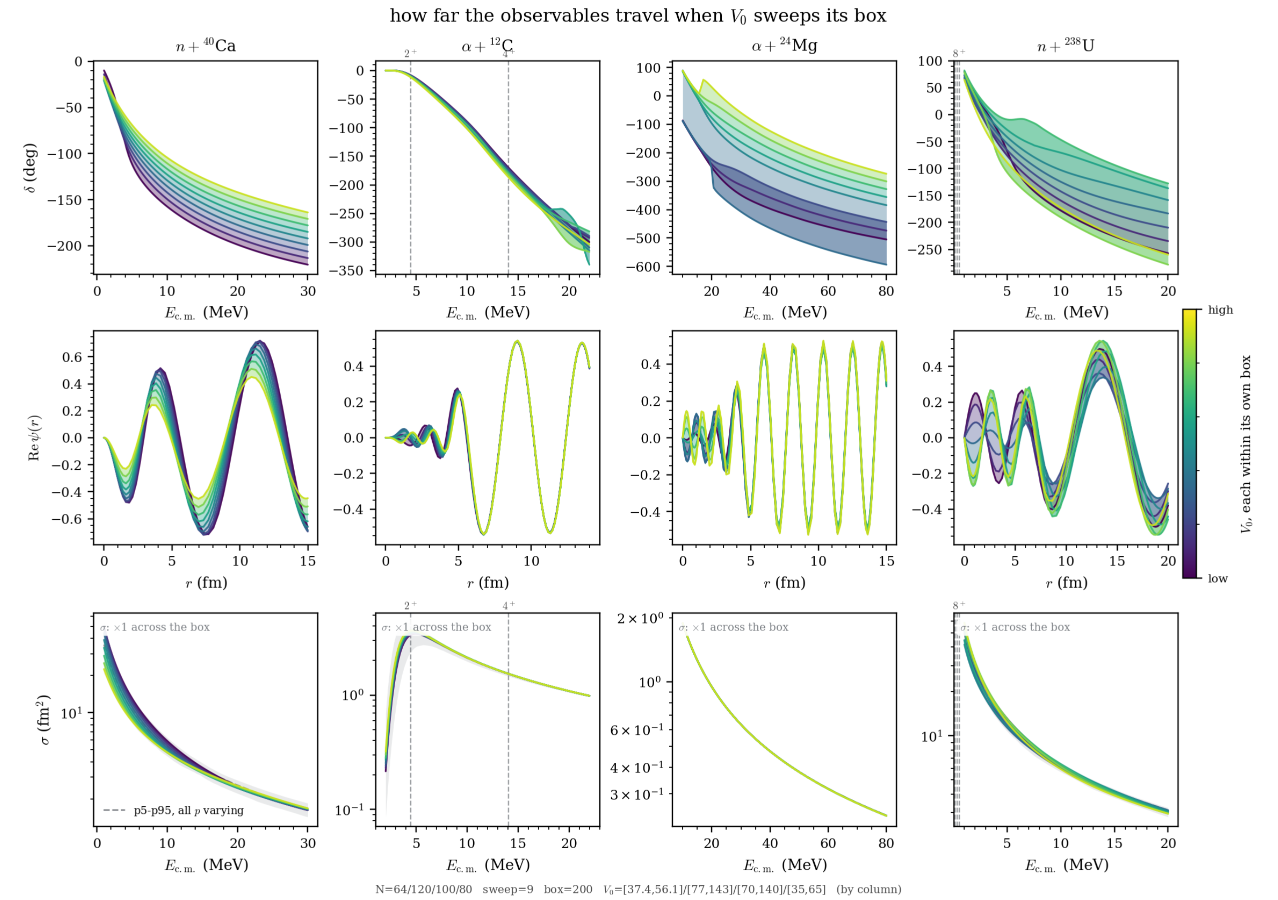

Figure C.3: How far the observables travel when a swept parameter crosses its box. The grey envelope is the p5–p95 spread over draws with every parameter varying at once, which is the range an emulator has to cover; the coloured ribbon varies one parameter only.

D Implementation

What a reader taking the code over needs, in the order they would need it: how it is laid out, how one run goes from a preset to a number, what each of the three objects does inside, how the hyperparameters are chosen, what was optimised and what that returned, and what it all ran on.

The code is a single Python package. It requires `numpy`, `scipy`, `matplotlib`, `pandas`, `mpmath` and `tqdm`. `python-flint` is optional in the strict sense and required in practice, since without it the charged Coulomb functions fall back to `mpmath` at ten times the cost, and `threadpoolctl` lets the thread policy be observed instead of assumed. `numba` is installed and deliberately unused: its mesh kernel is accurate to 2×10^{-12} where the default chain reaches machine precision, so it is not in the chain the results were produced with.

D.1 How the code is laid out

Two packages, separated by what each is allowed to know. `core/` is physics: channels, potentials, wave functions, the solver and the two emulators. It knows nothing about benchmarks, figures or files. `pipeline/` is the study: which systems are compared, under which hyperparameters, and where the results are kept. No module of `core/` imports anything from `pipeline/`, so the physics can be used without the study and a benchmark can be added without touching a solver.

A number in the text cannot disagree with the figure beside it, because of two further rules. No figure module imports a solver, and no compute module imports matplotlib. Every result therefore passes through one cached array on disk, which both the drawing layer and the table generator read.

The five classes a reader is most likely to want are `DBMMSolver` in `core/dbmm.py`, which carries the Lagrange mesh, the boundary rows and the flux-normalised extraction of **U**; `RBMEmulator` in `core/rbm.py`; `LROMEmulator` in `core/lrom_dbmm.py`, which holds the maxvol selection with its separate diagonal and coupling budgets; the calculable R-matrix reference in `core/python_rmatrix.py`, used only for validation; and the special-function chain in `core/specfun.py`, which tries SciPy, then `python-flint` (Arb), then mpmath, and records the library version behind every value.

The dependency graph

Twenty-nine modules, 13 300 lines. Figure D.1 is parsed from the import statements, so the acyclicity it shows is a property of the package and not of the drawing.

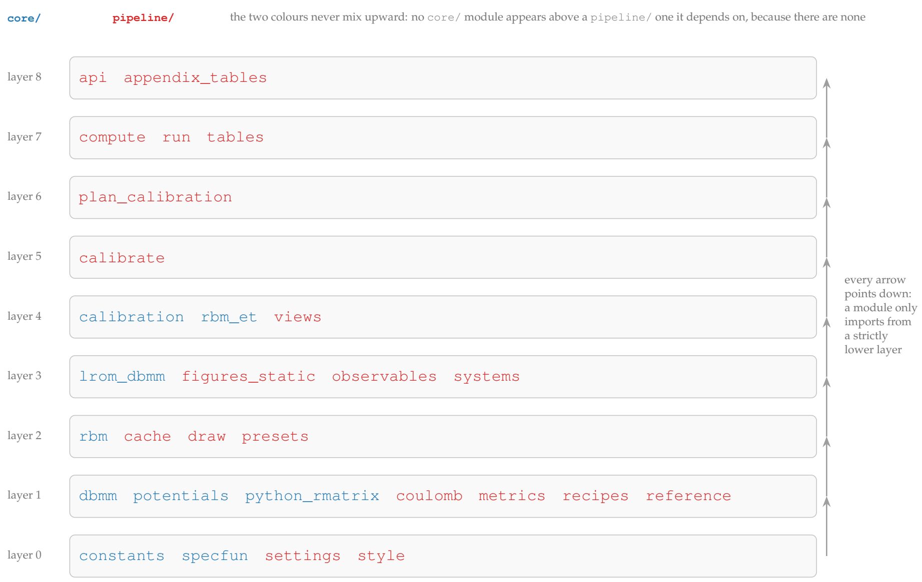

Figure D.1: The dependency layers of the twenty-nine modules, computed from the import statements as the longest path from each module to one that imports nothing. A module only ever imports from a strictly lower layer, so the graph is acyclic and can be read as a build order. Blue is `core/` and red is `pipeline/`, and the two are interleaved by layer and not by direction, because `pipeline/` calls `core/` and never the reverse. The full edge list is in the source.

One wart is visible and worth naming rather than hiding: `tables` imports `api` inside a function while `api` imports `tables` at module level. The cycle is broken by the deferred import and nothing else, so moving that import to the top of `tables` would break the package.

D.2 One path through the code

There is one entry point, the notebook `pipeline.ipynb`, and one path through it, drawn in Figure D.2. A `Run` names the systems, the settings, the emulators and the blocks. `plan()` reports what it would do and why before anything runs, comparing the settings and the mesh recorded in each cached block against the run being declared, so a recompute is incremental instead of total. `execute()` then runs what changed, and each block writes one `.npz`. Nothing downstream of that file calls a solver again, which is why regenerating every figure and table in this report costs seconds.

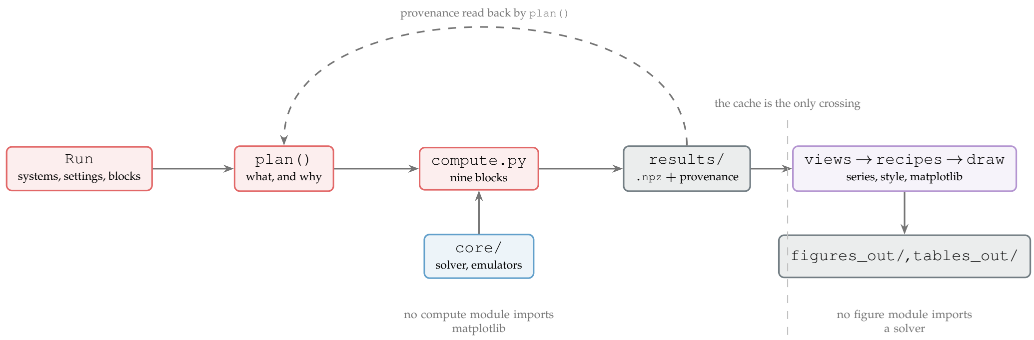

Figure D.2: One entry point, one path. Red is the study that drives the computation, blue the physics it calls, grey the cache, purple the drawing layer. The dashed loop is what makes a recompute incremental: each block records the settings and the mesh it was produced under, and the planner compares them against the run being declared. The vertical rule is the only place the two halves meet, which is why a number in the text cannot disagree with the figure beside it.

Everything else follows from the affine split of Section 3.4, directly or through what that split makes cheap. Table D.1 collects those decisions.

Table D.1: The five structuring decisions, what forces each one, and what each one makes possible.

Decision	What forces it	What it makes possible
`DBMMSolver` caches everything a change of θ leaves alone	$\mathbf{M}(E,\theta) = \mathbf{K}(E) + \mathbf{V}(\theta)$ with $\mathbf{V}$ diagonal on the mesh, so a new θ touches the diagonal only	The kinetic matrix, the Coulomb functions and the boundary parameters are built once per solver. The reduced operator can be precomputed offline, so no emulator needs an empirical-interpolation step
One interface over a chain of special-function backends	No single library covers the required (ℓ, η, kR) range reliably, and Coulomb and Whittaker functions dominate the cost on charged systems	`core/specfun.py` tries SciPy, then `python-flint` (Arb), then mpmath. Values are keyed on (ℓ, η, k, r), which no optical parameter enters, so one warm cache serves a whole campaign
Computing and drawing separated by a cache of arrays	A figure must not be able to disagree with the number quoted for it in the text	Blocks write results/<system>/<block>.npz and figures read nothing else. No figure module imports a solver, no compute module imports matplotlib
Provenance recorded per method, not per block	An RBM trained on 200 draws and an LROM trained on 400 can legitimately sit in one block	`plan()` compares the stored record against the run being declared and names *which* field moved, so a recompute is incremental instead of total
Figures in three layers	The physics of a panel, its appearance and the plotting calls are three different kinds of decision	A *view* turns arrays into labelled series and knows that a residual is a median with a p5–p95 band and never a mean; a *recipe* is data; only the renderer touches matplotlib

From a preset to a number, in ten lines

Listing 2 is the shortest program that exercises the whole chain, and it is the one to run first when taking the code over.

Listing 2: The shortest path through the package. It trains an emulator on the largest benchmark and queries it at a parameter it has never seen, printing the elastic element of the collision matrix from the solver and from the emulator.

```
from pipeline.systems import SYSTEMS
from pipeline.compute import solver_for
from core.rbm import RBMEmulator, latin_hypercube

```

```
system = SYSTEMS["alpha12C_band"]              # the four benchmarks are keys here
channels, model, theta_c, bounds = system.preset_fn()
E = system.E_demo                              # 16.0 MeV

solver = solver_for(system, theta_c, energies=[E])
emu = RBMEmulator(solver, E).fit(model, latin_hypercube(520, bounds, seed=0))

theta = 0.5 * (bounds[:, 0] + bounds[:, 1]) * 1.03       # never seen in training
print(solver_for(system, theta, energies=[E]).solve(E)[0, 0])    # DBMM
print(emu.predict(theta)[0, 0])                                  # emulator
```

Expected output, to the digits that reproduce across machines:

```
(0.0323824...+0.0121625...j)
(0.0323815...+0.0121617...j)
```

The two agree to 4×10^{-5} on this element, at the rank the default tolerance selects, $n_b = 98$. The training call is the expensive line, about forty seconds on the machine of Table D.11; every query after it is five milliseconds. Replacing `RBMEmulator` by `LROMEmulator` needs two more arguments, `theta_c` and the predictor budget, and Appendix D.4.3 gives that call.

D.3 One solve, end to end

Figure D.3 is one evaluation of the solver, with the size of the object each step hands on and what survives a change of θ; Algorithm D.1 is the same evaluation written as code, with the cost of each line.

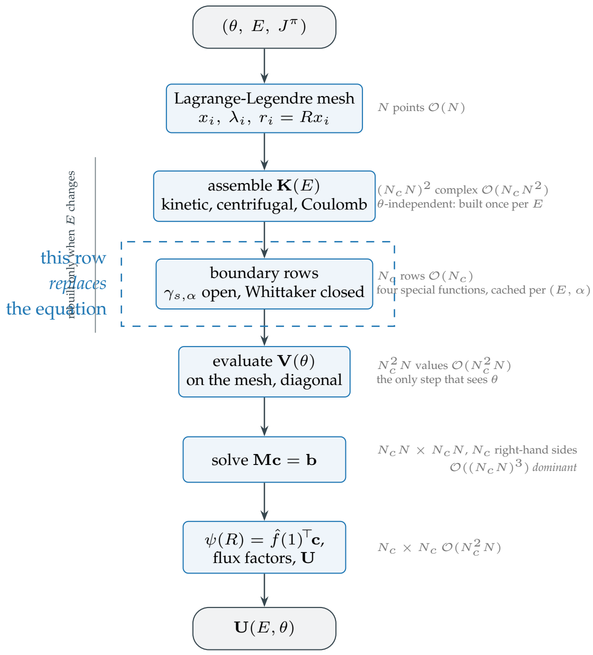

Figure D.3: One solve, end to end, with the size of the object each arrow carries and the dominant cost of each step. The affine split of Section 3.4 is visible as the boundary between the steps above the brace, which do not see θ, and the one below it, which does.

Algorithm D.1. One DBMM solve. The mesh, $\mathbf{K}(E)$ and the boundary rows are the parameter-free half of Eq. (12) and are cached per energy; only the potential and the source it feeds see θ.

---

input : θ, energy E, block J^π; channel list; mesh size N, radius R
output: collision matrix $\mathbf{U}(E, \theta)$, $N_c \times N_c$

$x_i, \lambda_i \leftarrow$ Gauss-Legendre nodes and weights; $r_i \leftarrow Rx_i$ — $\mathcal{O}(N)$
$\mathbf{K} \leftarrow T_{ij}$ of Eq. (45), centrifugal and Coulomb terms — $\mathcal{O}(N_c N^2)$
foreach *channel* α **do** row $N{-}1 \leftarrow \gamma_{s,\alpha}(E)$ if open, Whittaker if closed — $\mathcal{O}(N_c)$
$\mathbf{V} \leftarrow V_{\alpha\beta}(r_i, \theta)$ on the interior rows, boundary rows zero — $\mathcal{O}(N_c^2 N)$
$\mathbf{b} \leftarrow -V^{\text{eff}}_{\alpha\beta_0}(r_i)\, F_{\ell_{\beta_0}}(r_i)$, the Coulomb factor precomputed — $\mathcal{O}(N_c^2 N)$
solve $(\mathbf{K} + \mathbf{V})\, \mathbf{c} = \mathbf{b}$, one LU and N_c right-hand sides — $\mathcal{O}((N_c N)^3)$
$\chi_\alpha(R) \leftarrow \hat{f}(1)^\top \mathbf{c}_\alpha$; velocity factors Eq. (53); read $\mathbf{U}$ — $\mathcal{O}(N_c^2 N)$

D.4 The three objects, in code

D.4.1 The solver in code

Lei [9] specifies the method. The wave function is split into a known incident part and an unknown scattered one, which turns the homogeneous coupled system into an inhomogeneous one, and the outgoing-wave condition goes in the last row of each diagonal block instead of in a second matching step. The paper does not specify which parts of $\mathbf{M}$ survive a change of θ, which is not a question about the method. `DBMMSolver` is organised around that question, and Table D.2 states Eq. (12) in the terms of the code.

Table D.2: What is built when, in `core/dbmm.py`. The third column is why the object can be cached: an entry that no optical parameter reaches is built once per solver, whatever the campaign does afterwards.

Built	What it holds	Depends on
`_build_H0`, at construction	kinetic matrix, centrifugal barrier, full Coulomb term	the mesh and the channel list, nothing else
`_energy_part(E)`	the channel kinetic energies and the boundary rows carrying $\gamma_{s,\alpha}(E)$	E only, so it is computed once per energy node and cached
`K_matrix(E)`	the two above, assembled	E; this is $\mathbf{K}(E)$
`V_matrix()`	$V_{\alpha\alpha}(r_i)$ and $V_{\alpha\beta}(r_i)$ on the interior rows, boundary rows zero	θ; this is $\mathbf{V}(\theta)$, and it is diagonal

Listing 3: `K_matrix` is the parameter-free half of Eq. (12). It never touches the potential, which is what lets an emulator build its reduced operator from it once and reuse it at every θ. The boundary row overwrites row $N-1$ of each diagonal block rather than being appended, which is the whole of the DBMM in one line.

```
def K_matrix(self, E):
    """(Nc*N, Nc*N) parameter-independent DBMM matrix."""
    N, Nc = self.N, self.Nc
    K = self._K0_full.copy()
    erel, brows = self._energy_part(E)
    idx = np.arange(N - 1)

    for a in range(Nc):
        rs = a*N
        K[rs + idx, rs + idx] -= erel[a]       # interior rows: -(E - eps_a)
        K[rs+(N-1), rs:rs+N] = brows[a]        # last row: the outgoing condition
    return K
```

D.4.2 The reduced basis method in code

ROSE [14] decomposes the potential by empirical interpolation, because its solver propagates the radial equation and offers no matrix in which the parameter reaches a diagonal. Lei [10] shows that with a Lagrange mesh it need not: his Eq. (23) writes the projected potential as a sum over mesh points, which is exact for a local potential and needs no interpolation step. Our `RBMEmulator` takes that route, so the only approximation it makes is the Galerkin projection itself. Table D.3 sets the three possibilities side by side.

Table D.3: The three ways the reduced potential could be formed, and why the third is used. $N_{\text{tot}} = N_c N$.

Route	Cost per query	Why not, or why
Form $\mathbf{V}(\theta)$ dense, then project	$\mathcal{O}(N_{\text{tot}}^2 n_b)$	Defeats the purpose: the dense matrix is the object the reduction exists to avoid
Empirical interpolation, as ROSE	$\mathcal{O}(n_{\text{eim}} n_b^2)$	Adds a second approximation and an offline stage that the mesh makes unnecessary here
Radial sum, Lei's Eq. (23)	$\mathcal{O}(N_c N n_b^2)$	Exact for a local potential, no extra approximation; this is what `project_potential` does

Listing 4: The radial sum, as a batched product followed by one GEMM. The same expression written as a triple `einsum` leaves the n_b^2 contraction outside BLAS-3 and runs several times slower for the same flops; the two agree to 10^{-15}, the difference being the order of summation.

```
def project_potential(X_r, Vnuc, Vcoup, N, Nc):
    """V_r = sum_i X_r(i)^H V(r_i) X_r(i), Lei (2025) eq. (23)."""
    Xb, Xc = _mesh_blocks(X_r, N, Nc)                # (N-1, Nc, nb) each
    V = np.zeros((N - 1, Nc, Nc), dtype=complex)
    if Vcoup is not None:
        V[:] = np.transpose(Vcoup[:, :, :N - 1], (2, 0, 1))
    idx = np.arange(Nc)
    V[:, idx, idx] = Vnuc[:, :N - 1].T               # the diagonality property
    nb = X_r.shape[1]
    Y = np.matmul(V, Xb)                             # batched over mesh points
```

```
    return Xc.reshape(-1, nb).T @ Y.reshape(-1, nb)  # one gemm
```

Algorithm D.2 is the whole method, offline and online, with the cost of each line.

Algorithm D.2. The reduced basis method. The offline stage is paid once per energy node and per J^π block; the online stage is paid at every θ. Forming $\mathbf{V}_r$ is the step that carries the mesh size with the basis rank squared, and Algorithm D.3 removes it.

offline : box $\mathcal{P}$, training size N_s, tolerance ϵ_{tol}

Step	Cost
$\theta_1, \ldots, \theta_{N_s} \leftarrow$ Latin hypercube in $\mathcal{P}$	
foreach i **do** $\mathbf{c}_i \leftarrow$ Algorithm D.1 at θ_i	$N_s\,\mathcal{O}((N_c N)^3)$
$X\Sigma W^H \leftarrow \textsc{svd}([\mathbf{c}_1 \cdots \mathbf{c}_{N_s}])$	$\mathcal{O}(N_{\text{tot}} N_s^2)$
$n_b \leftarrow \min\{k : r(k) < \epsilon_{\text{tol}}\}$; $X_r \leftarrow$ first n_b columns of X	$\mathcal{O}(N_{\text{tot}})$
$\mathbf{K}_r \leftarrow X_r^H \mathbf{K} X_r$; $G \leftarrow \hat{f}(1)^\top X_r$	$\mathcal{O}(N_{\text{tot}}^2 n_b)$

online, at each θ:

Step	Cost
$V_{\alpha\beta}(r_i, \theta)$ at the $N-1$ interior points	$\mathcal{O}(N_c^2 N)$
$\mathbf{V}_r \leftarrow \sum_i (X_r^{(i)})^H V(r_i) X_r^{(i)}$, Eq. (17)	$\mathcal{O}(N_c^2 N n_b^2)$
$\mathbf{b}_r \leftarrow X_r^H \mathbf{b}(\theta)$, from the same mesh values	$\mathcal{O}(N_c^2 N n_b)$
$a \leftarrow (\mathbf{K}_r + \mathbf{V}_r)^{-1} \mathbf{b}_r$	$\mathcal{O}(n_b^3)$
$\chi(R) \leftarrow Ga$; velocity factors; $\mathbf{U}$	$\mathcal{O}(N_c^2 n_b)$

D.4.3 The learning reduced-order model in code

Giuliani [13] specifies the central-gauge residual fit. An operator $\mathbf{I} + \sum_j q_j \mathbf{M}_j$ is learned on the reduced coefficients, with predictors q_j centred so that the central parameter is reproduced exactly, and with the $\mathbf{M}_0 = \mathbf{I}$ gauge fixing the scale. The predictors are potential values at maxvol-selected radii (Appendix D.5.1). The reference implementation solves the fit as one linear system, and the fit here is split by row. Table D.4 measures what that is worth, and the ranks reported here would not be reachable without it.

Table D.4: The same fit, assembled two ways. The equation indexed (s, row) involves only row row of each $\mathbf{M}_j$, and its coefficients do not depend on row at all, so the n_b subproblems share one design matrix and differ only in their right-hand side. Measured at $n_{\text{samples}} = 400$, $K = 16$, $n_b = 16$.

Assembly	Design matrix	Time	Share of the block
One block, as in `simple_lrom.py`	$(n_{\text{samples}} n_b) \times K n_b (n_b + 1)$	47.8 s	97%
Decoupled by row, n_b right-hand sides	$n_{\text{samples}} \times K(n_b + 1)$	0.165 s	—

The two solutions agree to 10^{-13}, determined and underdetermined alike; that equality was measured rather than assumed, since `lstsq` returns the minimum-norm solution when the system is underdetermined and the two assemblies group the unknowns differently.

Listing 5: The online query. The learned operator is assembled from K scalars, the $\mathbf{M}_0 = \mathbf{I}$ gauge is added rather than stored, and one $n_b \times n_b$ solve per open channel gives the reduced coefficients. No special function and no mesh evaluation appear here, which is what the whole construction is for.

```
def predict_coefficients(self, theta):
    p_row = predictors(self._model, theta, self.points, self.center, self.scales)
    cols, M_kc, _v_kc, M_flat, v_flat = self._stacked_fits()
    nopen, nb = M_kc.shape[1], M_kc.shape[2]
    A = (p_row @ M_flat).reshape(nopen, nb, nb)
    A += np.eye(nb, dtype=A.dtype)                   # the M0 = I gauge
    rhs = (p_row @ v_flat).reshape(nopen, nb)
    a = np.linalg.solve(A, rhs[:, :, None])[:, :, 0]
    C = self.C_c.copy()
    C[:, cols] = self.C_c[:, cols] + self.X_r @ a.T   # lift, central gauge
    return C
```

Algorithm D.3 is the whole method. Read against Algorithm D.2 it differs in three places: the budget decision, the two MAXVOL passes, and an online stage in which the mesh does not appear.

Algorithm D.3. The learning reduced-order model. The snapshots are those of Algorithm D.2; the basis is not, since it decomposes their deviations from $\mathbf{c}_c$ rather than the snapshots themselves. The coupled case adds the branch on the diagonal and the separate MAXVOL pass per block, with the budget allocated by the rule of Section 4.4. The online stage never touches the mesh.

offline : as Algorithm D.2, plus the gauge point θ_c and the budget K

snapshots $\mathbf{c}_i$ as in Algorithm D.2, steps 1 to 2; $\quad \mathbf{c}_c \leftarrow$ Algorithm D.1 at θ_c
$X_r \leftarrow$ SVD of the deviations $[\,\mathbf{c}_i - \mathbf{c}_c\,]$, truncated at ϵ_{tol}; $\quad a_i \leftarrow X_r^H(\mathbf{c}_i - \mathbf{c}_c)$ $\quad \mathcal{O}(N_{\text{tot}} N_s^2)$
signature: stack $V_{\alpha\alpha}(r_j, \theta_i) - V_{\alpha\alpha}(r_j, \theta_c)$ and the same for $V_{\alpha\beta}$, one row per radius $\quad \mathcal{O}(N_s N_c^2 N)$
if *the diagonal does not move when the coupling parameter crosses its box* **then**
$\quad (K_{\text{diag}}, K_{\text{coup}}) \leftarrow$ the spectral split
else
$\quad (K_{\text{diag}}, K_{\text{coup}}) \leftarrow (K, 0)$
δ-MAXVOL on each signature separately, candidates at $r \geq 0.5$ fm, rows weighted by $|\chi_c(r)|$ $\quad \mathcal{O}(NK^2)$
$q_j(\theta_i) \leftarrow V$ at the K selected radii, centred so that $q_j(\theta_c) = 0$
$\{M_j, b_j\} \leftarrow$ least squares on the residual of Eq. (19): one system $N_s \times K(n_b{+}1)$ with n_b right-hand sides $\quad \mathcal{O}(N_s K^2 n_b^2)$

online, at each θ:
$q_j(\theta)$, $j = 1, \ldots, K$ $\quad \mathcal{O}(K)$
$A \leftarrow I + \sum_j q_j M_j$ $\quad \mathcal{O}(K n_b^2)$
$\rho \leftarrow \sum_j q_j b_j$ $\quad \mathcal{O}(K n_b)$
$a \leftarrow A^{-1}\rho$, one solve per open channel, no factorisation shared $\quad \mathcal{O}(N_c^{\text{open}} n_b^3)$
$\chi(R) \leftarrow \chi_c + Ga/\sqrt{R}$; velocity factors; $\mathbf{U}$ $\quad \mathcal{O}(N_c^2 n_b)$

What each step costs, predicted and measured

The three algorithms predict where the time goes; Table D.5 is that prediction put against a stopwatch. It is the arithmetic of Figure 9 with numbers in it.

Table D.5: The online stage, step by step, on $\alpha + {}^{12}$C at the calibrated $n_b = 104$ and $K = 15$, one core, each entry the minimum of forty repetitions. That rank is one rung above the operating configuration of Table 10, so the totals here are larger than the milliseconds quoted there. Forming $\mathbf{V}_r$ and $\mathbf{b}_r$ is four fifths of the RBM query and is where N enters; the LROM removes it, and its query is then dominated by no single term, so a further reduction of K would return little. The two solve entries are read at one rank and differ by a factor of seven because the LROM factorises once per open channel, its learned operator being different in each, where the RBM factorises once and applies N_c right-hand sides.

Step	Cost	DBMM	RBM ms	LROM
evaluate the potential	$\mathcal{O}(N_c^2 N)$ against $\mathcal{O}(K)$	9.66	0.32	0.03
form the reduced operator	$\mathcal{O}(N_c^2 N n_b^2)$ against $\mathcal{O}(K n_b^2)$		1.98	1.12
form the source	$\mathcal{O}(N_c^2 N n_b)$ against $\mathcal{O}(K n_b)$	—	0.23	0.003
solve	$\mathcal{O}((N_c N)^3)$; $\mathcal{O}(n_b^3)$; $\mathcal{O}(N_c^{\text{open}} n_b^3)$	60.08	0.11	0.77
extract $\mathbf{U}$	$\mathcal{O}(N_c^2 n_b)$	—	0.10	0.09
total query		69.7	2.74	2.01

The DBMM column carries two entries only, because its mesh evaluation and its source assembly happen in one call and its extraction is inside the solve. The comparison the report defends is the one between its total and the two emulator totals.

D.5 Autocalibration of the hyperparameters

Every emulator configuration quoted in Section 5 was chosen by the procedure described here, and none by hand. The four hyperparameters are the basis rank n_b, the two predictor budgets K_{diag} and K_{coup}, and the training size N_s.

`core/calibration.py` selects all four from spectra computed on a small probe sample, before any training. The delta matrices of the diagonal and coupling potential signatures (Section 4.3) are formed and their singular values normalised. K_{diag} and K_{coup} are the smallest indices at which each normalised spectrum falls below ϵ_{pot}, and n_b follows from the cumulative-energy criterion (14) at ϵ_{tol}. The number of predictor points is therefore governed by the intrinsic rank of the *potential* variation, as the basis rank is governed by that of the *solution* variation.

The spectra size the budget. Whether any of it goes to the coupling is decided by the rule of Section 4.4, and the calibration evaluates that condition rather than assuming it. Each parameter is pushed to the top of its box and the diagonal potential of every channel is re-evaluated. Among the parameters that reach the coupling, the rule is read on the one whose diagonal moves least, since on a rotational model R_0 moves the coupling form factor further than the deformation does while sitting on the diagonal as well. The budget is split only if that parameter leaves the diagonal exactly unchanged, at a threshold of 10^{-9}, far below the smallest non-zero travel on this suite and far above the noise of a quantity that is identically zero. Both allocations are kept, since the block that tests one against the other needs a genuine split whatever the calibration decides.

The training size is $N_s = 5n_b$, rounded up to a multiple of ten and never below 50. That choice makes the overfit ratio of Eq. (20) collapse to $r_{\mathrm{fit}} \simeq 5/K$, which is 0.29 to 0.48 across the suite: the residual-fit system is underdetermined by construction and LSTSQ returns the minimum-norm interpolant, which is an implicit regularisation rather than the absence of one. Reaching $r_{\mathrm{fit}} \geq 1$ would need $N_s \geq K(n_b + 1)$, an offline stage about four times larger. No such configuration was built, so every LROM accuracy in Section 5 is that of the minimum-norm interpolant and nothing here measures what the trade costs. One system run at $N_s \geq K(n_b + 1)$ would settle it, which on $n + {}^{40}\mathrm{Ca}$ is 230 snapshots against 110.

The probe also returns a *ladder* of six log-spaced ranks from $n_b/4$ to n_b, each rung carrying the budget in the proportion the calibration applied. One ladder serves every block, and Table 10 reports the cheapest rung that clears the posterior tolerance, the second from the top on three systems and the top rung on $\alpha + {}^{24}\mathrm{Mg}$. No accuracy measurement enters any of these choices. The frozen values are $n_b = 104$, 53, 37, 22 and $K = 15$, 17, 16, 10 on $\alpha + {}^{12}\mathrm{C}$, $n + {}^{238}\mathrm{U}$, $\alpha + {}^{24}\mathrm{Mg}$ and $n + {}^{40}\mathrm{Ca}$, every point on the diagonal; the spectra alone would have allocated 8+7, 6+11, 6+10 and 10+0, and Table 9 measures what those splits would have cost.

Why the break-even is not the training size

Section 5.7 reports break-even points N^* at which an emulator overtakes the split DBMM. If the offline stage were exactly the n_{train} solves, equating $n_{\mathrm{train}} t_{\mathrm{solve}} + N t_{\mathrm{on}}$ with $N t_{\mathrm{solve}}$ would give $N^* = n_{\mathrm{train}}/(1 - t_{\mathrm{on}}/t_{\mathrm{solve}})$. At the same rank and the same budget it returns 183, 559, 208 and 295 against 204, 698, 261 and 339 from the timed offline stage, low by 12 to 25 per cent, because that stage also carries the singular value decomposition, the Coulomb warm-up on the energy grid, and for the LROM the maxvol pass and the least-squares fit. The formula is a lower bound and not an estimate, and Section 5.7 quotes timed values.

Figure D.4 is the procedure in one picture, from the probe sample to the four hyperparameters and the ladder.

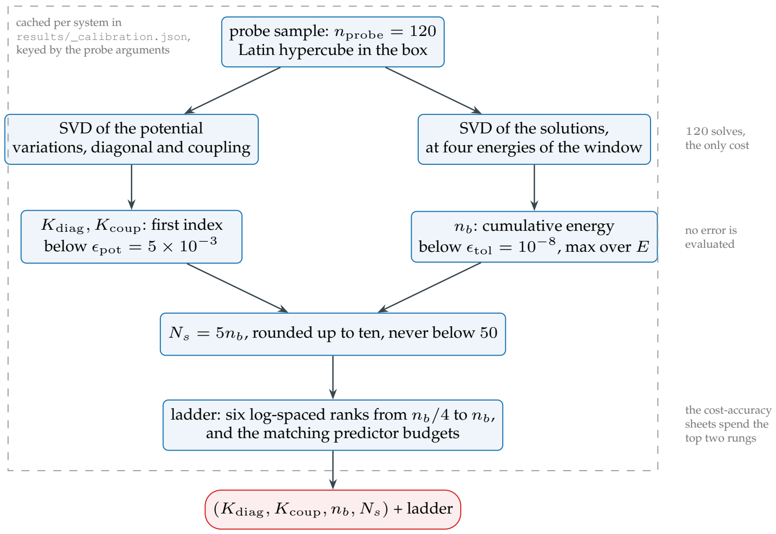

Figure D.4: The autocalibration loop. Two spectra decide the four hyperparameters, and neither of them is an error: the procedure never solves a validation problem, so a configuration cannot be tuned to the measurement it will be judged by.

D.5.1 Choosing the predictor points

The LROM replaces N mesh evaluations of the potential by $K \ll N$ scalar ones. Choosing which K radii is a selection problem. The candidate radii are ranked by how much information they carry about $\delta\mathbf{c}(\theta)$, and the K best are taken. Delta-maxvol does the ranking, on the singular vectors of the matrix of potential variations. Table D.6 lists what the implementation ranks on, and what each rule is worth.

Table D.6: The three rules the selection applies, and one that was tried and dropped. Effects are measured on $n + {}^{40}\mathrm{Ca}$, $\ell = 1$, $j = \frac{3}{2}$, at $n_b = 22$, as the median relative L^2 error on the radial wave function over 30 validation draws. Each rule is measured on top of the ones above it.

Rule	Why	Worth
One signature row per radius, kept complex	Stacking the real and imaginary parts gave the matrix $2N$ rows. Maxvol returned K indices among those $2N$, and the decoder mapped an index back to a radius, so two indices could land on the same one.	$K = 10$ returned 7 distinct radii instead of 10; the error at $K = 10$ falls from 1.9×10^{-2} to 5.2×10^{-4}
Candidates restricted to $r \geq 0.5$ fm	$\chi_\ell(r) \sim r^{\ell+1}$ at the origin, so the potential there cannot move the solution. The unrestricted selection spent one point at $r = 0.005$ fm at every budget.	$\times 6$ at $K = 6$, $\times 1.9$ at $K = 10$
Rows weighted by $\lvert\chi_c(r)\rvert$	The ranking is computed on δV, while the quantity of interest is $\delta\mathbf{c}$. A variation of the potential where the solution has no amplitude carries no information about it. $\lvert\chi_c\rvert^2$ overcorrects and is worse than no weighting at $K = 6$.	$\times 7$ at $K = 8$, $\times 3$ at $K = 10$
Dropped: candidates on a fine grid rather than the mesh	The points need not lie on the mesh, since the potential is called at an arbitrary radius. A 700-point grid offers maxvol ten times more candidates than the 64-point mesh.	No effect measured at equal cutoff; the candidate resolution is not what limits the selection

Together the three rules halve the budget: $K = 8$ with them reaches 4.4×10^{-4}, against 5.6×10^{-4} for $K = 16$ without them. Since the online assembly of the learned operator is $\mathcal{O}(Kn_b^2)$, and Table D.8 measures that step at 1.3 ms on the largest system, halving K halves it.

The weighting makes the selection depend on the energy, which it did not before: χ_c is the solution at θ_c and at the energy being fitted. Every energy node fits its own emulator and selects its own points, so nothing shares a selection across energies. The `predictor_points` argument of `LROMEmulator.fit` allows a caller to pass points in and skip the selection; those points are the ones chosen at whatever energy chose them.

What a trained emulator holds

Someone who wants to serialise a trained emulator and reuse it needs to know what is in it. Table D.7 is that inventory, at the largest system of the suite.

Table D.7: What a trained emulator holds, on $\alpha + {}^{12}$C at $N_c N = 960$, $n_b = 104$ and $K = 15$, in double-precision complex. In the last block, the fitted operator of the LROM is thirteen times the basis it acts on, because there is one $n_b \times n_b$ matrix per predictor and per open channel. That memory is the price of its online speed, and nothing in Section 5 counts it.

Object	Shape	Type	Size	What it is
shared				
X_r	$(N_c N,\ n_b)$	complex128	1.6 MB	the reduced basis
$G = \hat{f}(1)^\top X_r$	$(N_c,\ n_b)$	complex128	13 kB	the basis at the channel radius
σ_k	$(N_c N,)$	float64	7.7 kB	the spectrum, kept for the figures
RBM *only*				
$\mathbf{K}_r$	$(n_b,\ n_b)$	complex128	154 kB	the projected parameter-free block
LROM *only*				
$\mathbf{c}_c$	$(N_c N,\ N_c)$	complex128	123 kB	the solution at the central parameter
χ_c	$(N_c,\ N_c)$	complex128	1.0 kB	its boundary values
points	K indices	int	–	the selected radial points
centre, scales	$(K,)$ each	complex, float	0.6 kB	the predictor gauge
M_j	$(K,\ N_c^{\text{open}},\ n_b,\ n_b)$	complex128	21 MB	the fitted operator blocks
b_j	$(K,\ N_c^{\text{open}},\ n_b)$	complex128	200 kB	the fitted source blocks

Neither emulator is serialised to disk today. Everything above lives in memory for the duration of a run, and a campaign that wanted to train once and query from another process would have to add that. The arrays are plain `numpy`, so `numpy.savez` on the list above is the whole of it.

D.6 Optimisations and their measured effect

Each of these was made because a profile pointed at it, and each number was measured on the machine this work was done on.

Table D.8: Measured effect of each optimisation. The first three are configuration and scheduling; the rest are algebra and layout.

What	Why	Effect
One BLAS thread per process	the solves are small, numpy's threading loses on them and the workers oversubscribe the cores	38.8 s $\to$ 1.0 s on one validation loop
Arb backend for the Coulomb functions	mpmath was the fallback everywhere until `python-flint` was installed	$\times 11$ on charged systems
Coulomb cache warmed serially before any thread pool	on a cold cache twelve workers compute the same values twelve times over	a thirty-second block had become fifty minutes
One least-squares with n_b right-hand sides instead of one block system	the fit decouples by *row* of M_j: all n_b subproblems share one design matrix of size $N_s \times K(n_b+1)$	$\times 290$, and 10^{-13} on the solution
The source projection as a single GEMM	it is one matrix product once the mesh blocks are laid out contiguously	$\times 6.5$
The mesh blocks of the basis, and its contraction with the boundary row, kept against the identity of $\mathbf{X}_r$	both projections contract against the transpose of the basis and against its conjugate, and a numpy array carries no attribute of its own on which to keep them; a rank sweep rebinds $\mathbf{X}_r$, so the entry is dropped when the array it was derived from is	$5.11 \to 3.13$ ms on $\alpha + {}^{12}$C at $n_b = 104$, $\times 1.2$ on the other three, and 0 on $\mathbf{U}$
The K predictor points grouped by channel pair, one array call per group	they were evaluated one at a time, on one-element arrays; the LROM exists to replace N evaluations by $K \ll N$, and was paying K round trips through the interpreter	$0.285 \to 0.056$ ms on $n + {}^{40}$Ca, where that step was 91 per cent of the query
The learned operator assembled by GEMV on a K-contiguous layout	$\sum_j q_j M_j$ is a contraction over K; laid out channel-first and handed to `einsum` it read the operator, 32 MB at the budget then in use, at 9 GB/s with no BLAS kernel	$3.35 \to 1.32$ ms at $n_b = 104$, that is 24 GB/s
The entrance channels stacked into one solve and one GEMM	the operator differs per channel, so no factorisation is shareable, but its distribution is	5 per cent only: the row above was what mattered
Cumulative effect of the three rows above	at these sizes the LROM query is bound by call overhead and memory bandwidth, not by its arithmetic	at identical configuration, $0.36 \to 0.12$ ms on $n + {}^{40}$Ca, $3.79 \to 1.84$ on $\alpha + {}^{12}$C, $1.15 \to 0.50$ on $n + {}^{238}$U
Everything E-dependent and θ-independent cached on the solver, and the Coulomb mesh arrays shared between solvers	a campaign builds one solver per θ; the boundary rows and F_ℓ on the mesh depend on neither	rebuild $82 \to 75$ ms on $\alpha + {}^{12}$C
The energy nodes of the two per-energy blocks fitted concurrently	they are independent fits and numpy releases the global interpreter lock inside LAPACK	$\times 2.5$ on $\alpha + {}^{12}$C

Parallelism uses threads throughout, not processes, for two reasons. Numpy releases the global interpreter lock (GIL) inside LAPACK, where essentially all the arithmetic time goes, and the model closures would not pickle.

The first row of the table is the most fragile. The single-thread policy is set by environment variables that OpenBLAS reads only when its library is loaded: a script importing numpy, or anything that imports it, *before* the `pipeline` package gets the machine default, and the oversubscription that follows moves the timings by up to $\times 40$. Every block therefore records the observed thread count beside its timings, and writing a block whose whole content is a timing warns when the policy is not the expected one.

D.7 What the code is checked against

Eighty-two assertions run over four files in `tests/`, in eighty-five seconds. They fall into six kinds, and Table D.9 is what each kind would catch. The distinction that organises them is whether a check needs a reference answer: the first three kinds do not, which is what makes them usable on a system no other code can solve.

Table D.9: What the test suite asserts, and what each group would catch. The count is of parametrised instances, so one function tested on four systems counts four. The first three groups need no reference calculation and hold whatever code produced **U**; the last three are properties of this pipeline rather than of the physics.

Group	#	What is asserted	What it would catch
Invariants of **U**	13	symmetry under time reversal; unitarity with the absorption removed; the velocity factor of Eq. (53), separated from a discretisation error by refining the mesh	a formulation error in the asymptotics, on a system with no reference answer. The normalisation of Appendix B.3 is the fault this group exists for
Convergence and external agreement	2	the discretisation error falls with N; the entrance column of **U** matches the published Example 4 of the reference package above the highest threshold	a mesh chosen too coarse, and a coupling coefficient wrong by a phase, which no code-to-code comparison here can see
Emulator contracts	18	the LROM is exact at θ_c; the RBM error falls with rank; a prediction does not depend on the order of the queries; the boundary shortcut returns what the full-mesh lift returns, at every rank	a broken gauge, a basis assembled in the wrong order, hidden state carried between queries, and a shortcut that drifts from the construction it stands in for
One definition per quantity	3	the physical constants have a single definition; the solver's per-energy memo is transparent; an uncoupled system has no inelastic error	two values of $\hbar c$ in two modules, and a cache that returns something other than what a fresh call would
Pipeline consistency	43	every figure recipe has a unique stem; the blocks a view declares are the blocks it reads; timings appear only in the blocks marked machine-dependent	a figure silently drawn from the wrong block, and a timing frozen into a cache that is meant to be portable
Frozen reference	3	every cached array hashes to what it hashed at the last deliberate freeze	any change of value at all, whether a bug or a decision. Re-freezing is an explicit call, `api.freeze()`, so accepting a change is a recorded act

Whether an invariant test means anything is decided by two conditions, and both have to be checked when one is written. The fault it guards must produce a visible signal on the system it runs on, and the approximation error of whatever is being tested must not swamp that signal. On $\alpha + {}^{24}$Mg the three thresholds lie within 4 MeV of a 54 MeV demonstration energy, so $\sqrt{v_\alpha/v_\beta}$ reaches only 1.04 and removing the velocity factor moves the asymmetry of **U** from 8.6×10^{-7} to 5.5×10^{-6}, which is below the emulator's own 8.6×10^{-6}: the test would pass either way. The emulator symmetry check therefore runs on $\alpha + {}^{12}$C at its calibrated rank, where the same removal moves the asymmetry from 1.2×10^{-4} to 4.3×10^{-2}, a factor of 350. Both conditions are asserted in the test itself, so a later change of system or of rank that made it vacuous would fail rather than pass.

D.8 Differences from the reference LROM implementation

The LROM of Section 4.3 descends from Giuliani's repository ASCSN/LEARNING-REDUCED-ORDER-MODELS [13], [28], read in its state of August 2026. Only `simple_lrom.py` was usable: `cross_section_recompute.py` and `ae_wavefunction_extrapolation.py` import a module `phase3_implicit` that the repository does not ship.

Its ten-parameter configuration is $(V_v, W_v, W_d, V_{\rm so}, R_v, R_d, R_{\rm so}, a_v, a_d, a_{\rm so})$, term for term the vector θ of $n + {}^{40}$Ca. This benchmark was therefore moved onto his values, so that both codes run one problem: Koning-Delaroche on ^{40}Ca, a ± 20 per cent box, Latin hypercube sampling, $E_{\rm lab} = 14.1$ MeV (Appendix C).

Three conventions had to be reconciled first, and the sign of the two imaginary parts was measured rather than read: both parts absorptive give $|U| = 0.51$ at the central point, both emissive give 1.95, and only the first conserves flux. Table D.10 gives the three.

Table D.10: Mapping his potential onto ours. Two of the three are exact; the third is the one entry of the preset that is not the number written in the repository.

Term	His form	Ours	Mapping
Volume	$(1\mathrm{j}\, w_v - v_v) f$	$-(V_v + \mathrm{i}W_v) f$	exact, $W_v = \lvert w_v \rvert$
Surface	$-(4\mathrm{j}\, a_d w_d) f'$	$-\mathrm{i}W_d\, g$	exact, since $g \equiv -4a\, \mathrm{d}f/\mathrm{d}r$
Spin-orbit	carries $\lambda_\pi^2 \simeq 2$ fm^2	`thomas_so`, without it	his strength multiplied by λ_π^2; otherwise half strength and dimensionally short

The partial-wave content differs too, and not by convention: the repository solves $\ell = 0$ to 10 with both spin-orbit partners, each with its own basis, and sums them into a differential cross section, where this report keeps $j = \ell \pm \frac{1}{2}$ at $\ell = 1$ in one system matrix. Neither couples the partial waves; the second puts them in one algebraic object, which the three coupled systems require.

Section 5.3 reports the two measurements, taken under these conditions. The solvers are compared on the diagonal element of $\mathbf{U}$, at the central values and over twenty draws, after one further convention is aligned. ROSE applies the relativistic correction of Ingemarsson [31], so its reduced mass depends on the energy and is 929.34 MeV$/c^2$ at the $E_{\rm lab} = 14.1$ MeV of this benchmark, where the non-relativistic $m_1 m_2/(m_1 + m_2)$ gives 916.43 from measured masses and 916.02 from the mass numbers used here. Those 1.4 per cent move k by 0.7 per cent and $\mathbf{U}$ by 5 per cent, more than every other difference between the two codes. The mass-number construction accounts for 0.05 of those 1.4 per cent on this system and for 0.8 per cent on the two α systems, where both fragments are composite. One solve costs the DBMM 1.3 ms against 43 ms for ROSE at its Runge-Kutta tolerance of 10^{-9}.

The emulators are compared on the radial wave function, in relative L^2 norm against each code's own solver, the quantity both actually predict. Both errors are read on one radial grid, 0.1 to 15 fm, and on the total wave function: the reference mesh runs to 31 fm in 700 uniform points and this one to 15 fm in 64 Gauss-Legendre points, and the emulator here returns the scattered wave by default. Only one timing is comparable, the reduced solve itself, at 0.10 to 0.20 ms here against 0.017 to 0.091 ms there, so that query is about five times faster at comparable accuracy; the lifting onto the full mesh sits outside the reference timer and cannot be set against anything.

What is his, and what the coupled case forced

The form of the reduced model and its gauge are the repository's: the operator is learned as $\mathbf{I} + \sum_j q_j \mathbf{M}_j$, the predictors are centred so that $q(\theta_c) = 0$, and the central point is reproduced exactly by construction. The maxvol selection is ported from `cross_section_recompute.py`, swap loop included, rather than from the purely greedy version in `simple_lrom.py`.

The solver forces two differences, which are not choices. The repository decomposes the potential by empirical interpolation, as ROSE does, because a Runge-Kutta propagator offers no object in which the parameter reaches a diagonal and nothing else; the quadrature property (8) gives that split exactly here. And with N_c coupled channels $V_{\alpha\alpha}$ and $V_{\alpha\beta}$ differ by orders of magnitude, so a joint selection puts every point on the diagonal and a parameter acting only off it becomes invisible whatever the rank (Table 9). Hence the separate budgets $K_{\rm diag}$ and $K_{\rm coup}$.

The rank changes order too: $n_\phi = 4$ per partial wave there, 22 to 104 here, and the fit cost grows with its square. `simple_lrom.py` assembles one system of size $(N_s n_b) \times K n_b(n_b + 1)$; the fit decouples by row of $\mathbf{M}_j$, so a single `lstsq` with n_b right-hand sides of size $N_s \times K(n_b + 1)$ gives the same solution to 10^{-13} (Appendix D.6).

Neither code regularises, and the retained solution is the minimum-norm interpolant `lstsq` returns when the system is underdetermined.

The comparison does not reproduce two of the reference studies. Notebook 04 extrapolates in energy, 10–22 trained to 8–30 MeV tested, and in isotope, $A = 32$–48 trained to $A = 30$–60 tested, on a single basis built at the central point. Here one emulator is fitted per energy node, the joint-basis variant is unvalidated (Appendix E.10), and no extrapolation in mass is attempted. Its errors are quoted on the differential cross section; here on the integrated one and on the off-diagonal elements of $\mathbf{U}$, as a median and a p95 over the box. The two sets of figures are not comparable term by term.

D.9 Tools used

Table D.11 is what the work was made with and what it ran on. Every figure separates its series by dash pattern and marker as well as by hue, so no statement in this report depends on colour being reproduced.

Table D.11: The software and the machine. Every block of the cache records the interpreter, the numpy version, the special-function chain with its library versions and the observed BLAS thread count it ran under, so a result can be asked what produced it. The thread count is the entry that matters. The package sets it to one before numpy loads, so an online query is a single-threaded measurement whatever the machine offers, and the offline stage spends the twenty logical processors. A different machine moves every absolute time in this report and, to the extent that both columns move together, not the ratios.

Tool	Role	Component	Value
Python 3.14	everything numerical	Processor	Intel Core i7-13700H
numpy 2.3	arrays and dense linear algebra	Cores	14 physical, 20 logical
scipy 1.16	closed-form special functions, Latin hypercube sampler	Base clock	2.4 GHz
python-flint 0.9	Coulomb and Whittaker functions outside scipy's reach	L2, L3 cache	11.5, 24 MiB
mpmath 1.4	arbitrary-precision reference for the other two	Memory	13.7 GiB
matplotlib 3.10	every figure	Operating system	Windows 11, build 26200
pandas	every table the pipeline returns	BLAS	OpenBLAS 0.3.30
numba	installed, out of the active chain: its mesh kernel reaches 2×10^{-12} where the default reaches machine precision	Threads when timing	1, set at package import and read back with `threadpoolctl`
Jupyter, VS Code	one notebook drives the whole computational surface (Appendix D.2)		
Claude, Qwen	engineering assistants on the code base, under the terms stated below		
DeepL	checking formulations and translating passages drafted in French		

The L3 figure is the one to keep in mind when reading the mesh scan of Section 5.7. The largest reduced basis stored here, X_r for $\alpha + {}^{12}\mathrm{C}$ at $n_b = 104$, is 960×104 complex in double precision, that is 1.6 MiB, and the full system matrix at that size is 14 MiB. The first sits in L3 comfortably and the second does not, which is where the online cost stops scaling with arithmetic alone.

Language models and translation

The last two rows of Table D.11 are declared here. The physics, the derivations, the choice of benchmarks, the identification of the affine split and its use, the diagnosis of the flux-factor error and every number in this report are my own. Claude and Qwen were used as engineering assistants: refactoring the first procedural version of `core/` into the object-oriented structure described in Appendix D.1, and building and then cleaning up the figure-generation pipeline, which is infrastructure. Every suggestion was read, tested and in several cases rejected.

The report is written in English, which is not my first language. DeepL was used to check formulations and to translate passages drafted in French. The technical content, the argument and the final wording are mine.

E Measurements

Every measurement the body states and does not print. Each subsection names the claim it supports, and Table E.1 is the index the other way round, from the claim to the measurement. The six tables here that no cached block produces are written by `pipeline/appendix_tables.py`, which runs its own measurement and stamps each file with what produced it.

Table E.1: What is measured here, and which claim of the body it supports. Every row is a measurement taken on the same cache and under the same protocol as the figures of Section 5: validation parameters never seen in training, against the DBMM except where the two solvers are the two things compared.

The claim, and where it is made	What is measured	Here
The solver floor on $\alpha + {}^{24}$Mg is the mesh and not the implementation (§5.3)	both codes against a converged mesh, at four mesh sizes	Table E.2
ϵ_{tol} cannot be relaxed (§5.4)	rank, error and online time at three tolerances	Table E.3
More snapshots at fixed rank change nothing (§5.4)	error against training size, at the calibrated rank	Figure E.1
Two systems stop improving, and the conditioning jumps at that step (§5.4)	$\mathrm{cond}(\mathbf{M}_r)$ and the error at every rank of the ladder	Table E.4
The choice of scored quantity moves the answer by orders of magnitude (§5.2)	error by quantity against rank, with the magnitudes it is relative to	Figure E.2
The agreement is not confined to the cross section (§5.5)	the phase shift from both emulators over the window	Figure E.5, Table E.5
Outside the box the degradation is gradual (§5.8)	error at three box widths, emulators trained on the nominal one	Table E.6
A channel threshold costs the emulators nothing (§5.5)	error at three offsets above each threshold of $\alpha + {}^{12}$C	Table E.7
The LROM removes the only term carrying the mesh size (§5.7)	online cost against N at fixed physics, rank and budget	Figure E.6
The advantage should be larger at six hundred channels than at eight (§6.3)	cost, rank and gain swept over the channel count in one family	Figure E.7
The solver is correct on the two features the benchmarks conflate (§5.3)	cross section and interior solution on $p + {}^{12}$C and $\alpha + d$	Figure E.8
Two variants exist and are not part of the comparison (§6.2)	what each does, and what validating it would take	Table E.8

E.1 The solver against a refined mesh

Section 5.3 attributes the 5×10^{-3} between the two codes on $\alpha + {}^{24}$Mg to the DBMM not being converged at its working mesh rather than to a difference of formulation.

Table E.2: The two solvers against a refined mesh, on $\alpha + {}^{24}$Mg at its central parameters and its demonstration energy, worst relative difference over the elements of $\mathbf{U}$. Each code is measured against its own answer at $N = 320$, so the first two columns are convergence and the third is the disagreement between implementations. The gap between column one and column three at the working mesh is how far the DBMM is from converged.

N	DBMM against $N = 320$	R-matrix against $N = 320$	code against code
100	4.9×10^{-3}	3.0×10^{-5}	5.0×10^{-3}
140	1.3×10^{-3}	9.7×10^{-6}	1.2×10^{-3}
180	6.2×10^{-4}	4.6×10^{-6}	4.7×10^{-4}
240	2.3×10^{-4}	9.0×10^{-7}	1.3×10^{-4}

Between $N = 100$ and $N = 240$ the DBMM error falls as $N^{-3.4}$ and the R-matrix error as $N^{-3.9}$, so the two converge at comparable order and what separates them is the constant, about 160 times larger for the DBMM. Replacing the radial equation at the last mesh point by the boundary condition therefore costs a prefactor and not an order, and the price is paid once, as a mesh size. At its central parameters the solver also reproduces the published $\alpha + {}^{12}$C values of the reference R-matrix package to 4.6×10^{-4} at 16 MeV and 1.0×10^{-3} at 20 MeV, both above the highest threshold, computed there at a channel radius of 11 fm against the 14 fm used here, which is a convergence check the internal comparison cannot provide.

E.2 Truncation tolerance and training size

Section 5.4 states that ϵ_{tol} cannot be relaxed and that more snapshots at fixed rank change nothing.

The truncation tolerance $\epsilon_{\text{tol}} = 10^{-8}$ is the strictest number in the calibration, and it is what sets the rank, so a looser one is worth testing. The argument for relaxing it is short. The criterion thresholds the discarded energy $r(k)$, and a subspace that discards a fraction r of the snapshot energy is at a distance of about $\sqrt{r}$ from them, so $\epsilon_{\text{tol}} = 10^{-8}$ buys a representation error of 10^{-4}, nearly two decades below the operating target $\epsilon_{\text{post}} = 5 \times 10^{-3}$. On that reasoning $\epsilon_{\text{tol}} = 10^{-5}$ would buy 3×10^{-3}, just inside the target, at a substantially smaller rank and therefore a substantially cheaper online stage. The rank fractions n_b/N_cN of 10 to 16 per cent are large for a reduced model, and this would be the way to bring them down.

Table E.3 is that experiment, on 200 validation parameters never seen in training.

Table E.3: Measured effect of a looser truncation tolerance. Rank, rank fraction, median and 95th percentile of the relative error on $\mathbf{U}$, which is the quantity the Eckart-Young bound applies to, and online time, over 200 validation parameters. The rank is read at the demonstration energy, whereas the calibration takes the maximum over the energy window, which is why these ranks differ by a few from Table 7. The operating target is $\epsilon_{\text{post}} = 5 \times 10^{-3}$; entries above it are configurations that no longer clear it at the median.

System	ϵ_{tol}	n_b	n_b/N_cN	median error	p95 error	online (ms)
$n + {}^{40}\text{Ca}$	10^{-8}	20	16%	4.8×10^{-4}	1.7×10^{-3}	0.12
	10^{-6}	13	10%	3.9×10^{-3}	8.8×10^{-3}	0.12
	10^{-5}	10	8%	9.5×10^{-3}	2.2×10^{-2}	0.11
$\alpha + {}^{12}\text{C}$	10^{-8}	99	10%	9.6×10^{-5}	2.6×10^{-4}	2.56
	10^{-6}	71	7%	9.7×10^{-4}	3.2×10^{-3}	1.76
	10^{-5}	56	6%	5.1×10^{-3}	1.6×10^{-2}	1.23
$\alpha + {}^{24}\text{Mg}$	10^{-8}	37	12%	5.9×10^{-4}	2.7×10^{-3}	0.24
	10^{-6}	26	9%	1.2×10^{-2}	7.0×10^{-2}	0.21
	10^{-5}	20	7%	6.9×10^{-2}	2.2×10^{-1}	0.18
$n + {}^{238}\text{U}$	10^{-8}	46	12%	3.7×10^{-4}	9.7×10^{-4}	0.45
	10^{-6}	31	8%	1.6×10^{-3}	4.1×10^{-3}	0.42
	10^{-5}	26	6%	3.9×10^{-3}	1.1×10^{-2}	0.35

The $\sqrt{\epsilon_{\text{tol}}}$ estimate is optimistic out of sample, and by an amount that grows as the tolerance is loosened. At $\epsilon_{\text{tol}} = 10^{-8}$ it predicts 10^{-4} and the measured median runs from 9.6×10^{-5} to 5.9×10^{-4}, within a factor of six. At $\epsilon_{\text{tol}} = 10^{-5}$ it predicts 3×10^{-3} and the measured median runs from 3.9×10^{-3} on $n + {}^{238}\text{U}$ to 6.9×10^{-2} on $\alpha + {}^{24}\text{Mg}$, up to twenty-three times the prediction, and three systems of the four no longer clear ϵ_{post} at the median. The reason is the one Figure 8 states: the discarded singular values bound the distance from a *snapshot* to the subspace, and not the distance from a query point the basis never saw. The gap between the two is the subject of Section 5.5.

What the tolerance does buy is small. Going from 10^{-8} to 10^{-5} halves the online time on $\alpha + {}^{12}\text{C}$, cuts it by a fifth to a quarter on $\alpha + {}^{24}\text{Mg}$ and $n + {}^{238}\text{U}$, and does nothing on $n + {}^{40}\text{Ca}$, because at that size the cost is not in the reduced solve. Four to eight points of rank fraction are not worth one to two decades of accuracy.

The rank fractions therefore stand as a property of these problems rather than as a tuning mistake. A scattering solution over the boxes of Appendix C is not compressible to one per cent of its dimension at the accuracy a posterior needs, and the report should not pretend otherwise by loosening a criterion until the number looks better.

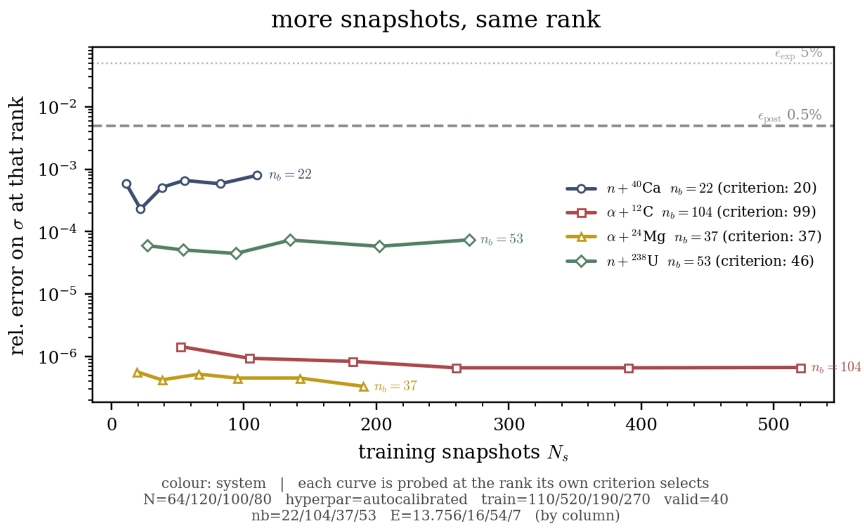

Figure E.1: At the rank each system's own criterion selects, the RBM error against the number of training snapshots. It flattens within a few tens of draws on all four systems, so its offline stage is a fixed cost sized in advance rather than something to be grown until the error gives way. The LROM is not measured here, and no measurement in this report resolves its own dependence on N_s (Appendix D.5).

E.3 Conditioning of the reduced solve

Section 5.4 acquits the subspace, and then the conditioning of the reduced solve, as the cause of the $n + {}^{40}$Ca and $n + {}^{238}$U error floors, at every rank of the calibrated ladder on all four systems.

Table E.4: Conditioning of the reduced solve against rank, median and worst over 30 validation draws, beside the distance from an unseen solution to the subspace and the median error on $\mathbf{U}$, all at the same rank. The rank ladder is the calibrated one of Appendix D.5. The best-approximation column is $\|\mathbf{c} - X_r X_r^H \mathbf{c}\| / \|\mathbf{c}\|$ at the validation draws, so it is what the subspace can do before any equation is projected onto it, where the column beside it is what the projected solve does instead. $n + {}^{40}$Ca and $n + {}^{238}$U are the two systems whose error stops falling.

System	n_b	median cond	worst cond	best approximation	median error
$n + {}^{40}$Ca	5	11	21	3.7×10^{-2}	1.4×10^{-1}
	7	14	23	1.2×10^{-2}	7.0×10^{-2}
	9	23	26	4.0×10^{-3}	2.0×10^{-2}
	12	39	43	1.0×10^{-3}	4.0×10^{-3}
	16	49	68	2.2×10^{-4}	1.1×10^{-3}
	22	364	397	3.7×10^{-5}	1.7×10^{-3}
$\alpha + {}^{12}$C	26	54	81	4.6×10^{-2}	6.8×10^{-2}
	34	76	123	2.4×10^{-2}	4.0×10^{-2}
	45	111	155	1.1×10^{-2}	1.9×10^{-2}
	60	158	188	3.5×10^{-3}	3.3×10^{-3}
	79	271	345	8.9×10^{-4}	5.0×10^{-4}
	104	534	658	1.0×10^{-4}	7.8×10^{-5}
$\alpha + {}^{24}$Mg	9	4	5	2.2×10^{-2}	7.5×10^{-1}
	12	8	17	1.1×10^{-2}	5.6×10^{-1}
	16	21	35	5.3×10^{-3}	2.0×10^{-1}
	21	26	36	1.6×10^{-3}	4.5×10^{-2}
	28	38	51	4.2×10^{-4}	7.3×10^{-3}
	37	49	60	5.2×10^{-5}	6.5×10^{-4}
$n + {}^{238}$U	13	20	32	3.0×10^{-2}	9.8×10^{-2}
	17	23	36	1.7×10^{-2}	3.3×10^{-2}
	23	31	42	4.4×10^{-3}	5.3×10^{-3}
	30	51	64	1.1×10^{-3}	1.8×10^{-3}
	40	100	112	2.0×10^{-4}	3.6×10^{-4}
	53	506	550	3.1×10^{-5}	4.0×10^{-4}

Two systems stop improving at the last rank of their ladder, $n + {}^{40}$Ca from 1.1×10^{-3} to 1.7×10^{-3} and $n + {}^{238}$U from 3.6×10^{-4} to 4.0×10^{-4}. Over that same step the distance from an unseen solution to the subspace falls by a factor of six on both, from 2.2×10^{-4} to 3.7×10^{-5} and from 2.0×10^{-4} to 3.1×10^{-5}: the added modes carry information the solution needs and the answer gets worse anyway.

The two columns measure different things, a relative distance in the coefficient vector against a relative error on

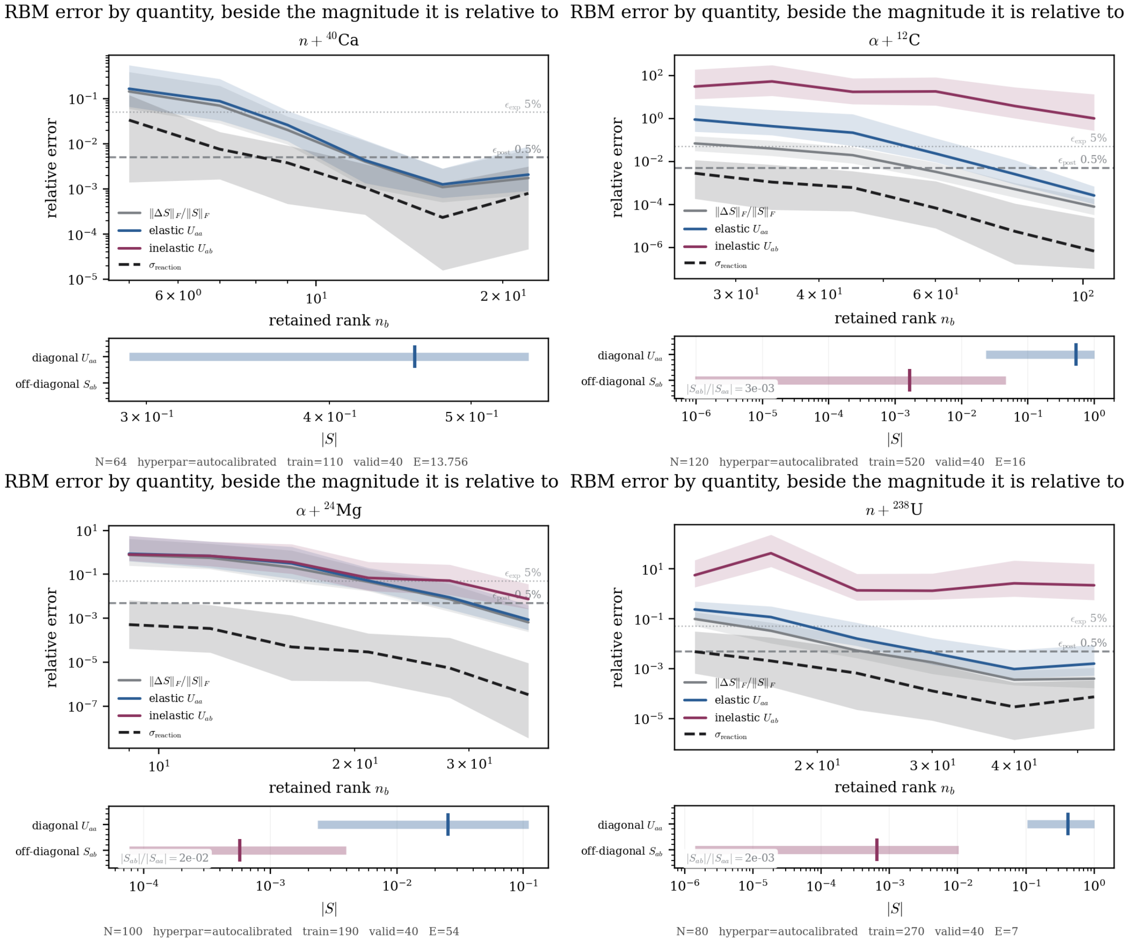

Figure E.2: Emulator error by quantity against retained rank, one sheet per system, and beneath each the magnitudes those errors are relative to. The second row is what makes the first readable.

U, so their ratio carries the stability of the projection together with the sensitivity of **U** to the coefficients. It stays nearly constant along each ladder, at 4 to 6 on $n + {}^{40}$Ca, below 2 on $\alpha + {}^{12}$C, 1 to 3 on $n + {}^{238}$U and 12 to 51 on $\alpha + {}^{24}$Mg, and jumps by a factor of seven to nine at the two rungs where a system floors, to 46 and to 13. That is where the floor sits: in the step from the subspace to **U**, which is the quantity Section 4.5 names and this report does not otherwise estimate.

The median condition number jumps at those same two rungs, from 49 to 364 and from 100 to 506, and it is not sufficient on its own: $\alpha + {}^{12}$C reaches 534 at its retained rank with both columns still falling. The two cases differ in that the jump is abrupt on $n + {}^{40}$Ca and $n + {}^{238}$U and gradual on $\alpha + {}^{12}$C. The standard remedy degrades the result. A least-squares Petrov-Galerkin projection, which is unconditionally stable, loses a factor of 1.1 to 9 against Galerkin on $n + {}^{238}$U at the three highest ranks and 3 to 33 on $\alpha + {}^{24}$Mg, which has no floor. The reason is structural and specific to this system matrix: the DBMM reads **U** from one row per channel, and an unweighted least squares over all N_{tot} rows weights that row like any interior mesh point. A projection in a norm that gives the boundary rows their weight is the test that remains.

E.4 Error by quantity, system by system

Section 5.2 states that the choice of scored quantity moves the answer by orders of magnitude. The four sheets below carry the magnitudes the errors are relative to beneath each.

The same statement lands on the cost-accuracy plane. Figure 16 is drawn on the elastic element, which every system of the suite has. The three coupled benchmarks also carry an off-diagonal element, and that is the one Table 10 and Figure 18 are read on, since it is what a coupled-channel campaign propagates. Figure E.4 is the same plane on that element. The horizontal positions do not move, because the cost does not depend on which element is scored. The vertical ones move by a factor that is neither uniform across systems nor the same for the two methods: at the operating configurations of Table 10 the off-diagonal element is worse than the elastic one by 1.0 and 1.4 on $\alpha + {}^{12}$C, by 2.0 and 15 on $\alpha + {}^{24}$Mg and by 4.8 and 7.3 on $n + {}^{238}$U, RBM then LROM. That is why the ranking of Section 5.9 is stated per system and per quantity.

The same draws seen as distributions are wide, by up to two decades over the box, and part of each falls to the

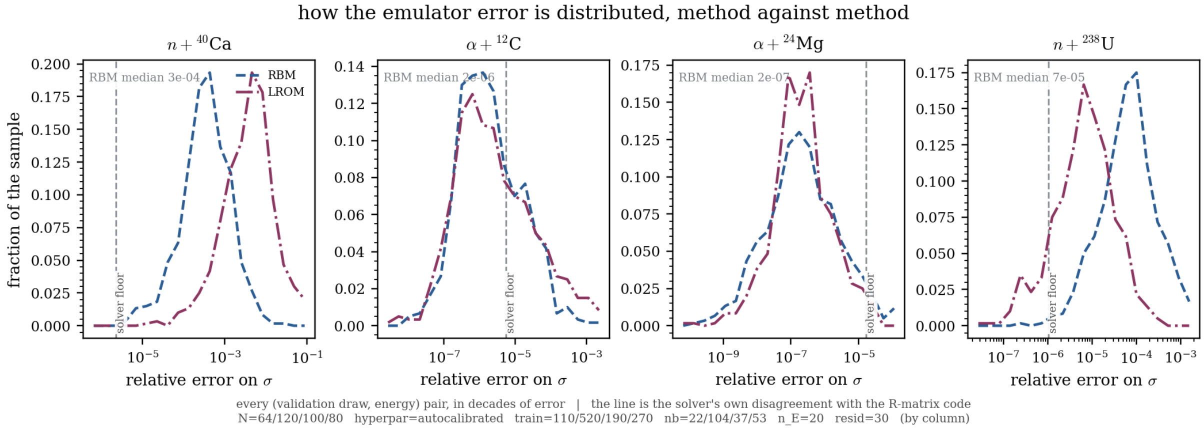

Figure E.3: The distribution of the emulator error over the parameter box, method against method. The vertical rule is the DBMM measured against the R-matrix code, that is the disagreement between solvers, below which an emulator error no longer means anything.

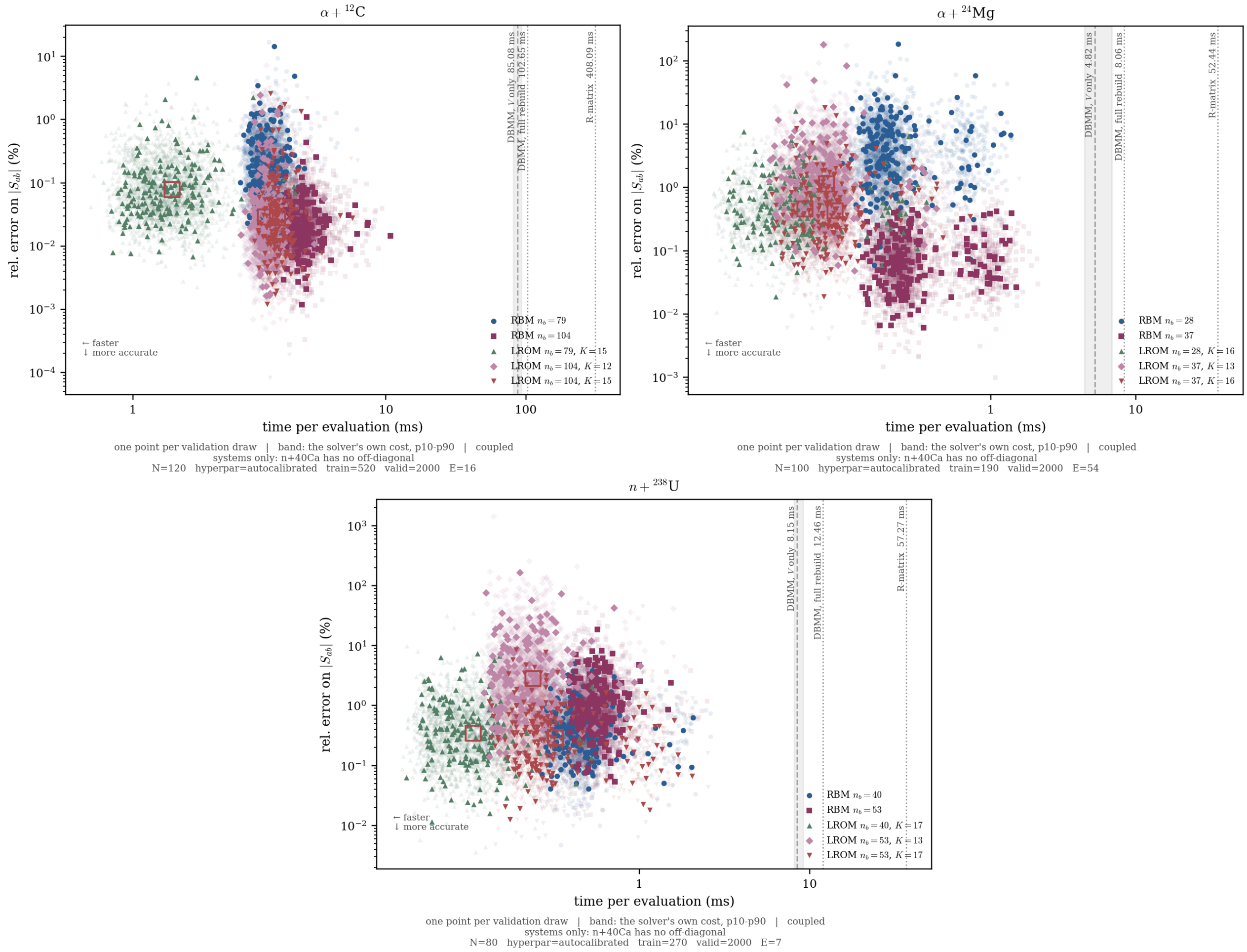

Figure E.4: Cost against accuracy on the off-diagonal element, one point per validation draw, on the three coupled benchmarks; $n + {}^{40}$Ca has no off-diagonal element and does not appear. Rules and rings are those of Figure 16. This is the quantity Table 10 and Figure 18 score.

left of the solver's own disagreement with the R-matrix code, where the emulator is closer to this solver than this solver is to the reference.

E.5 The interior solution and the phase shift

Section 5.5 scores the emulators on the cross section and on the interior solution. These are the two quantities the cross section is drawn from, measured against the solver at the same parameters: $\psi(r)$ pointwise, and the phase shift over the whole window.

Table E.5: The interior solution and the phase shift against the solver, median over 30 validation draws outside every training set. The deviation on ψ is pointwise, as a fraction of the peak amplitude; that on δ is absolute, since a phase shift passing through zero makes a relative error unbounded.

	$\psi(r)$, fraction of peak		$\delta(E)$, degrees	
System	RBM	LROM	RBM	LROM
n+^{40}Ca	2.6×10^{-4}	3.1×10^{-3}	0.0169	0.2095
α+^{12}C	1.4×10^{-5}	9.7×10^{-6}	0.0010	0.0013
α+^{24}Mg	5.5×10^{-5}	2.2×10^{-5}	0.0050	0.0045
n+^{238}U	1.5×10^{-4}	1.6×10^{-5}	0.0176	0.0020

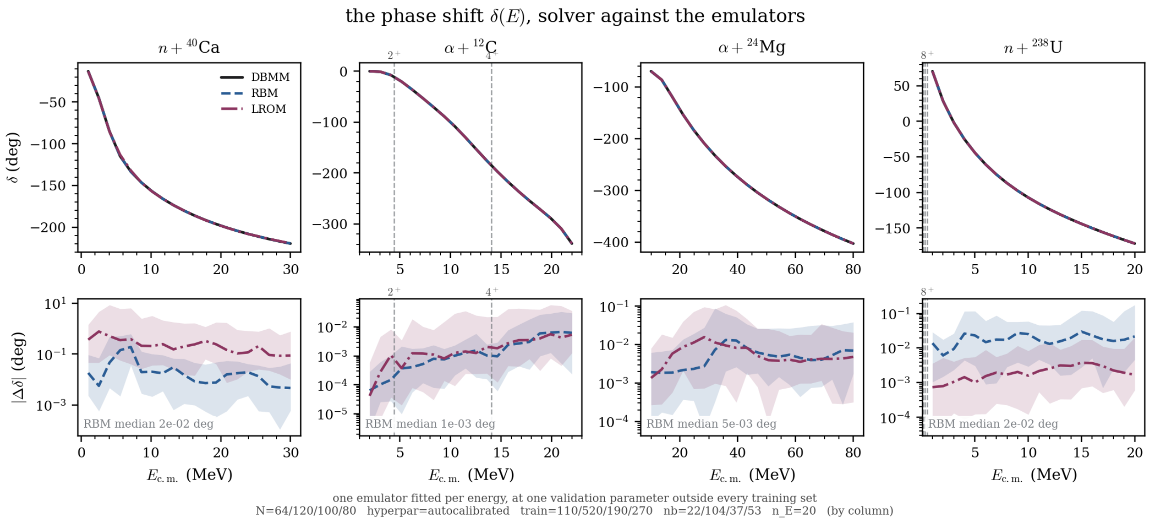

Figure E.5: The phase shift $\delta(E)$, solver against both emulators, one emulator fitted per energy node. The deviation is absolute and in degrees, since a phase shift passing through zero makes a relative error unbounded.

E.6 Domain of validity

Section 5.8 reports the degradation outside the training box, and Section 5.5 the behaviour at a channel threshold.

Table E.6: Error outside the training box. The emulators are those trained on the nominal box; only the sampling box is widened, by the factor in column two. Median and 95th percentile of the relative error on σ at the demonstration energy, over 40 draws none of which is in any training set.

		RBM		LROM	
System	width	median	p95	median	p95
$n + {}^{40}$Ca	1.0	7.6×10^{-4}	3.6×10^{-3}	2.5×10^{-3}	1.2×10^{-2}
	1.2	6.6×10^{-4}	5.3×10^{-3}	4.8×10^{-3}	2.7×10^{-2}
	1.5	1.1×10^{-3}	7.8×10^{-3}	1.2×10^{-2}	1.2×10^{-1}
$\alpha + {}^{12}$C	1.0	5.4×10^{-7}	2.6×10^{-5}	7.8×10^{-7}	1.6×10^{-5}
	1.2	1.7×10^{-6}	3.9×10^{-5}	8.9×10^{-7}	1.1×10^{-4}
	1.5	2.5×10^{-6}	2.5×10^{-4}	9.8×10^{-6}	1.4×10^{-3}
$\alpha + {}^{24}$Mg	1.0	9.7×10^{-8}	1.2×10^{-5}	1.4×10^{-7}	2.0×10^{-6}
	1.2	2.0×10^{-7}	4.5×10^{-5}	3.1×10^{-7}	1.1×10^{-5}
	1.5	3.9×10^{-7}	2.1×10^{-4}	9.9×10^{-7}	6.1×10^{-5}
$n + {}^{238}$U	1.0	5.5×10^{-5}	3.7×10^{-4}	4.4×10^{-6}	4.8×10^{-5}
	1.2	9.1×10^{-5}	5.6×10^{-4}	1.2×10^{-5}	1.8×10^{-4}
	1.5	1.4×10^{-4}	2.2×10^{-3}	2.4×10^{-5}	1.3×10^{-3}

Table E.7: Emulator error close to a channel threshold, on $\alpha + {}^{12}\mathrm{C}$, median over 30 validation draws. The two left columns are the median over the elements of $\mathbf{U}$ that exist; the two right columns are the elastic element alone. Within tens of keV of a threshold the first metric collapses, and that collapse is the physics rather than a fault: the channels that have just opened carry almost no flux, so solver and emulator return the same numerical zero for them and a relative error on those elements means nothing, so anything under 10^{-16} is printed as a bound. The elastic element is of order unity at every energy and is what can be compared across the table.

			all elements		elastic element	
Energy	E (MeV)	open	RBM	LROM	RBM	LROM
demonstration	16.000	8	3.1×10^{-3}	1.2×10^{-3}	2.1×10^{-4}	3.2×10^{-4}
$\epsilon_2 + 10$ keV	4.450	4	$< 10^{-16}$	$< 10^{-16}$	1.1×10^{-5}	7.2×10^{-5}
$\epsilon_2 + 50$ keV	4.490	4	$< 10^{-16}$	$< 10^{-16}$	1.1×10^{-5}	7.4×10^{-5}
$\epsilon_2 + 200$ keV	4.640	4	$< 10^{-16}$	$< 10^{-16}$	1.2×10^{-5}	8.1×10^{-5}
$\epsilon_4 + 10$ keV	14.090	8	1.1×10^{-4}	1.1×10^{-4}	1.4×10^{-4}	1.8×10^{-4}
$\epsilon_4 + 50$ keV	14.130	8	1.1×10^{-4}	1.2×10^{-4}	1.5×10^{-4}	1.8×10^{-4}
$\epsilon_4 + 200$ keV	14.280	8	1.6×10^{-4}	1.7×10^{-4}	1.5×10^{-4}	1.8×10^{-4}

Approaching a threshold from above, $k_\alpha \to 0$, the velocity factor of Eq. (53) goes with it and $\mathbf{U}$ acquires a square-root branch point in the energy, the Wigner cusp, at which the family of solutions is not smooth in E. The measurement returns a null result. On the elastic element the error is flat to within fifteen per cent across all three offsets at either threshold, and four to twenty times better at the lower one than at the demonstration energy. The explanation is the protocol, since the cusp is a non-smoothness in E and these emulators are fitted at fixed E, one per node, so what each represents is the variation with θ at an energy that never moves. The corollary lands on the construction this report does not validate: an emulator carrying E as a coordinate would have to represent the branch point itself.

E.7 Online cost against mesh size

Section 5.7 argues that the LROM removes the only term carrying the mesh size, so its advantage should grow with N, over the range the suite spans.

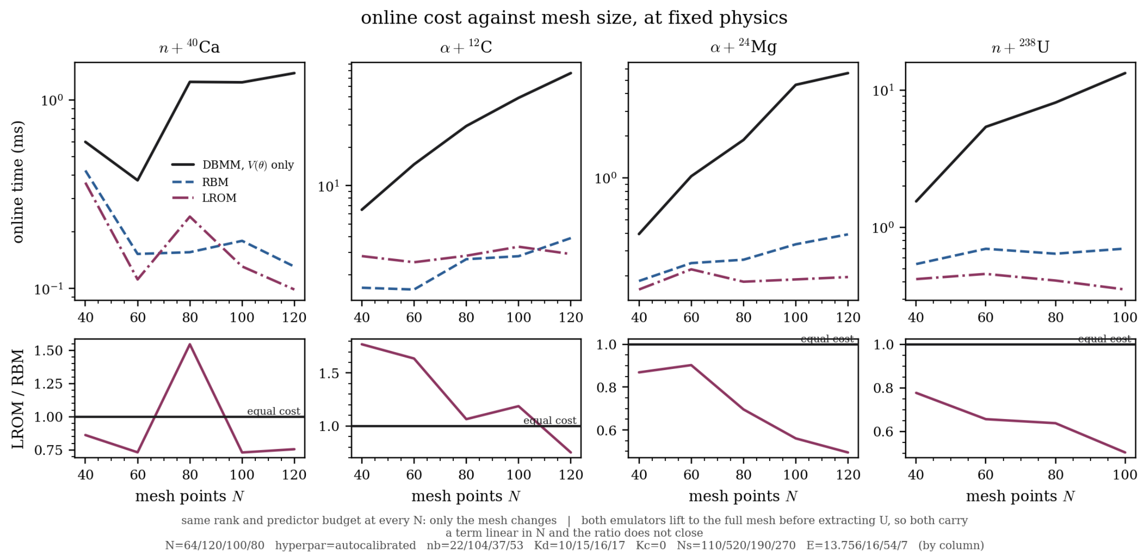

Figure E.6: Online cost against mesh size, at fixed physics, rank and predictor budget. Top row: the two emulators against the split DBMM. Bottom row: their ratio, which falls with N and sits below unity at every working mesh, which is the asymptotic argument of Section 4.3 put to the test.

E.8 How the gain scales with the channel count

Section 6 argues that the advantage should be larger on a problem with many more channels than any benchmark here. This is the only measurement bearing on that, taken by sweeping the channel count within one physical family at fixed everything else.

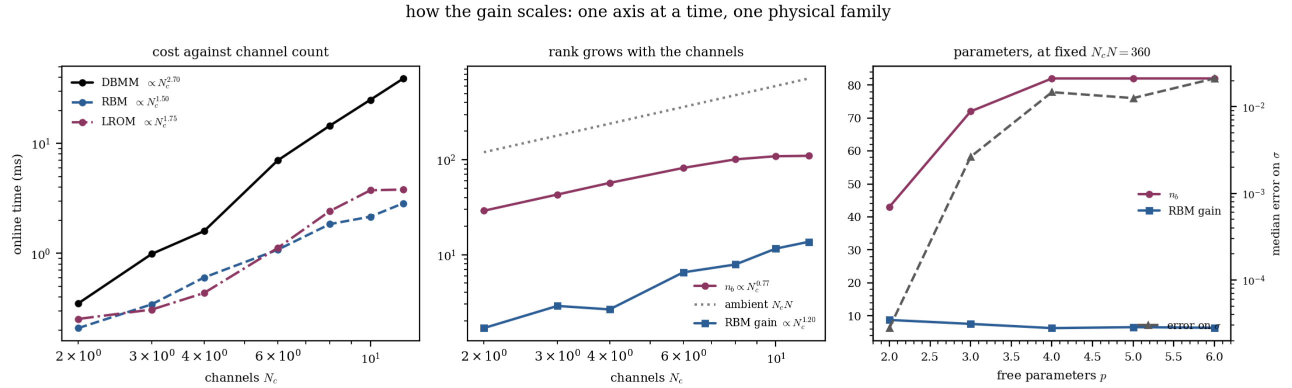

Figure E.7: How the gain scales, one axis at a time, within one physical family. *Left:* online cost against channel count. The direct solve grows as $N_c^{2.70}$ and the two reduced ones as $N_c^{1.50}$ and $N_c^{1.75}$. *Centre:* the retained rank grows as $N_c^{0.77}$ while the ambient dimension grows linearly, so the RBM gain grows as $N_c^{1.20}$. *Right:* at fixed problem size the rank saturates in the number of free parameters, and the gain does not depend on p.

The gain grows with the channel count and does not with the parameter count, over one family at these sizes. The two emulators do not gain at the same rate. The LROM online cost grows as $N_c^{1.75}$ against $N_c^{1.50}$ for the RBM, so its advantage narrows as channels are added, and at the four largest sizes tested the RBM is the cheaper of the two. The reason is the one Section 6.3 gives: the learned operator differs from one entrance channel to the next, so the reduced solve is N_c^{open} factorisations where the RBM performs one. Section 6.3 carries this to six hundred channels.

E.9 The control benchmarks

$p+{}^{12}\mathrm{C}$ and $\alpha+d$ are the systems the machinery was first shown correct on. They carry no result of the comparison, and their sheets are collected here.

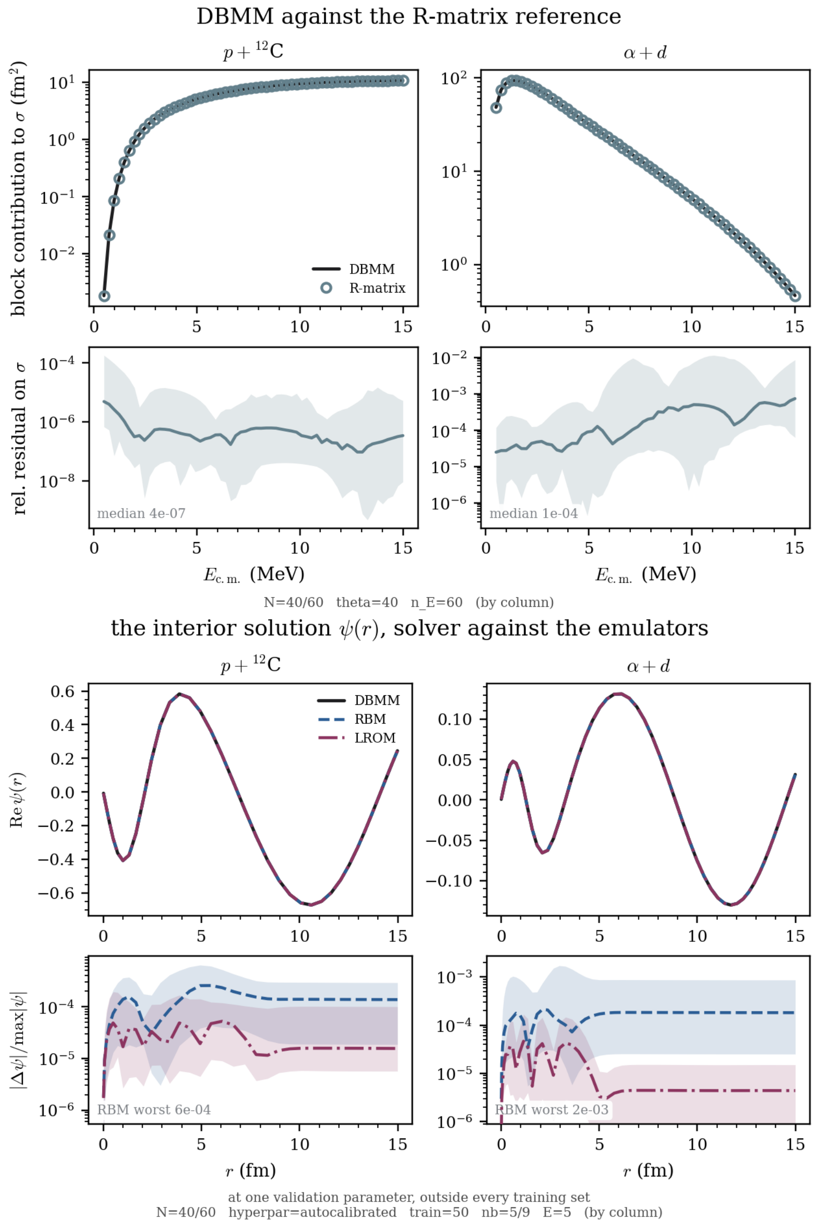

Figure E.8: The two control benchmarks against the R-matrix code. *Above:* the cross section, with the relative residual beneath, median and p5–p95 over 40 parameter draws. *Below:* the interior solution, solver against both emulators. The two components of the tensor-coupled $\alpha+d$ system are the harder of the two, since the D wave sits behind a centrifugal barrier.

E.10 Exploratory extensions

Two variants on the RBM branch are operational but outside the comparison of Section 5, which measures four codes on the four benchmarks and the two controls. These were exercised on two; Table E.8 says what each does.

Table E.8: The two variants, what each does, and what validating it would take. A speedup measured on two systems is not reported as though it held on six, which is why neither appears in Section 5.

Variant	What it does	Exercised on	What validating needs
RBM-ET, joint (θ, E) basis `core/rbm_et.py`	$\mathbf{K}$ is affine in E, so one basis covers an energy window instead of one per node. $\gamma_{s,\alpha}(E)$ and the source $\mathbf{b}(E)$ are not affine and are tabulated offline; the source is the onerous one, a full mesh per channel. One basis per interval between thresholds, since crossing one changes the dimension represented.	$n + {}^{40}\mathrm{Ca}$, $\alpha + {}^{24}\mathrm{Mg}$	The suite, the out-of-sample protocol and the timing of Section 5, plus a check that the dispatcher introduces no discontinuity at a window boundary
RBM-ETP, predictor points on the RBM projection	Replaces the mesh loop of Eq. (17) by evaluations at maxvol-selected points, inside the Galerkin projection rather than instead of it.	$n + {}^{40}\mathrm{Ca}$	The same, plus a measurement separating what the LROM's lead owes to K points from what it owes to its learned operator

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